

A Concise Guide to Medical Research

Michael Brimacombe

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A Concise Guide to Medical Research

This book is a concise beginner's introduction to medical research written for biomedical researchers, medical students, young faculty, and clinicians who would like a readable guide to the research process, in particular, the medical and biomedical research process.

In medical research, where treatments for illnesses and diseases are developed for the human population, a variety of genetic backgrounds, diets, and lifestyles often limit the ability of researchers to control extraneous sources of variation and clearly observe treatment effects. Probability-based statistical models are often used to control these secondary variables after the experiment has ended.

The chapters of this book provide guidance on many aspects in the process of doing medical research. From defining a useful idea and developing a proof-of-concept pilot study to formally testing a new treatment, this book provides an overview of how to develop and carry out a basic research project, analyze, and report the data. Throughout, the importance of planning, support structure, and research ethics is emphasized, and several basic examples are used to motivate the discussion.

Key Features:

- Easy to read with examples.
- Easy to understand.
- Provides a simple outline that can be followed in many practical settings.
- Provides a simple guide to basic statistics useful in medicine.
- Provides a pathway into medical research.

Dr. Michael Brimacombe is a biostatistician, author, and medical researcher who has published over 115 articles in peer-reviewed journals. He has taught many undergraduate and graduate courses in statistics, biostatistics, and computational statistics at several U.S. academic institutions. He has been Director of NIH-funded Data Management and Biostatistics Core and Director of CDC-funded FAS Educational Center. He is currently Senior Advisory Biostatistician at CT Children's and Associate Professor at the University of Connecticut Medical School. He is a previously published author with Taylor & Francis (Brimacombe M. (2019). *Likelihood Methods in Biology and Ecology: A Modern Approach to Statistics*. CRC Press, New York).



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Author Preface and Overview of the Book

This book is a concise beginner's introduction to medical research, and research in general, written for biomedical researchers, medical students, young faculty, scientists, and clinicians who would like a readable guide to the research process, in particular, the medical and biomedical research process.

Research in health, medical, and biomedical areas is important and rewarding but can be challenging to organize and conduct.

Conducting research is a process, usually with a timeline and limited funding. So it is important to plan carefully the steps necessary to be successful. This book takes the reader through these multiple steps, one at a time.

The level of the book is introductory and provides an overview of the research process. The various sections can also be read on a stand-alone basis.

While each research setting can differ, the steps tend to be similar in many settings, such as research labs, universities, medical schools, or industrial research. The overall plan often follows the same basic steps:

- Identify research goals and ideas worth pursuing.
- Find a mentor and a research support team for the project.
- Find sources of pilot data that you can use to demonstrate the basic proof of your concept.
- Develop a study design to guide the collection of data that is appropriate to your concept and related hypotheses.
- Implement and carry out your study.
- Carry out statistical and data analyses showing that your hypothesis is supported by the data.
- Summarize and present the results.

These steps are laid out and discussed in this book in some detail for the reader.

As much research requires some understanding of statistics as part of the planning of studies and the analysis of data, a brief introduction to basic statistical ideas and their application in study designs is given, as well as sections addressing basic epidemiology and several newer research areas.

Discussion points are available at the end of most chapters.

Medical Research Studies

Research is rarely taught from a book. It is usually learned in the context of time spent as a graduate student or research fellow or through mentoring by established researchers. What is learned is often specific to the area of application and discipline being studied.

But many areas share basic structures and approaches when it comes to skills required for research. This is especially true in medical research.

The standard practices that guide research are meant to aid in the development of experiments and studies that will yield clearly interpretable results. These practices do not guarantee success. In fact, often, as in the case of the discovery of antibiotics, a new result can be due as much to luck as to serious study. But they provide a set of methods and approaches that, if a useful result is obtained, can be understood by other peer researchers and thus replicated and verified.

If research practices are not held to a high standard, results supporting new treatments cannot be trusted and may not be verifiable or replicated.

A carefully developed study design helps to develop a very long list of medical, public health, and industrial advances and results in improved health and longevity in many parts of the world. Most pharmaceuticals and industrial processes use formal methods of testing, often applying the statistical design and analysis of experiments.

Hospitals and medical schools provide useful settings for biomedical research. They provide a natural environment for such research since on a daily basis, they experience treatments, medicines, various types of patients, the application of medical knowledge, and a variety of medical experts all concentrated in the same place.

In medical research, where treatments for illnesses and diseases are developed for the human population, a variety of patient genetic backgrounds, diets, and lifestyles often limit the ability of researchers to clearly observe treatment effects. Probability-based statistical models are often used to control these secondary variables after the experiment has ended.

The chapters of this book provide guidance on many aspects of the process of doing medical research. From defining a useful idea and developing a proof-of-concept pilot study to formally testing a new treatment, this book provides an overview of how to develop and carry out a basic research project and analyze and report the data.

Throughout, the importance of planning, support structure, and research ethics is emphasized, and several basic examples are used to motivate the discussion.

Developing Medical Research Ideas and Defining Goals

Research ideas are as varied as research itself. In the early stages of a career, they are often reflective of the interests of a young researcher's mentor. This should not remain the case as the researcher grows in independence and maturity. It is important to develop your own interests and ideas.

Research ideas can be as simple as reading the research literature in one's own field and replicating interesting studies done in other settings in your own research environment. This can also include demonstrating the use of new types of equipment, new testing methods, and new treatments for illnesses.

It can also involve the questioning of the existing treatments and methods, seeking to improve and make more efficient the existing medical treatments.

Research ideas can also be developed from within the community, examining the care-giver burden of patient populations such as HIV-positive patients or individuals with autism or the effects of COVID on community health. Disparities in access to medical services across socioeconomic groups are also of interest.

If research interests are more process driven, research ideas may, for example, relate to developing more effective clinical treatments or lowering the length of stay by increasing the speed of recovery. Access to the existing internal clinical databases may allow for the study of patterns of resource usage by various types of patients.

The specific research project that can be accessed often depends on the specific combination of resources provided by a mentor, research support team, institution, and training program.

Pilot Data, Mentor, and Research Team Support

The development of a research project, once you have an idea, requires the development of a series of steps. These will often culminate in the carrying out of a research study, the presentation of results to the research community, and the eventual publication of the findings in a peer-reviewed journal.

The most important step, in a formal medical school or academic setting, is the obtaining of mentor support. Access to a lab, computing environment, data collection, and the opportunity to interact with the expertise necessary to discuss and guide the research process and final results can help along in this way.

Support should also include access to data, either data in the form of already collected electronic databases that can be used for retrospective cohort studies or data collected from ongoing clinical studies.

A small initial dataset, pilot data, is often necessary to give proof of concept for your research idea. At a minimum, pilot data should be representative of the population of interest, and there should be adequate sample size to provide accuracy in estimating population characteristics and testing hypotheses.

Pilot data allows you to show that the research concept is sound and the carrying out of a full research project is possible.

Example

A basic case study example with data is used to motivate the discussion throughout the book. Other data is also examined. The example reflects various common data structures and study designs. The dataset includes medical and biomedical responses of interest, explanatory variables, demographics, clinical measures, genetics, and other variables.

The study to be discussed includes:

- A cohort study.
- A case-control study.

Study Design and Basic Statistical Questions

The most important practical challenge once your research idea is clear and basic support is found is to choose the appropriate scientific design for your study and data collection. The design of your study provides a context within which you can test the hypotheses of interest to you.

The design also relates to basic issues such as the obtaining of patient consent, so it should be addressed early in the research process.

The study design will determine both the research personnel and patient sample size required. The budget for your study will have to reflect these as well.

Statistical designs appropriate for your study can be developed to give accuracy in modeling patterns of interest in the data, seeing through the clutter of random variation in measurements.

Some initial statistical questions include:

- What statistical design is appropriate for the study you will carry out?
- Is there enough sample size to properly test the specific hypotheses of interest or estimate clinical treatment effects?
- Do you have the support to recruit and consent a sample of patients, collect their information, and enter it into a database?
- Do you have the computational expertise and support to develop the database and carry out the various required statistical and data analyses?

Statistical issues often reflect the nature of the data (sample size, scale of data, correlation among the measurements), the way in which the data is collected (randomization, the type of study), the hypotheses of interest, and the set of reasonable models or data analytic settings within which the hypotheses can be validated with real-world observations and data.

There are different statistical perspectives available (frequentist (standard), non-parametric, Bayesian, or machine learning). Here, the standard frequentist approach is mostly discussed.

Deciding on a Specific Study Design

There are many possible study design templates for the collection and analysis of data. It is important to select the one that is most practically relevant to your research project.

Some of these include (and are discussed in later chapters):

- Laboratory-based studies, basic science research, often mouse-based research.
- Randomized control studies using consented patients.
- Simple case reports – for example, the first few times COVID or AIDS was observed.
- Retrospective descriptive studies – often using data pulled from large electronic databases.
- Prospective cohort studies – for example, used in treatment development, where a cohort of infected patients is arranged, all of them inoculated with the new treatment, and all are observed to see who recovers and who does not.

- Case-control studies – a group of cases (infected or ill) is compared to a control or reference group.
- Community-based research – this examines public health access, the existing health disparities, and the epidemiology of illness.
- Meta-analysis and systematic reviews of the literature – these may not require an actual database but may require lengthy searches of the published literature, transforming the results obtained in each study into an overall summary.

Statistical Issues Arising in Designing a Research Study

Each study design has its own set of probability distributions, assumptions, and techniques for assessing hypotheses and estimating population characteristics. As a lead researcher, you should be aware of some of these issues.

There are a few key common aspects:

- The need to limit bias in patient selection and the measurement process: The sample should not favor healthier subjects, for example.
- Randomization of patient selection.
- Representative sampling.
- Sample size and the accuracy of estimation.
- Timelines for the study and follow-up issues.
- Missing data: Sometimes missing data patterns can induce a nonrepresentative or biased sample.
- Clinical effect size versus statistical significance: When the sample size is very large, it is possible that very small clinical effects will be deemed statistically significant. Sample size justification-based calculations can limit this possibility by determining reasonable sample sizes that will detect the desired clinical effect, if present.
- Reporting results: Model summary, tests of significance, and confidence intervals.

A typical design-related consideration applicable to most studies is the matching of randomly selected subjects in the control and treatment groups by age and gender. This limits secondary variation due to differences in age and gender and should be used when those variables are possible sources of variation.

Basic Probability, Statistical Modeling, and Data Analysis for Medical Studies

It is important to learn enough about probability and statistical models to understand how they are used in relation to study design and data collection. This includes outputs such as p-values and confidence intervals and also the assumptions that underlie them.

Understanding these concepts will allow you to ask the right questions when interacting with colleagues who are computer and data oriented and speak (as much as possible) the language of data analysis and statistical inference. It will also help you discuss statistical designs with statisticians and those helping you plan your study.

Basic reviews of the essential concepts are provided in this book.

Large Databases

Even with good research practices, some areas advance slowly. Most research builds step-by-step on the existing knowledge. Occasionally, a new paradigm surfaces, providing insights into the onset of a disease. The creation of very large, integrated electronic health databases available for analysis is such an advance.

Genomics, for example, reflects this advance. This research has led to the use of massive datasets in the collection of gene expression data and attempts to link patterns of gene expression with the onset of a disease. This has led to the growing use of genetic considerations in the study of diseases, including incidence, prevalence, and immunity.

An advance related to genomics is the study of large health-related databases. Here the idea is that keeping track of many connected bits of health data may allow for the development of individual health profiles that will better explain the onset, prevention, and treatment of diseases.

This is a growing field, and the use of data-based analysis techniques, for example, in the form of artificial intelligence (AI) and machine learning techniques such as recursive partitioning and predictive neural networks, is becoming common. The term "evidence-based medicine" is sometimes mentioned. Note the related issues of personal privacy.

All this data is leading to a broader set of data analytic techniques. Often, older techniques are being re-developed to deal with the new data-oriented reality. New approaches include recursive partitioning (looking for clusters and structures in the data) and artificial neural networks (data-based predictive classification methods). These can often be integrated into the available set of statistical and data analyses that help identify and model results in medical research.

Review of Basic Concepts for Infectious Diseases and Epidemiology

Infectious diseases are an old enemy. The use of probability and statistical models to examine the patterns of infectious diseases are a large part of epidemiology. The SARS-Covid pandemic was a harsh wake-up call to the inevitability of infectious diseases in the human population. The extent of the impact an epidemic is almost always related to social and national structures, and thus politics, economics, and ecology are often closely related to epidemics.

For example, the political collapse of the Soviet Union (HIV and antibiotic-resistant TB, along with much shorter expected life spans) or the international structures of global economic supply chains (increasing the global impact of local epidemics).

Locally, epidemiology is closely linked to the many aspects of public health, for example, proper vaccinations and early childhood health and the use of statistical methods. Research in this area often means outreach into the community or the collection of data in outreach and pediatric clinics.

Background data from local state and federal public health databases may also be required to assess baseline levels of infection and infection rates.

Administrative Issues: IRB, HIPAA, Consent Guidelines

The paperwork necessary to carry out research in an accredited institution can be daunting. Because of past abuses and poor record keeping in regard to research, all current research projects in a formal medical setting are reviewed by various internal boards and committees. The particular set will depend on the institution in which you are conducting research.

Often there is a Scientific Review Committee (SRC) and/or a formal Institutional Review Board (IRB) which reviews all research projects.

If you intend to access electronic health records, you must follow (in the United States) the federal HIPAA guidelines. You will also need to take some training in how to develop and administer consent forms to subjects participating in the research project. These issues and procedures are briefly reviewed here.

This careful approach to documentation and consent is the by-product of past abuses. As such, ethics in research is very important, and these aspects of research need to be focused on and discussed with your local IRB and research administrators of your institution.

Each public or private research institution recognized by the federal government, public and private, will have administrators and training courses to guide you.

Topics for Future Consideration

A brief overview is provided of some newer and interesting areas of possible research.

Genetic and Epigenetic-Related Studies

Genetics in relation to health research is both old and new. Genetics has been a component of animal breeding and related research for many years, and there exist statistical techniques that parse out genetic versus environmental effects on a phenotype of interest. But the recent development of genomics and Affymetrix chips that contain data on 500,000 single nucleotide polymorphisms (SNPs) or portions of chromosomes has led to the incorporation of gene expression data in almost all areas of medical research.

The meaning of all these detected gene expressions in relation to medical outcomes has yet to be sorted out. But advances have been made, especially in relation to basic processes in the context of basic biological research.

Epigenetics, the study of short-term alterations in the genes and their expression due to alterations in the environment, is a source of ongoing discussion. And the general question of whether genetic factors truly explain a large portion of the variation in an outcome or phenotype of interest or are less important than environmental exposure and the presence of virus, bacteria, and other infectious causes of illness is an open one.

The relationship between standard clinical assessments and genetic information is of growing interest. Genetic assessments may improve the understanding of potential outcomes of the disease in question and help indicate optimal treatments.

From a practical research perspective, genetics is now affordable but presents a serious issue in terms of subject privacy and truly informed consent. Many clinical assessments might benefit from the addition of a formal genetic test, but this must be incorporated carefully into your study design and analysis.

Introductory Aspects of Machine Learning

The presence of very large health-related databases in the clinic, hospital, and medical school context offers opportunities for data-oriented research using data analytic methods. These are related to probability and statistical methods but are often very focused on the actual sample in hand. Drawing data samples from the existing databases must be done carefully if they are to be representative of the study database and subject population as a whole.

Retrospective cohort designs are typical when using the existing data. The methods for pattern recognition in relation, for example, to unplanned

seven-day readmissions in a hospital setting are very computing intensive and include machine learning methods such as neural nets (comparable to logistic regression), recursive partitioning (finding clusters in the dataset) and more general AI-based assessments.

This is a new field, and the methodology is still developing, but many early-stage medical researchers may have access to clinical databases and many initial projects may be data analytic in nature. This is discussed in a separate chapter.

Personalized Medicine: Microbiome

Personalized medicine is becoming a possibility and perhaps a necessity in the study of diseases such as populations of cancer cells and conditions related to the microbiome – a collection of bacteria (as many as 40 million varieties) that help the human digestive and metabolic systems to function. The idea that such phenomena are often unique to the person being studied has led to the possibility of medicines modified and developed especially for each individual.

As standard methods seem to be only slowly developing, this area is growing and, for lab-oriented medical research studies, may provide interesting research opportunities.

Summary

These topics reflect the tools necessary for designing, developing, and successfully completing a medical research project. Much of the material can be applied to biomedical research projects in general.

Each chapter contains a checklist of suggested actions focusing on making the material here relevant to a specific research project. Some assignable questions, projects, and discussion points are made available at the end of several chapters.

At the end of this book, readers should have an overview of how to develop a basic research project, carry out the project and collect relevant data, analyze and report the data, and have the level of commitment necessary to carry forward an initial idea to a completed peer-reviewed research paper or grant application.

Michael Brimacombe, Ph.D.

Teaching, Learning, and, Using This Book

Overview

The many steps necessary to conceive, develop, and carry out a research project are challenging. But like all complex tasks, they are primarily composed of a series of basic steps. A basic general guideline is provided here for new researchers looking to begin developing research projects.

Research as a Process

Research projects are best carried out in the context of a research career focus. If possible, research projects should reflect the interests of the researcher. Research is always best pursued if it is motivated by a desire to learn and investigate new possibilities and applications. The challenge is to develop an interesting idea that can find support and be developed in a formal setting where your research project can be appreciated, mentored, and brought to life.

The first series of chapters provide guidance on the practical challenges that arise when beginning a research project. These include:

- Developing a research team.
- Accessing pilot data.
- Developing a research goal and set of ideas.
- Finding a mentor.
- Dealing with administrative issues such as protocol development and Institutional Review Board evaluation.
- Accessing the resources and expertise needed, such as laboratory support and statistical support.
- Setting a timeline for completing the research project.

How these are developed will reflect the institution within which you are working. They are all necessary first steps in developing the research team that can support your efforts and achieve a successful outcome within the time frame and funding available.

As a Textbook

The material in this book may serve as a useful reference for presentations and short courses focused on the "how-to" of research and the challenges of beginning research projects. It may also be used in the context of training programs related to HIPAA, IRB, and other aspects of research in various institutional settings.

As a suggestion, each student can be tasked with developing a research concept or idea and the writing of a formal protocol, developing the project as the various chapters are reviewed and discussed.

The administrative aspects covered in the book can be modified to reflect the actual research structures that each student has to deal with, especially the local IRB.

The dataset that is relevant to Chapter 2 and provides examples throughout the book is available upon request.

As a Reference Guide

The chapters in this book can be read in various ways.

The first six chapters deal with the important issues of how to develop research ideas; concepts and projects; and the supporting effort, team, and administrative work required to make it all happen. They also examine possible study designs and related goals and hypotheses.

Chapter 7 gives an overview of the two basic approaches to collecting and interpreting data values: Quantitative and Qualitative data analyses. Qualitative data allows for a more free-form approach to obtaining data. This type of data can be used to examine self-reported well-being in the patient, as well as how they feel they are being treated in the healthcare system.

Chapters 8–11 give a basic review of the core concepts related to quantitative data analysis: basic probability concepts, an introduction to probability sampling distributions, statistical estimation and hypothesis testing, and the application of predictive models – statistical and machine learning based.

Chapter 12 examines the development of research projects that employ large electronic databases to develop research data. The use of such resources involves their own set of challenges.

The book ends with a brief overview and discussion of several newer areas of research that may be of interest: genomics/genetics, machine learning, and personalized medicine. The chapters emphasize the investigation of how new perspectives and ideas can be brought into the existing areas of medical research. These various aspects and areas of focus in the book can be organized:

Table of Chapters

Getting Started	Statistical and Data Analysis Topics	Other Topics
• Developing Research Goals • Developing a Research Team • Mentoring • Initial Pilot Data • Administrative Issues • Defining a Research Timeline • Study Design and Research Protocol Development	• Basic Data Analysis and Data Summaries • Basic Probability • Sampling Distributions, p-values, and Confidence Intervals • Statistical Inference • Statistical Models • Linear Models • Logistic Regression • Survival Analysis and Time to Event Models	• Machine Learning/AI • Personalized Medicine • Qualitative and Quantitative Research

Learning the Material for Practical Application

The example study with dataset given in Chapter 2 is examined throughout various chapters in the book. This provides a practical focus for study, mostly related to data analysis. This can be used as a thread to guide progress through most of the book. The Chapters 1–6 define general, practical principles to guide the development of research ideas and related research practices: including a focus on ethical conduct and clearly defined hypotheses. Other examples are introduced as appropriate throughout the book.

The importance of carefully defining the context for a research project should be emphasized. This includes the need for access to data, access to specific clinical settings relevant to data collection, the recruitment of subjects, and the obtaining of proper consent. An underlying context is timeline. It is often very useful to identify a finishing point (time of graduation for students, endpoint of the grant, etc.) to guide the development and speed of the project. A formal timeline, giving monthly goals, is often a very useful resource to keep all members of the research team on the same page.

Several templates and checklists are given throughout the book.

Teaching Concepts

The material in the book can also be used for the teaching of specific concepts. These include:

- Developing a research idea and related project.
- The need for a clear study design.
- Developing a research team.

- The definition of a related set of goals and hypotheses.
- Qualitative or quantitative data modules and collection.
- Basic data science and related statistical analyses, including numerical and graphical summaries, tables of results, statistical models for hypothesis testing, and possible prediction of future values.
- Statistical estimation and testing and related probability and statistical concepts, including the need for careful follow-up and a need to limit missing data values.
- Basic statistical models.
- Basic epidemiological ideas and analyses.
- An introduction to the basic context of machine learning and AI-related data analysis.

Many of the chapters related to the basic ideas of probability and statistical inference associated with medical research are stand-alone chapters written to minimize mathematical detail.

Discussion Points

1. At the end of almost all chapters, there are a series of discussion points that are given to help guide discussion, study, and related development of a research project on the part of the student or researcher.
2. These discussion points can be the basis of a set of questions assigned as part of developing a research project by the student or researcher.
3. Many are open-ended questions, relevant to the material presented in the relevant chapter, to encourage discussion and application to a wide variety of potential research topics.

1

Medical Research: Goals and Ideas

Before you can work on the details of a research project, you need research ideas and related research goals. These need to be reasonable within the context of where you are studying or working.

Conceptualizing, organizing, and carrying out a research project can be challenging.

Research requires an understanding of a particular area of medicine or science, and an idea regarding a treatment or procedure or health outcome that can be measured and assessed using a study design and related research methodology.

- It requires the development of a formal overall research strategy or protocol.
- It will often require the collecting of data organized into databases that can be analyzed.
- The development of research questions and a set of related specific aims to investigate is a particular challenge.
- The question of where to find good research ideas is an interesting and challenging one.
- Ideas can be simple, for example the reporting of adherence to good clinical practice criteria in a hospital clinic.
- They can also be complicated, for example, a detailed, multi-arm study of the efficacy of a new pharmaceutical treatment regime in relation to the treatment of a specific ailment (diabetes, hypertension, obesity, cancer etc.).

The research idea you finally choose should reflect both your interests and what is practically do-able in your institutional setting.

1.1 Searching for Research Project Ideas and Possibilities

Ideally, medical research is conducted under a mentor's guidance or in a mentor's lab. The research questions of interest are well-defined and

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practically do-able within the research setting. This is often the context for funded research and much biomedical research. The research questions and resources have been obtained and are already present when the young researcher, medical student or graduate student is recruited.

However ideal situations may not occur.

Or students may wish to begin pursuing their own ideas.

In settings where resources are limited, there are some standard approaches and places to begin developing research goals and ideas within a given field of research.

This often requires accessing existing published research and may involve the use of publicly available sources of data, for example the medical research listing called PubMed [1], or U.S. census data [2] or the National Survey of Children's Health [3]. Clinical trials can be found listed at clinicaltrials.gov.

Types of possible of very general research projects requiring relatively few resources other than time include:

- Systematic literature reviews.
- Meta-analyses (literature reviews along with meta-data).
- Electronic Data reviews (What datasets are available and what information do they represent).
- Validation studies of existing research results (part of health outcomes research and clinical practice improvement).
- Clinical trials review (really a specific meta-analysis).
- Small internal surveys, (using a local clinic or an internal mailing list).

1.2 Meta-Analysis/Literature Review

To understand the research ongoing in a specific area of medicine or science it is very helpful to review the published research over the past 3–5 years in that field. There are several database collections that are helpful.

- PubMed is a very user-friendly collection of general medical research publications that can be searched using author or keywords.
- Other disease-specific literature databases are also useful. See, for example, the Department of Defense Breast Cancer research listing [4].
- This type of resource will give you abstracts or summaries of the papers corresponding to your keyword search parameters. If the papers themselves can be accessed, then you can develop a systematic review of the literature.

- If there are reported statistics relevant to your interests in the set of collected research papers, your review might take the form of a meta-analysis, with the data for the paper being the set of summary statistics collected from each of the set of published papers. This will require specialized statistical analysis.

This is often an excellent place to begin developing ideas for a research program as it provides you with an opportunity to review an entire area of research and may give you a publishable review paper.

1.3 Electronic Data

- If you have a specific medical interest and want to begin examining databases to see if there is sufficient data with which to build a research study, many institutions such as medical schools and hospitals have internal databases that can provide enough detail to be used in research studies.
- Many hospitals contribute to the EPIC [5] and EPIC Cosmos [6] hospital utilization databases.
- In addition to basic data, these databases can guide you, when reviewing published research, in identifying and choosing a set of key measurement variables relevant to the outcomes you are interested in studying.
- There are also listings of federally funded clinical trials [7], that may be accessed online.
- If you have some expertise in genetics, there is NCBI [8] which gives access to publicly available genetic data.
- CDC (WONDER) [9] public health-oriented results can be accessed if community-based analysis are to be studied.
- The COI database (Children's Opportunity Index) [10] can be linked to zip codes and can be useful for examining disparities in socioeconomic related aspects of health research.
- The U.S. Census site [2] is also very useful for a wide variety of socioeconomic background data.

1.4 Studies Validating Existing Studies

- An important area that is sometimes applicable is the assessment of newly reported treatments.

- Many initial pilot studies are often under-powered, meaning the sample size is small.
- As such many newly reported treatments require assessment in both similar settings and related areas of application.
- These types of studies may be reported in PubMed and clinicaltrials.gov and other internet summary sites. For example, a new COVID treatment or pilot study cancer treatment.

1.5 Clinical Trials

- In a specific area of research there are often many past and ongoing clinical trials of new treatments. Some are successful and some are not.
- It is difficult to track the results of many studies as they tend to be industry funded and are usually pilot studies with smaller sample sizes.
- The Cochrane Collaboration [11] is a useful site for investigating ongoing, reported clinical trials.
- These are also useful resources for possible ideas, treatments, and developing modifications suitable for the institution and area of medicine in which you are working.

1.6 Administer Your Own Survey

- Studies focused on internal aspects of the institution, such as quality of care, are also a possibility.
- Surveying faculty, staff or students themselves is often a useful setting for focused, internal studies. This can then be compared to similar studies at other institutions and more generally.
- This approach is often very useful if you are at an institution such as a hospital which has a clinical database of patients and family.
- Many treatments require follow-up and may carry serious caregiver burden responsibilities.
- These are often of interest to the institution and state departments of health and federal health and human services.

- Note that there may be a need for survey validation via expert review. Usually, a small group of experts or colleagues are asked to review a survey for completeness and relevance.
- It is important always to standardize, as much as possible, the entry of study data. Use a standardized data entry tool such as REDCap [12] if possible.

1.7 Spatial Analyses (GIS Patterns)

- If you have access to GIS spatial related information, a growing field of data, and adequate census tract or community-based reporting, then you can examine the spread of a disease through time using spatial analysis [13].
- This is often fairly simple and is primarily graphical, using, for example, heat maps to depict the spread of a disease.
- The programs to carry out such analyses may require some statistical expertise, but many are fairly easy to learn.

1.8 Data Sources Summary

Curated and standardized resources exist in which you can access literature and possibly initial data for your review:

- PubMed: Keyword searches may allow you to review ongoing research very close to your own interests.
- clinicaltrials.gov: An active listing of clinical trials.
- U.S. census data: census tract background data
- Databases in your field: (e.g., DOD Breast cancer, COI)
- Genetic databases at NIH: GenBank at NCBI.
- Electronic National and International Databases: primarily national databases such as EPIC, PHIS [14], KID [15], NCHS, CDC WONDER, and international databases such as WHO [16], and World Bank [17]).
- Local Clinical files: An optimal source of data, if it can be accessed in a manner that allows for detailed study. These may involve a review of older paper files. These are rare but not all records have been digitized.

- Small local studies may require entering data directly into an Excel or other database. It is unfortunate, but survey data, even if the results of electronic data entry, needs to be placed in a standardized file format to be analyzed. The REDCap data collection system is very helpful in this regard.

Some databases require permission and perhaps training before you can use them. Depending on your location and access, these may include, for example, EPIC and PHIS.

1.9 Developing Research Goals and Hypotheses

The development of a primary research goal, along with specific aims and precise hypotheses can take time and is usually the result of insight into the scientific area in question. This insight may arise from carrying out a comprehensive literature review in the area, or experience in the specific area. Or it may be developed in the context of the continuing research program of a mentor.

Once a basic research idea is determined, developing an appropriate experiment or study which can test the basic hypothesis and related ideas is a major challenge.

There are several steps to think about:

Goals: These tend to be over-arching, related to the general area of medicine in which you are working. For example, Improving cancer treatments for patients with a specific form of cancer.

Specific Aims: These are much more focused, including both primary and secondary aims. For example, carrying out a cohort study to examine how well patients with various levels of cancer severity respond to a new drug, perhaps across a set of dosage levels. A secondary aim might be to assess possible side effects of the treatment.

Primary Hypothesis: Should be written so they can be formally tested in the context of a specific research design and related statistical analysis. For example, the use of a new pharmaceutical treatment for breast cancer will reduce tumor size by an average of 90%, compared to baseline.

You may have more than one specific interest. If so, in addition to the primary goal, list a series of related secondary hypotheses.

Secondary Hypotheses: Some standard secondary analysis include sub-group analysis, caregiver burden, the ability of patients to adhere to the treatment over time, and quality of life measures for patients.

Some additional databases are listed below.

1.9.1 Example 1

A study is to be conducted to examine the use of pharmaceutical based intervention versus surgery in the treatment of Crohn's disease.

Goal: To compare long-term pharmaceutical treatment of Crohn's disease versus surgery.

Specific Aims: For age and gender matched treatment groups, the success of pharmaceutical treatment is evaluated and compared to successful outcomes using surgery.

Primary Hypotheses: The two-year success rate in the pharmaceutical treatment groups will be 20% higher than in the surgical treatment group.

Secondary Hypotheses: Measures of caregiver burden and patient quality of life will be better in the pharmaceutical treatment group.

1.9.2 Example 2

A new, less expensive vaccine to deal with COVID and related respiratory illnesses has been developed and requires testing and verification.

Goal: To develop a new less expensive vaccine that provides similar levels of successful treatment to the current gold standard.

Specific Aims: To develop a randomized control trial to compare the use of two vaccines in similar cohorts of people and examine the relative properties of the new vaccine versus the gold standard.

Primary Hypotheses: The sensitivity, specificity, and false positive rate of the new vaccine in regard to the gold standard will be, respectively, 85%, 95%, and 2%.

Secondary Hypotheses: The results obtained will be examined by age quartile and gender. The number of extreme negative reactions, for example, blood clots and heart related problems, will be lower in the new treatment compared to the gold standard.

1.9.3 Example 3

Program evaluation of new approach to teaching health professionals about Fetal Alcohol Syndrome (FAS) is to be carried out.

Goal: Develop and initiate a new educational program for medical and health professionals to better educate in regard to FAS.

Specific Aims: To increase the knowledge of medical and health professionals by developing educational interventions and programs.

Primary Hypotheses: To show that educational presentations increase the knowledge among health professional of the diagnosis and treatment of FAS using pre and post comparison of an FAS based assessment instrument.

Secondary Hypotheses: Assess the extent of caregiver burden related to FAS, identify any differences in FAS occurrence across socioeconomic and gender groups.

In the end, the results of your research project will have to convince peer reviewers in your field. It is in your interest to be as clear as possible at each step of the research process, focusing on the specific set of ideas you have chosen to assess and interpret in the context of the medical literature.

1.10 Practical Issues

- The feasibility of the resulting research project should be assessed.
- Do the resources exist to carry out the research in the chosen setting?
- Can adequate numbers of cases and controls be identified in a reasonable time frame?
- Are experts accessible to you (biostatistics, genetic analysis of samples, pathology, physiology) and can they be drawn together to form a research team for the purpose of the grant? Much modern science requires contributions from experts in several scientific areas.
- The difficulty of addressing these issues successfully should not be underestimated.

1.11 Research Project Checklist

In this book various aspects of the research process are reviewed in a step-by-step manner.

An initial research To-Do checklist might look like the following:

- Define your basic idea; what to study, what changes are of interest, what patterns are to be assessed.
- Population to be studied.

- Setting Goals.
- Specifying Specific Aims and Related Hypotheses.
- Pilot Data.
- Research Team.
- Designing the Experiment or Study. Once a feasible research goal has been specified, and this can be very flexible, a set of steps should be carried out to shape the idea into a form that can be investigated using the standard templates of scientific research. This typically involves defining a study design for the project. Some basic study designs include:
  - **Cohort studies**: This is primarily when you have a group of patients that are exposed to a potentially infectious agent and you follow them through time to see if, indeed, they become ill. The same design can be used to examine the use of a new drug treatment on a group of patients who are ill, and you follow them through time to see how many become healthy.
  - **Case-control studies**: When you already have a set of cases and want to compare them to a set of controls. The challenge is to randomly select a set of controls (usually 2 controls for each case) being sure to match the controls to the cases on relevant secondary variables (age, gender, etc.).
  - **Randomized control studies**: This is often viewed as the ideal design for testing a new medical treatment. This is a prospective study. Each new patient consented into the study is randomly assigned to either treatment or control groups. The treatment group receives the drug and the control group receives a placebo. They are then compared for differences at the end of the study.
- Feasibility of the study timeline. Set a deadline at some point in the future. Now work backwards to set your timeline for developing the various aspects of your research project.
- Administration and Paperwork.
- Sample and Data Collection.
- Data Analysis, Statistical Models, Estimation, and Hypothesis Testing.
- Reporting Results and Conclusions.

While this listing and brief survey of how to get started in developing a research goal or idea is general, it is often in simply reading the literature in your medical field of interest that possible research questions arise.

Where a result is not clear or seems to need further investigation, that is where a possible research project may lie.

As we move through the various sections of the book each element on this list is described and discussed.

1.12 Example: Case Study

A sample case study is outlined in Chapter 2 along with a simulated dataset. This is referred to throughout the various chapters of the book.

The case study examines obesity, focused on changes in fat content for various parts of the body that are measured before and after participation in a diet and exercise regime.

There is also a small related study examining the lowering of blood glucose levels to help fight the early onset of diabetes.

The specific aims, hypotheses, secondary aims, and data-related aspects of the study are given in Chapter 2.

1.13 Discussion Points

1. Any research project will require a detailed review of recent literature regarding your topic of interest. It is also possible to utilize such a systematic review to develop an initial overview research summary for possible publication. Discuss how this may be applicable to your project, outlining how you would carry out the required literature review.
2. If an electronic database is not available and the research project will use a survey of a cohort of subjects or patients, develop a pilot survey instrument. It should collect basic demographics, quantitative, and qualitative responses to questions directly related to your research topics. Remember that there is often a time restriction regarding such data collection. The longer the survey, the more missing data, or unfinished surveys, you may collect.
3. For your research project, develop a list of primary and secondary aims that you would like to study. For each aim, develop specific hypotheses you will be able to test. For each give the required variables and the respective statistical tests that will be used.
4. For your research project, fill in an initial research checklist, as given above.
5. Outline the possible use of GIS based spatial assessments and related graphics in your project. For example, heat maps of the regions within which the subjects will be recruited showing different levels of the response variables. For example, the percent of individuals having a specific illness.

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2

Sample Case Study

Almost all research projects are centered around the planning, collection, and analysis of data. The development of the study database is a key aspect of the study.

A sample case study is given here, along with a simulated dataset, that is referred to as a source of examples throughout the various chapters of this book.

A cohort study is to be conducted. The same group of female subjects are to be assessed before and after an intervention, a diet and exercise regime, that is assumed to lower body mass and related measures of obesity. The observed differences in overall body weight and other weight related measurements are the key responses of interest.

As well, a smaller case-control study is to be carried out for a group of cases having a specific gene present. This group is compared to a group of matched controls that also have the gene present. This case-study also examines obesity, focusing on changes in fat content for various parts of the body that are measured before and after participation in a diet and exercise regime. The observed differences in overall body weight is the key responses of interest, controlling for the presence of a specific gene.

The various elements of the overall study are described here, such as population of interest, study design, specific aims and hypotheses, list of possible variables to be measured and collected, data collection, data dictionary, and statistical/data analysis.

2.1 Overview

- The primary goal of the study is to assess if an intervention in the form of a dietary based regime will lower the baseline weight of participants over a six-month period.
- The study is based in a large urban hospital and participants are recruited and consented into the study using a network of participating clinical (mostly internal medicine) offices if they meet study inclusion criteria.
- **Primary Study**: **Cohort Study Design** is used in the context of a lifestyle based intervention (diet, exercise, meditation and meetings with a psychologist) to reduce obesity. The basic time period is one

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year and the primary goal for each participant is a 20% reduction in overall body fat. A non-obese control group provides a reference group (given only general health information), with patients drawn from a similar demographic (gender, age, body-mass index (bmi)) and socioeconomic profile as the treatment group.

- **Primary Response**: **Cohort Study**: Changes from the initial baseline measurement of body fat to measurement at six-months is compared within the treatment group itself, and between treatment and control groups, controlling for secondary variables.
- **Possible Secondary and Mediating Variables:** Apart from the responses of interest, the set of variables includes demographics, clinical measurements such as diabetes and hypertension, detailed body fat related measurements, psychometric assessments.
- **Case-Control Study:** The study also contains a separate smaller randomized clinical trial with a case-control design. An osteopathic non-pharmaceutical treatment for lowering blood glucose in obese individuals is to be tested. The cases in question all have gene A and meet the definition for obesity. For each member of the case/treatment group, age matched controls are obtained from a clinical patient database. The average improvement in body fat is compared between treatment and control groups.
- **Case-Control Study Primary Responses**: Comparison of differences between the treatment group and a matched control group, controlling for the presence of gene A.
- **Possible Secondary and Mediating Variables**: Apart from the responses of interest, the set of variables includes demographics, clinical measurements, psychometric assessment (depression z-score), gene (Gene A or not A). Categorical (high, low) scales are also developed for the response.

2.2 Basic Cohort Study Details

- Two sets of patients, with similar gender and age distributions, are to be assessed at each of three time points. The two cohorts will be given different diet treatments and expected to adhere to the protocol for the entire time period.
- **Timeline**: Measurements to be taken at baseline and six-month follow-up.
- Adherence to the prescribed diet treatment via ongoing app usage and monthly cell phone based contact.

- Only complete cases will be examined in the final statistical analysis. There may be a need to be aware of drop-outs and missing data effects in the final analysis.

2.2.1 Population of Interest

- Obesity in the adult population is a serious issue. The population of interest here are adults with obesity challenges.

2.2.2 Specific Aims

- To show that the various fat density measures will decrease over time due to the prescribed dietary based intervention.

2.2.3 Secondary Aims

- Assess improvements within demographic sub-groups.

2.2.4 Sampling and Recruitment Approach

- Patients will be recruited at local hospital based clinics.
- Local pediatric offices will also be contacted.
- Standardized data intake will be used throughout the recruitment and study process (Redcap).
- The study will use in-person consent, or consent to contact via phone.
- There are always challenges regarding follow-up and limiting missing data and study drop-outs.

2.2.5 Complete Listing of Measured Variables

- The key responses here, apart from overall body weight, are density of fat and overall measurement of various parts of the body at each of three time points and how the six-month and one-year measurements differ from baseline. These include (note that not all variables are relevant to the analysis):
  - Abdomen
  - Hip
  - Knee
  - Ankle
  - Forearm
  - Biceps

  - Thigh
  - Chest
- Demographics:
  - Ethnicity
  - Race
  - Baseline age, height, BMI
  - Gender
  - Socioeconomic status
- Genetic assessment (Gene A and B: Yes/No).
- Standard clinical assessments and blood work at each of the three time points.
  - Family history
  - Diabetes
  - HbA1c
  - Hypertension
  - Blood glucose
  - Other relevant pre-existing conditions
- Psychometric instruments reflecting measures of depression (e.g., the Beck Depression Index (BDI)) and other related conditions.

2.2.6 Sample Size Issues and Justification

- The case-control study sample size (n = 100) provides adequate statistical power (greater than 80%) to detect a mean difference of 20% using a pairwise t-test with Type I error = 0.05.
- For continuous variables, the clinical effect size in a study design is defined as the mean difference from baseline (improvement). This is usually set at 20% for sample size justification calculations.

2.2.7 Expected Statistical Analyses

- Estimation, testing and predictive modeling.
- Two-sample t-tests, ANOVA based hypothesis testing, linear models, logistic regressions.
- Nonparametric tests where sample size is small or outliers.
- Linear model: Predict change in bodyfat density as a function of several body measurements from baseline to final measurement.
- Logistic regression: Predict change in bodyfat density as a function of several body measurements from baseline to final measurement.

2.3 Basic Case-Control Study Details

- This sub-study will examine the effects of a new diet based treatment within a set of Gene A female patients who are seen as being at high risk of obesity.
- **Timeline**: Measurements to be taken at baseline and six-month follow-up.
- Controls will be matched on age.
- Same overall population as cohort study.
- Same basic measurements and responses as cohort study.
- Smaller sample size.
- Controls chosen to represent general population without Gene A.

2.3.1 Sampling and Recruitment Approach

- In-person consent or consent obtained to contact via phone.
- Recruitment sites include
  - local weight treatment centers.
  - Pediatric offices.
  - Hospital.
- For each case, a matching control.

2.3.2 Collected Data

- Same basic intake process (REDCap data entry) as cohort study above.
- Same basic set of variables.

2.3.3 Specific Aims

- Obesity improvement over time between obese cases (Gene A positive) and non-obese controls (Gene A positive).

2.3.4 Secondary Aims

- Compare improvements within demographic sub-groups.
- Compare improvements within age quartiles.

2.3.5 Sample Size Issues and Justification

- The case-control study sample size (number of cases = 35, number of controls = 40) provides adequate statistical power (greater than 80%)

to detect a mean difference of 20% using a two-sample t-test with Type I error = 0.05.

- For continuous variables, the clinical effect size in a study design is defined as the mean difference from baseline (improvement). This is usually set at 20% for sample size justification calculations.

2.3.6 Expected Statistical Analyses

- Estimation, testing and predictive modeling.
- Two-sample t-test, ANOVA based hypothesis testing, linear model, logistic regression.
- Nonparametric tests where sample size is small or outliers.
- Linear model: Predict change in bodyfat density as a function of several body measurements from baseline to final measurement.

2.4 Data Tables

- The database for each study has been collected into an Excel file. A sample data dictionary listing of the variables is given in Table 2.1.

TABLE 2.1

Data Dictionary for Key Variables

Variable	Dictionary
Body Fat	Continuous index values from 0 to 100
Log(Body Fat)	Log(Body Fat)
Within subject difference in body fat (post – pre)	Continuous difference of index values
Gene A	Yes/No evaluated from initial genomic expression values
Gender	All females
Ethnicity	Latino (yes/no)
Age	Months
Socioeconomic status	High/Low (1/0) variable
Age Median split	High/Low (1/0) variable
Age quartiles	Categorical variable (1, 2, 3, 4)
BDI Depression index	Continuous index values
Weight	Continuous variable measured in inches
Height	Continuous variable measured in lbs.
BMI	Continuous variable

(*Continued*)

TABLE 2.1 *(Continued)*

Data Dictionary for Key Variables

Variable	Dictionary
Wrist	Continuous index values from 0 to 100
Neck	Continuous index values from 0 to 100
Abdomen	Continuous index values from 0 to 100
Chest	Continuous index values from 0 to 100
Hip	Continuous index values from 0 to 100
Thigh	Continuous index values from 0 to 100

The datasets in the form of Excel data files can be found in the supplementary materials for this book.

3

Ethics in Medical Research: All Medical Research Should Be Ethical

As such, it is important to know the basic ethical principles that now guide medical research, before planning and actual project. A brief review is provided here. More detail can be found in [1].

These principles were developed over the years in response to experiences with research in various settings, and specific past situations that were not always ethical.

3.1 Nuremberg Code (1947)

In the twentieth century, the judgment of the trial of the Nazi doctors at Nuremberg is recognized as the beginning of modern research ethics. It contained ten paragraphs, referred to as the Nuremberg Code [2]. This founding document of modern research ethics emphasized sound scientific research protocol and consent.

3.2 Declaration of Helsinki (1964)

In 1964, the Declaration of Helsinki [3] was published by the World Medical Association. It introduced the need for prior review of any kind of human research. It emphasized scientific standards to guide scholarly research, but allowed physicians to omit consent procedures in special circumstances. A revision in 1975 further required the review of research protocols by an independent research ethics committee.

3.3 Belmont Commission

In the United States the Belmont commission (1979) [4] reviewed many of the ethical issues related to medical and human subjects based research. It

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provided, in its final report, a set of recommendations that form the basis of most ethically based current medical research practices. The report suggested three core principles to guide medical research: respect for persons, beneficence, and justice. See also [5, 6].

3.4 Respect for Persons

The principle of respect for persons includes two basic components:

- Individuals should be treated as autonomous agents.
- Those who have diminished autonomy are entitled to additional protections.

Rules derived from this principle include:

- The requirement to obtain informed consent from each research subject.
- The requirement to respect the privacy of research subjects.

3.5 Beneficence

This principle implies minimizing harms and maximizing benefits for research subjects within a sound research design.

Rules derived from this principle include:

- The requirement to use procedures that increase benefits and decrease harms when they are consistent with good research design.
- The requirement to make sure researchers are adequately trained.
- The restriction of research lacking a favorable risk-benefit relationship.

3.6 Justice

The principle of justice requires that people be treated fairly and research projects are designed so that the burdens and benefits are shared equitably in society.

Rules derived from this principle include:

- The requirement to select research subjects equitably.
- The requirement to avoid exploitation of vulnerable populations.

3.7 Balancing the Principles

The Belmont Commission's intention was that each principle should have equal moral force. This means that in some situations, the three principles might be in conflict with one another. It is important to consider each case on its own merits in light of all three principles [7].

These three principles have guided the development of the rules governing medical research and how human subjects based research should be conducted.

Practically:

- In each institution the IRB provides oversight and practical guidance for all human medical research projects.
- An Internal Animal Care and Use Committee (IACUC) is essentially an IRB for basic animal research. Here the need to test lab based results in living organisms intersects with the need to limit cruelty to animals. Often, in the development of new pharmaceutical treatments, the initial phases of research are based on animal subjects.

3.8 Example

For the case study outlined in Chapter 2, the balancing of the principles is relevant.

Note that as all the subjects are human, an IRB will have to review the project.

- Respect for persons is addressed by carefully reviewing the project with each patient before they consent to participate in the research study.
- Beneficence is incorporated as the treatment in question may help individuals across all ethnic groups lose weight and achieve higher levels of health, including a lower chance of diabetes.
- The principle of justice is respected here by having an ethnically mixed group of patients.

3.9 Applying Basic Research Principles

Following the Belmont Report [4], there are several basic standard research principles that focus on protecting the rights of medical research human subjects.

These are:

- **Social Value**: The research should provide insight and information that has value to society.
- **Scientific Validity**: The basic science of the research project must be valid.
- **Fair Subject Selection**: The research project should have as wide a demographic for the study as reasonably possible.
- **Favorable Risk-Benefit** ratio: The riskier the research, the more benefit it should offer to be considered ethical.
- **Independent Review**: This is usually a component of the IRB assessment of the research project.
- **Informed Consent**: All research projects must obtain informed consent from study participants.
- **Respect for Enrolled Subject**: The research protocols consented to by participants should be followed. Also, participants can leave the study at any time.

We can also add *Sample size justification and Study design implications* to this list. The study you carry out should use the smallest possible sample that allows you to detect meaningful clinical differences. This is an ethical consideration affecting both the scientific validity of your study and ensuring a favorable risk-benefit ratio.

These principles are worth reviewing in greater detail as they apply to research in general, not just human subjects research.

3.10 Social Value

Some basic social value principles that should guide your study design include:

- Your research study should help improve people's health or well-being.
- Your research can do this by providing results that support or guide the development of better clinical tests and improved treatments for disease.

- It is unethical to place people at risk of harm or even discomfort when neither they nor society benefits.

Example: In the example study described in Chapter 2, there are several possible benefits that the individual participants and society in general may realize from the proposed research. These include (i) improved treatment for obesity, (ii) limitation of secondary conditions such as diabetes and hypertension, (iii) improved knowledge of whether a specific gene is relevant to obesity.

3.11 Scientific Validity

Research should be focused on improving knowledge:

- Research is expected to produce useful results that will increase knowledge.
- Research that is not scientifically valid is a waste of resources.
- A research study should be carefully planned to answer a specific question.
- There should be a clearly stated, formal hypothesis to be tested, a control or reference group to compare against, and well-defined, clearly measured variables.
- The result should be replicated, if possible.
- Researchers must design their experiments to avoid conscious or unconscious bias. For example, a researcher testing a new cancer drug might subconsciously choose healthier subjects for the treatment group. This might make the drug look more effective than it truly is. Thus, the need for a double-blinded study.

Example: In the example study described in Chapter 2, the scientific validity of the overall study is clearly defined: a well-known cohort study design is to be applied when recruiting and collecting data from research participants. An additional case-control study design is to be applied to a smaller group of patients with matched controls (similar age and gender distribution). Measurements taken from each participant include validated blood chemical or physical measurement. Demographics and response variables are clearly defined. The statistical tests to be applied are listed in the study protocol.

3.12 Fair Subject Selection

- The subjects recruited into the study should reflect a broad demographic of society (various age groups, women, children, various ethnic groups), unless the disease is specific to a group or demographic.
- This consideration reflects past over-sampling of white, adult male subjects in many research areas.

Example: In the example described in Chapter 2, the patient cohort and both cases and controls are to be drawn randomly from an ethnically mixed population of patients.

3.13 Favorable Risk-Benefit Ratio

- The riskier the research, the more benefit it must offer to be considered ethical.
- Cutting edge research, if it exposes patients to higher levels of risk, must have a higher level of benefit to be justified.
- An example would be high-risk research on new cancer treatments for terminally ill patients.

Example: In the example described in Chapter 2, the benefits from the weight reduction program will be far greater in terms of improved health than the discomfort of the participants while in the program.

3.14 Informed Consent

You typically cannot involve a subject in a research study without their consent [8].

- Potential subjects in a research study must be told the details of the study for informed consent.
- The subject must be competent, capable of understanding the research and making a decision based on them.
- Researchers must tell subjects the benefits of the study and any risks the subjects may face.

- Researchers must tell subjects what they will have to do as part of the study and options they have besides being in the study. No one has to participate, it is a choice. Subjects can drop out anytime.
- Subjects must understand what the researcher tells them.
- The subject's decision to participate must be voluntary.
- Typically, informed consent forms should be signed by the participant. In some studies, however, there may an exception. For example, a telephone interview often has minimal risk, it is enough for subjects to say they consent at the beginning of the interview.
- If research subjects cannot give informed consent, it may still possible for them to participate if a competent family member or guardian gives the informed consent.
- All subjects should agree to participate, which is called assent, if they are capable of giving it, even young children.

Note that if the giving of consent intimidates specific subsets of the population of interest, it will lead to a biased, non-representative sample.

Example: In the example study outlined in Chapter 2, the informed consent document would have to have cover the following issues: possible benefits of participation in the study, the benefits of losing weight in a healthy manner, the use of the information (such as publications, presentations), the ability to drop out of the study at any time, possible risks of participation in the program.

3.15 Respect for Subjects

To show respect for human subjects, researchers should:

- Remove subjects from study if the study becomes too risky or harmful.
- Keep information about subjects confidential.
- Allow subjects to quit the study any time they want.
- Subjects should be informed how their data will be used.
- Data integrity and privacy considerations must be part of the study protocol.
- Data sharing (eventually) is expected in many NIH funded studies and this should be explained to participants.
- Share the results of the study with the subjects.

Example: In the example described in Chapter 2, the issues described here are relevant. The informed consent document should detail how subject data will be collected into electronic databases solely under control of the principal investigator, keeping their information private. De-identified data will be forwarded to the NIH to be included in the relevant data collection.

3.16 Sample Size Justification and Ethical Design Implications

The study you carry out should use the smallest possible sample size that allows for the detection of meaningful clinical differences. This is an ethical consideration.

Sample size selection is typically a function of:

- The **clinical effect size** of interest. For example, a treatment is successful if the subjects average weight decreases by 20% over the course of a twelve-month period. The amount of the change expected needs to be precisely defined, even if it is only an estimate. Sometimes a change of one standard deviation is used as an effect size.
- **Type I error**, the probability of incorrectly rejecting the null hypothesis needs to be defined in advance, as part of the protocol. It is usually set at 0.05. We typically reject the null hypothesis of no change if the p-value is less than 0.05.
- The **level of variation**, typically estimated by a standard deviation taken from the existing literature.
- The **statistical power concept**, the probability of rejecting the null hypothesis of interest correctly, is used as a guide here to sample size selection. Typically, for a 20% change or clinical effect, a power of at least 80% should be attained. *Note that if the power of the test is high, that is good, but does not guarantee a small p-value.* It supports the idea that the overall process, including the observed p-value, can be trusted.

Some details are further discussed in Chapter 8. See also [9].

Example: In the example described in Chapter 2, assessment of the average change in weight is achieved by looking at average (last – first measurements) within each subject. These differences are used to test a null hypothesis of zero average difference (no change). Using a paired t-test with Type I error = 0.05, an effect size

of 20%, an assumed standard deviation for the observed differences (from the literature), a sample size of 60 per group achieves a power of 80%.

3.17 Discussion Points

1. Using the literature, carry out a review of past experiments in your research area of interest that now appear to have questionable study designs, applying the three basic Belmont principles.
2. For you research project, discuss briefly how each of the seven standard research criteria are satisfied by your study design and related protocol.
3. Discuss sample size issues in relation to your research project. Note that the goal in selecting a sample size is to choose the least number of subjects in each treatment group that will give meaningful and clinically relevant information.

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4

Getting Started: Project Mentors, Support Team and Pilot Data

Once you have been able to identify a research goal and related set of hypotheses, the question arises as to whether the goal is feasible in your research setting and what kind of support you will need to move it forward.

Typically, this involves:

- Finding mentors who can help provide a context for the project, give you some guidance and perhaps help you obtain protected time in which to carry out your research project.
- Developing a research support team to help with the various pieces of the project. These may include, in addition to your mentor:
  - Lab based support.
  - Computer and database support.
  - Administrative support, especially useful if enrolling subjects into a study.
  - Statistical, biostatistical, and other data analytic support.
  - Research assistant support.
  - IRB guidance and possible evaluation of your project.

With this support, you will need to conceptualize your project: plan it out, fund it, carry it out and analyze the results. Then present the results.

So, you need supportive individuals on your team that can help you do these tasks.

- Finding some initial "pilot" data with which to show proof of concept for your basic idea. Some specific questions that arise include:
  - Can you find initial pilot data that can support proof of your concept (initial results) for reviewers and peers who may review the proposed research?
  - Can you identify a local resource that can support your area of research (a clinic with patients, an electronic database)?
  - Is there an internal granting mechanism that might be a resource to get your project started? Pilot projects are usually easier to get funded.

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- Write up a draft outline or protocol of your project. Often there is a committee that will review this formally. Have your mentor(s) review it for you before having others comment.

4.1 Research Project Mentors

- It is useful and sometimes essential to have the support of mentors and more senior investigators. See Figure 4.1.
- In some formal settings, for example academic medicine, this may be required.
- Mentors help guide the selection of research problems and the planning of studies.
- They can also help access datasets for analysis.

The best mentors will show the young researcher:

- How to develop research-oriented ideas for investigation
- How to develop hypotheses for testing

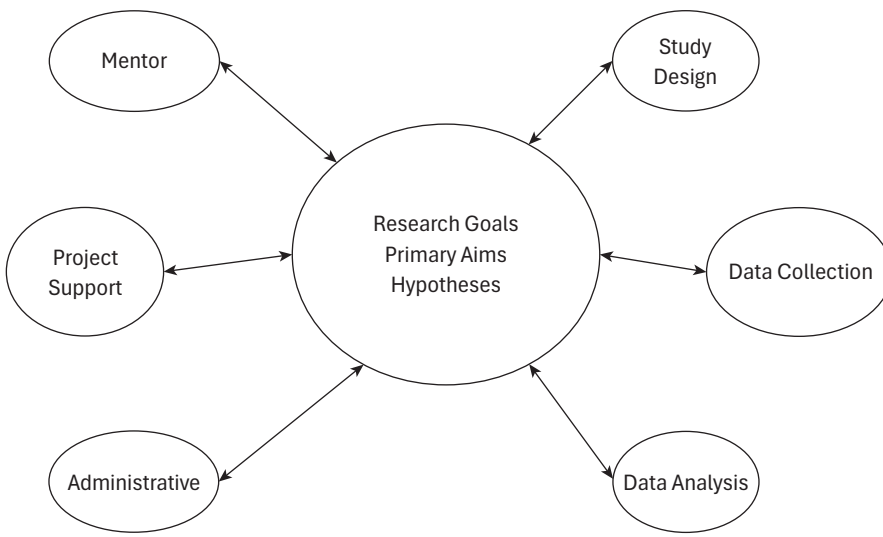


FIGURE 4.1
Creating a Research Team and Support.

- How to develop study designs within which related hypotheses can be tested
- How to write and structure grant proposals to find the necessary funding to pursue these interests and knowledge.

4.2 How to Choose a Mentor

- Choose someone whose career you would like to emulate, who inspires you. But also remember that you may not want to go through all your research life following someone else's interests.
- Be sure the person who trains you shows you how to actually define and carry out a research project.
- Mentors are essential in finding pilot data and initial levels of support, especially if you are or are working as an M.D. active in clinical research or a lab-oriented researcher (M.D. or Ph.D.).
- They provide guidance, potential collaborators with experience in the research process, and help in defining and developing research ideas, pilot data and potential grant applications.

Science is a process in a modern institutional setting. There are permissions to obtain, support from research administrators and colleagues.

Often there are also available resources, for example research cores which provide access to specific analyses, for example genetic analysis of biomedical samples.

This is especially useful in clinical research where often patients are recruited through a specific clinic and support from lab studies is necessary.

Research projects oriented toward lab science can be very detailed and focused on the demonstration of specific pathways or pathologies.

Many diseases can arise in more than one scenario and careful testing is often required to understand the onset and characteristics of the disease agent, for example parasite, virus, bacteria or fungi.

These require a mentor who has a lab or clinic and often a grant or financial support to carry out the research.

Given the large amount of private health information available in clinical research settings, it is important to be aware of regulatory limits (e.g., HIPAA, see below) and obtain required training and oversight.

Again, this is much easier with a mentor and research support team.

4.3 Support Team

More details are given here regarding the type of support you will need to develop your research idea into a formal project.

- Data entry and cleaning (wrangling). This is often essential to develop the data into a useable database for data analysis.
- Statistical design and planning. Most studies will require formal sample size justification and some form of randomization in the selection of subjects. As well as an assumed probability model from which the required p-values, confidence intervals and related inferences can be drawn.
- Research assistants and students and others to carry out contacting potential subjects, obtaining consent, aiding in data collection.
- Lab support if relevant. This can be the most expensive aspect of the study and may require skilled technicians.
- Data Science for analysis. Often graphical depictions of the data, statistical summaries and various fitted models and inferences need to be calculated using various computer software. This typically requires the participation of a statistician or data scientist.
- Clinical Mentors for help interpreting the various measurements and how they impact the outcomes of interest. The goal is to always translate the numerical information from the statistical and data analysis into clinical information and inference.
- Formal timeline to guide your progress through the project. if you are student with a graduation date, or you have financial support that has a specific end date, then you need to work backwards to plan the overall timeline; leaving time for planning, administrative review, organizing your support team, recruitment, carrying out the study, data analysis, writing up results, presentation and publication of results. When outlining this, work backwards from your desired completion date.

4.4 Research as a Process

When planning a research project it is important to realize that it is a process.

It is also useful to see any particular project as a step in the development of the research aspect of your career. Not just a single paper or grant.

TABLE 4.1

Typical Timeline for a 24-Month Project.

	Planning		Study Ongoing				Results	
Time	1–3 months	3–6 months	6–9 months	9–12 months	12–15 months	15–18 months	18–21 months	21–24 months
Activity	• Project begun • RAs hired • Oversight Committee organized • Monthly meetings scheduled • Project schedule developed • IRB review of protocol completed	• Patient recruitment and consent begun • Materials and lab supplies obtained • Databases developed • Study begun	Data collected Monthly Meetings Data entry	Data collected Monthly meetings Data entry	Data collected Monthly meetings Data entry	Data collected Monthly meetings Data entry Data cleaning	• Data analysis begun • Initial reports written and presented	• Results developed • Database finalized, shared with NIH if funded by NIH • Papers written • Conference presentations

Apart from a careful review of the literature which will guide the selection of your research project and study design, it is important to be aware of more recent projects and areas of research.

- Go to conferences (whether you present or not) and observe what areas of research are being emphasized in your field of study.
- Accept opportunities to peer review papers.
- Read journals in your field and conference abstracts.

Success in research work tends to be measured in terms of conference abstracts and presentations, peer-reviewed publications and the obtaining of research grants.

- It is all a process.
- You need a plan and realize it takes time and dedication.
- If you take your efforts seriously, so will others.

4.5 Accessing Pilot Data and Databases

Pilot data is often drawn from existing databases, so it is often retrospective in nature, or the outcome of a small prospective study. It allows for initial examination of the research question.

- It can often be seen as "proof of concept" and usually will have to be carried out with limited funding.
- Calls for proposals in this regard are often open to younger researchers.
- Modern research is sometimes based entirely on the examination of existing databases, for example health services, informatics, aspects of epidemiology, retrospective study of clinical records where patterns of usage and treatment are of interest.
- Such databases are also potentially very useful for the identification of subject cohorts who might be recruited into future studies.
- The large amount of information on each subject makes it possible to carefully draw a useful sample for study. But carefully.

Some *databases and sources are mentioned in* **Chapter 1**.

4.6 Database Issues

- Accessing data is never easy and the procedure may differ in each institution.
- Often there are clinics within a hospital or medical school and these will have their own datasets that may be available for research purposes if you or your mentor work in the clinic.
- Downloading the data should be done carefully. Very large databases may require additional memory or the use of cloud based storage.

Some standard questions can be asked:

- Is the Data Biased?
- What Does it Represent?
- Is there missing data? Does it follow a non-random pattern?
- The resulting dataset will often have to be cleaned.
- May have to generate index values or re-categorize the data etc. . . . and often go back for additional data, additional variables of interest or additional data as the dataset is developed over time. Many versions of the dataset may have to be generated for analysis.

4.7 EPIC

The EPIC database [1], for example, is a wide-ranging collection of data that is collected from the hospital system of the United States and integrated into a large database that can be searched using various keywords and ICD codes.

Many academic medicine institutions and hospitals have access to this database. The data can be viewed by hospital or across the entire system of participating hospitals.

The EPIC Cosmos [2] overall database currently contains data on more than 195 million Americans. The variables are mostly hospital usage related with much collected clinical and demographic information. This database collects data through time and is very useful for longitudinal studies.

4.8 Pediatric Health Information System (PHIS)

The PHIS [3] is offered through the Children's Hospital Association (CHA)

- The database is focused on pediatric medicine.
- Researchers looking to access the database must undergo training online (offered in the summer) or in person (2-day course) to have direct access.
- There is analytic support to help you with your specific database requests.
- Over 50 participating children's hospitals across the county submit data to this database (both medical record & billing/utilization data).
- This database can provide a useful benchmarking tool to better compare hospital performance versus peer pediatric hospitals across a range of metrics.
- Examples of the types of data accessible in this database include: LOS, resource utilization, readmissions, CMI, payor contract benchmarks, researching certain patient populations

4.9 Genomic Databases (NIH) Repositories of Data (That Must be Shared)

- The dbGaP [4] data repository contains databases of publicly available genetic data of various types. These have been shared from various NIH funded research projects. These include basic nucleotide sequences.
- How to interpret this data?
- To interpret this type of data, you will need a background in genetics/genomics and genetic data in relation to the specific context/science/medicine.
- The statistics for this type of data can be non-standard and require some background in the analysis of genetic databases.
- It is challenging to handle this type of data which is typically not in the form of Excel type files, but rather very long text files, for example of CTAG sequence data.
- There is often a need for bioinformatics support to examine this type of data.

4.10 Additional Challenges that Often Arise from Electronic Data

Retrospective datasets in general can have challenges and these challenges can be magnified in small datasets. It is important to be aware of their potential shortcomings.

- Missing data components may not be missing at random, causing biases in the remaining dataset.
- There may be entry errors in the data, and again these may have a pattern, causing biases in the data.
- In a large, integrated database, not all the sub-datasets may be interacting as expected, giving misleading data.
- Each variable in the dataset needs to be carefully defined and understood.
- Many variables such as Age or BMI can be reported on several scales, typically as a continuous measure, or a derived categorical measure,

  for example (NewAge = 1 if age > 25 , NewAge = 2 if age ≤ 25).
- Real life and the structure of organizations can affect data collection. The actual database may be culmination of many years of data collection, with entire years being collected from different clinics, using potentially different scales of measurement and reporting techniques. Be aware of these issues.
- Data can be entered daily, weekly or monthly, or some combination, depending on the reporting practices. If time to an event (a recurrence of cancer, a readmission to a hospital) is of interest, this may be affected by the reporting practice.
- Ask as many questions as possible about the data. It will only help.

It is important to note that "I have data" may not be a very useful approach to research.

It is always important to first understand the medical or scientific context within which the information was collected. Also the particular questions this information was expected to answer, and how the data is or was collected. Having a useful hypothesis before examining the data is

also helpful in developing a focused and well guided analysis with useful results.

4.11 Statistical Planning and Data Analytic Support

The planning of a study or experiment requires not just medical or scientific insight, but once the basic goal is established, it also requires a study design and related statistical insight.

- This technical support needs to be available to develop a high quality research protocol.
- The research team you need will require someone versed in the application of related statistical methods. The relevant methods need to be listed out and a strategy for analyzing the data outlined.
- Once the data is collected, you will need support to analyze your collected data.

4.12 Example

In the example considered in Chapter 2, a cohort based randomized clinical study is to be conducted, that would require substantial support. Assessments of each subject are to be carried out at baseline, 6-month and, if available, one-year follow-up. Assistance will be needed to support the follow-up nature of the study. Scheduling is required and clinical support.

As well, a smaller case-control study will be conducted. Here the cases have been identified and a set of matching controls (perhaps two controls for each case) need to be found. This will require assistance to carefully review electronic hospital records and identify the set of controls. This is can be challenging.

This type of effort requires a dedicated team of individuals to recruit and consent participants for both cohort and case-control study, schedule assessments, organize the data collection and entry, develop the databases, carry out analyses and all other IRB related reporting requirements as the study moves forward.

On the part of the lead researcher, this also involves finding resources to train and support the research assistants carrying out the work.

4.13 Summary

- In summary, the challenges to developing and carrying out research are many and the support of mentors and a research team very helpful.
- Team building is a very useful skill for a researcher to develop.
- Time management and related resources and funding are often underlying issues you will have to deal with in developing a research team.
- Accessing research-level data is often not easy without institutional support.
- Finding pilot data that has all relevant medical and scientific variables for your study may be challenging if you do not have a mentor.

4.14 Discussion Points

1. With regard to your own idea or project, which database or set of databases seems most relevant? This discussion should include a listing of the various types of data that are of relevance to your project. Some possibilities include demographics, hospital utilization data, clinical data, blood work related data, genomic data and data from peer institutions that may serve as reference data.
2. Organize the set of data variables you wish to collect by type and by time. Create a table listing out the different sets of variables.
3. Data often changes over time. New variables are defined and added to databases. New sources of insurance and other governmental program support alter healthcare access and related patterns in the data and the patients whose data will appear. This affects the ability of the database to be a representative sample of the underlying population of interest. Are these issues relevant to your study?
4. Genomic data is difficult to obtain and challenging to analyze. Is this type of data to be considered? How will you maintain subject confidentiality?
5. How will you assess missing data? If these patterns related to missing data are occurring at random, then it is often assumed that the effect is averaging out over the various demographics and treatment groups. What might cause missing data in the databases you are considering for your project?

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5

Research Studies and Study Design

5.1 Introduction

Good science requires planning experiments or observational studies carefully. It is very easy to bias subject recruitment and related data collection procedures, and if the dataset is biased, then the summary statistics, data analysis, and conclusions based on the data will also be biased.

Quality data for analysis is the result of planning and careful collection of the subjects themselves and related measurements; clinical measurements, environmental and demographic contextual data, exposures, medical responses of interest, and any other measurements of interest.

Each basic study design has its own structure and related set of probability distributions, assumptions, and techniques for assessing the hypotheses of interest and estimating population characteristics.

As a lead researcher you should be aware of some practical issues related to the implementation of a study design and the quality of the subsequent data analysis.

These include:

- The need to develop a *consent procedure* that is respectful of patient rights. This is discussed in several chapters of this book.
- The development of a *data collection procedure*. This may include paper survey intake, online data entry, the use of apps on cell phones, the downloading of electronic data files from an existing database, or all of the above.
- The need to **limit bias** in patient selection and the measurement process. The sample should not favor healthier subjects, for example.
- **Randomization** of patient selection will help limit bias.
- **Representative sampling** of the population of interest will provide relevant results.
- **Sample size and the resulting accuracy** of statistical or machine-learning based estimation.
- Time and **follow-up issues**. Many studies require follow-up with patients over time. This is often challenging as some patients will drop out of the study for various reasons.

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- **Missing data**. Sometimes missing data patterns can induce a non-representative or biased sample.
- **Clinical effect size** versus statistical significance. When the sample size is very large it is possible that very small clinical effects will be deemed statistically significant. Sample size justification calculations can limit this possibility by determining reasonable sample sizes that will detect the desired clinical effect, if it is present.
- **Matching of subjects** in treatment and control groups: A typical design related consideration applicable to most studies is the matching of randomly selected subjects in the controls and treatment groups on age and gender. This limits secondary variation due to differences in age and gender and should be used when those variables are not of key interest.
- **Reporting results:** Model summary, tests of significance and confidence intervals, other.

5.2 Study Protocol Consider ations

If you are required to develop a grant application or a research study protocol related to your specific research aims, most research proposals written in relation to medicine and biomedical disciplines often follow very direct and focused templates. These are usually outlined in the request for proposals of the funding agency, or the instructions of the internal review committee or board.

- This often reflects federal and other funding organizations priorities and their pre-defined application templates.
- Even if your research can be viewed as primarily internal improvement of clinical practice, it is very useful to write up a formal protocol for your study.
- This will force you to define and focus on primary and secondary goals, data to be collected, data analysis to be conducted, expected results and where you intend to present your results.
- If the research is to be conducted in a formal setting such as a University, medical school or other institution, then administrative review may be required before you are able to move forward in the project.
- These practical issues will affect the type of research you may be able to conduct.

Often the goal in the initial stage of a research project is to conceptualize and describe the project in such a manner that permissions and initial support for the research will be given. This is easier as the researcher develops a track record.

5.3 Statistical Aspects of Study Design

The theory of the design of experiments is a detailed study of how to incorporate sampling related randomness and probability into the planning of experiments and studies.

- A goal is to use randomization in the design to average out potential bias.
- As noted above we also want to reduce overall variation where possible and identify independent sources of variation among variables.
- Replication of the experimental or study results and follow-up time are also important issues in designing experiments.
- These issues are all pre-experimental, before the actual study is conducted and relate to the planning stage of an experiment.
- Since we are we are often using randomness to limit the effects of design issues such as bias, we can apply probability based assessments to the resulting data.

A challenge, however, is that often the fundamental properties and justification of the resulting statistical methods have little to do with the data actually observed, but rather the underlying random procedure used to collect the data.

Statistical ideas apply primarily to (i) the planning of the study and (ii) the data analysis after all the data is collected.

- The study design principles referred to above should be integrated with the science and lab concepts and any related techniques employed.
- Designing an experiment or study in the real world is not an attempt at perfection, it is doing the best possible with the tools and contexts available.
- As long as any limitations are acknowledged and weaknesses in the design of an experiment acknowledged, they can be incorporated into the analysis and interpretation of the analysis.
- The sample is often based on the sequential collection of data from a set of carefully chosen consented subjects.
- This choice however has to be at random and made subject to a carefully developed list of inclusion and exclusion criteria.

- If randomization is not applied carefully, biased samples may result and affect the entire subsequent data analysis.
- Ensuring randomness in the selection of subjects is difficult. As all subjects must be consented, it is only those subjects agreeing to participate who will contribute data to the study for analysis. The set of actually consented patients will form the sample of interest.

5.3.1 Limiting Sample-Based Variation

- For most health related responses (blood pressure, glucose level, etc.) there is always subject-to-subject sample variation in the measured response. This random variation makes seeing group differences and trends in the data challenging. We use statistical and probability based methods to model this variation and see any differences or trends more clearly.
- A goal of study design is to lower the overall sample variation in the observed data.
- This is accomplished, for example in a case-control study, by matching cases and controls using gender, age or other variables. This lowers any variation in the response related to these variables.
- The use of matching makes the cases and control groups more comparable, with any dis-similarity caused primarily by the treatment in question (the cases group receives the treatment and the controls do not).
- In some settings there may be limitations on the sampling scheme available to be used while selecting the sample.
- If there complex limitations are placed on the statistical design, for example spatial sampling issues when looking at census tracts, then the goal of sampling is to provide a sample which remains random, unbiased and representative, with low levels of sample variation, but also reflects the spatial aspect of the data.

5.4 Statistical Power and Sample Size Selection

The optimal type of study is a randomized clinical trial, where you prospectively enroll new subjects into the trial and observe and analyze their outcomes.

Whether you are using such a study design or not, there are some planning issues related to linking up clinical and statistical significance.

- It is important to plan your study in advance so that you have adequate levels of information to answer the specific questions you are studying.
- If you have a control group and intervention/treatment group, this often requires writing down what you would consider a *clinically meaningful difference* between the two, on average. This is often called the *effect size*.
- You will need to have a sample size that will allow you to detect differences of this size "most of the time" (usually at least 80% of the time).
- This does *not* guarantee that you will see a difference (have a small p-value), but if there is a difference in the actual data, you will probably detect it.

5.4.1 The Concept of Interest Here Is the Statistical Power of the Test

This is usually defined in terms of Power = Probability that if you reject, you reject correctly. Having high power does not imply that you will get a significant result (a small p-value). But if you do obtain a significant result, you can believe it. Note that power > 80% is the rule of thumb.

We leave out the mathematical details here, but often you will have a table of possible power values corresponding to various assumptions on sample size, effect size, and underlying variation.

Some standard guidelines:

(i) As the standard error estimate of sampling variation decreases, it is easier to detect significant differences.

(ii) As the effect size increases [1, 2], it is often easier to detect significant differences with moderate sample sizes.

(iii) As sample size increases, the relevant amount of information increases and it is easier to detect significant differences. This is technically due to the standard error becoming smaller.

(iv) A challenge is that if the sample size is too large, the p-value will be small for almost any effect size [3]. This is why we use power calculations as a formal guide to sample size assessment. To find, in a sense, the smallest useful sample size.

Your practical ability to carry out your study will often depend on your ability to fund enough sample size to reach adequate power in your study.

An example of a power table is given here, see Table 5.1, for a hypothesis looking for a difference in mean values between a treatment group versus a control group. For given effect size = 1.5 here, power is given as a function of sample size and control group standard deviation. Acceptable sample sizes give power > 0.80.

Table 5.1 Example Power Table

n = 20		n = 40		n = 60	
σ	power	σ	power	σ	power
1.7	.57	1.7	.66	1.7	.82
1.6	.64	1.6	.76	1.6	.88
1.5	.71	1.5	.81	1.5	.92

(1)

5.5 Basic Statistical Format for Estimation and Testing

In terms of actual analysis, estimation and testing, the standard linear model is usually applied:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_p x_p + \varepsilon \tag{2}$$

The response y is explained by a linear combination of explanatory variables x_j, plus random error ε. This error is typically assumed to follow a normal distribution [4].

Much statistical analysis corresponds to testing specific values (usually zero) for the parameters β_j defined within the above linear model. If the parameter $\beta_j = 0$, then the associated variable x_j can be removed from the model equation.

The ANOVA table often summarizes hypothesis testing for this type of basic model. A wide variety of basic study designs can be placed in this simple format, subject to various restrictions.

More information is available in Chapters 7 and 8.

5.6 Building a Study Design Protocol

In all research studies, the context for your study design is provided by the medical or scientific literature, practical knowledge and the specific medical or scientific hypotheses to be examined.

Once the context is well understood, some simple, yet crucial issues need to be considered when choosing an appropriate study design and related statistical approach for your research project. This is especially important in the context of pilot data where resources are limited.

These issues include:

- Define a primary goal at the beginning of the study, clearly stating the central aim or goal of the study. Define also as well as any secondary aims.

- The chosen study design needs to provide a context within which your key hypotheses can then be formally tested and population characteristics of interest estimated.
- It is sometimes not obvious how to get from your initial research ideas and general topic of interest to a carefully defined research hypothesis that can be placed in a designed data collection based setting and statistically tested.
- Part of the study design is the development of a recruitment process for subjects which will yield an unbiased sample that is representative of the population of interest.
- When defining your study protocol, it is important to define clearly the set of relevant outcome and predictor variables.
- This includes how the variables are to be scaled: for example, as continuous variables, or categorized into discrete variables with levels A,B,C, . . . etc. and how this scaling is to be incorporated into the statistical model and related statistical estimation and testing.

5.7 Statistically Related Questions

Some initial basic statistical questions useful in developing your research project include:

- What variables are necessary to answer the specific research questions of interest?
- What variables can be measured in a meaningful manner?
- On what scale(s) should the response(s) and other variables be assessed?
- What is the time frame of the study?
- What are the overall goals of the experiment or study?
- What are the key hypotheses to be investigated?

Once these basic questions are answered, you will have enough of a context to develop the particular study design you will need.

5.7.1 Further Study Design Issues

The following should be carefully considered when designing an experiment and form a list of issues to address with a consulting statistician or collaborator:

- The need for controls and how they are used (e.g., as a reference group or matched to individual cases) in the experiment or study. Note that cases can be their own controls when baseline measures are available and changes from baseline are the responses of interest.
- The level of variation expected to be present in the data (e.g., obtained from a literature review of pilot data) and the resulting expected sampling variation of the statistics to be employed.
- The need to justify the sample size, usually through defining the statistical power of the set of primary hypothesis tests and in relation to an assumed clinical effect sizes (usually the average differences between the treatment and control group). As discussed in elsewhere in this book, a standard rule of thumb is to achieve a level of 80% power for a 20% difference between treatment and control averages.
- Issues related to replication and controlling variation if using designed replication (blocking for example [5]) or within subject follow-up. See below.
- The need to consider the practical complexities of the collecting your dataset. Structural issues affecting the collection of the dataset such as hierarchical or multi-site data collection sites, often an issue if you are studying a rare disease which requires, for example, several hospital sites to collect enough patients.
- Choosing a mathematical model, for example a linear model or logistic regression model, that formally models the response and incorporates sources of variation, treatment groups, exposures and other secondary explanatory variables of interest.

Some of these questions are detailed and some are very basic, reflecting the technical details of the study design you have chosen. They may require the aid of a statistician or biostatistician to help you organize the data analysis and study design you are going to employ.

5.8 Types of Research Studies

There are several standard design templates for the collection and analysis of research data. It is important to select the one that is most practically relevant to your research project and that helps you address the issues discussed above. This will often depend, as a first step, on the overall type of research study you will be carrying out.

Medical research, for example, is often generally classified [6] into several overlapping types:

- **Primary: Clinical/experimental studies.** Three main areas are usually defined: basic medical research (laboratory based research), clinical research, and epidemiological research. Clinical research includes both observational studies and interventional studies. Interventional clinical studies are performed to study or demonstrate clinical or pharmacological properties of drugs/devices, side effects and establish efficacy or safety.
- **Secondary:** Literature reviews, systematic literature reviews, and statistical meta-analyses.
- **Observational clinical study:** Here, knowledge from the practical treatment of persons with an illness is analyzed using basic statistical methods. In this setting, the diagnosis, treatment, and monitoring of patients are performed exclusively according to medical practice and not according to a specified study protocol.
- **Epidemiological studies:** Usually there are two basic approaches, interventional, and observational. Clinicians are more familiar with interventional research, whereas epidemiologists usually perform observational research (the Framingham study for example [7]).

These categories, while useful, are not that important. What is important is finding a reasonable and useful design that you can practically apply to your research question and the context within which you are working.

5.9 Brief Review of Specific Some Study Designs

5.9.1 Lab-Based Research

- This type of research is usually very particular to the specific biomedical questions and materials being studied.
- This research tends to be focused on individual variable analysis, for example biomarkers, genetic expression.
- This research often reflects and requires very specific techniques and types of measurement and related interpretation.

5.9.2 Case Report

- This research is usually the reporting of interesting cases observed by clinicians. To submit this type of result, follow the outline given by the journal in which you would like to report the case of interest.
- The case being reported must typically must be unique and somewhat rare to be of interest.

5.9.3 Cohort Study

Given a defined population and time frame for observation, study subjects are observed continuously from a point in time and forward. Some are exposed to variables that may cause the onset of a medical condition, some are not. The response of interest is usually the onset of an illness, infection or disease, with a comparison between the exposed group versus the non-exposed group. Also of interest may be a comparison of the time to onset in each group.

Some properties of this type of study [8] are:

- Exposure status is present from the outset for all subjects.
- Disease occurrences are accumulated until the end of the trial.
- Sometimes the initial measurement of a given response is taken as the baseline and differences from baseline are of interest (pre/post). An example is the study of newborn growth patterns, for example, weight gain over time.
- This type of study can be difficult to conduct as they require time to observe the cases occurring. Also, if the cohort has a wide range of subject ages, it may be necessary to age standardize the final results before reporting and comparing incidence rates.
- While these studies may also include a formal control group, they often include variables measured as baseline or the change in variables pre and post exposure.
- Note that if using an electronic database, cohort studies can also be retrospective, using data collected, for example, ten years ago and comparing to outcomes recorded five years ago.

5.9.3.1 Example

In the example considered in Chapter 2, the cohort study is the primary study design. The study uses a pre-post approach, using the pre-treatment measures for each individual as their own control. Differences from this baseline measure are then examined for patterns and used to test hypotheses regarding improved health.

5.9.4 Case-Control Study

For the case-control study design [9] we typically have identified the set of cases. The risk factor or exposure variable to be studied is also known and exposure information is obtained for each of the cases. This is often obtained using electronic databases or survey data. The challenge here is in the selection of a set of representative controls to compare against. The exposure distribution among the controls should reflect the population distribution

of exposures, the controls being the baseline reference group against which cases are compared.

- Often it is challenging to find a good control group. The exposure distribution in the controls must match the general population.
- To improve comparability, and lower variation, controls should be chosen to match to cases on age, gender, and other secondary variables.
- The odds ratio is used as an estimate of relative risk, the relative risk of becoming a case when exposed. For large samples, you can use the odds ratio to estimate the relative risk.

5.9.4.1 Example

In the case-control study outlined in Chapter 2, the set of cases obtained is matched with a set of controls. The cases have Gene A and the controls do not. Lifestyle, age, and gender are used as matching variables. Differences in various health outcomes are then compared between the groups.

5.9.5 Community-Based Research

Many institutions have relationships with nearby community health organizations and this may be a very useful place to carry out community related research [10]. Some examples include:

- Access to care
- Caregiver burden
- Effects of school based interventions for the young
- Older population health
- Assessing health care outreach
- Program evaluation
- Survey based research.
- Develop your own.

5.9.6 Systematic Reviews

Systematic reviews methodically and comprehensively identify studies focused on a specified topic, appraise the methodology, summarize results, identify key findings and reasons for differences across studies, and cite limitations of current knowledge [11].

- It is necessary to only include studies with reproducible research methods and appear to be of good quality. The methods used to

compile data should be transparent, allowing the reader to assess quality of review and potential for bias.

- In graphic form, data can be plotted with the point estimate and 95% confidence interval for each study.
- A properly conducted systematic review presents the best available research evidence for a focused clinical question.

5.9.7 Meta-Analysis

Systematic reviews that incorporate summary data from each study and analyze these as data points collectively are often very useful if a treatment or disease has been studied in a series of clinical trials in many different hospitals or countries.

The meta-analytic review [12] compiles aggregate-level data from a set of primary studies (lit review). The pooled results are referred to as the meta-analysis.

- Basic idea: Pooling the samples of individual studies increases overall sample size, enhances statistical analysis power, reduces confidence interval and provides an estimate of the true effect.
- The set of studies included in the review should have been conducted methodically.
- Systematic review of several randomized clinical trials are less likely to mislead.
- Health practice guidelines supported by large randomized clinical trials and meta-analyses are considered the "gold standard" in Evidence Based Medicine.
- The statistics used in reporting a meta-analysis differ slightly from standard statistical analysis. This is due to the summary statistics being reported in each paper and not the raw data. If there are 20 studies included in the meta-analysis, you will have 20 sample means, 20 odds ratios, 20 standard deviations, etc. This require careful consideration. See [12].

5.10 NIH Flowchart for Clinical Trials

For studies where you will recruit and consent new subjects, it is often necessary to outline how the subjects of the study will be recruited, consented, and studied.

This needs to be done before the trial begins and is often a requirement of the IRB and NIH if a grant proposal is to be written to support the research. The CONSORT diagram is often suggested along with a suggested list of entries [13].

5.11 Typical Study Design Research Protocol

The research protocol document will typically have several sections. As a whole they provide an outline of the steps that will comprise your research project. Included in the protocol will be a requirement to provide primary and secondary hypotheses.

As well, a detailed description of the various components of the study design will be required and a discussion of the statistical models to be used. In relation to the primary hypotheses, sample size justification will be requested. See Table 5.2.

Copies of the patient consent form may also be required, as well as a clear description of how the corresponding research database will be maintained and kept secure. A description of the subject recruitment process should be included, including perhaps a Consort flowchart.

Ethics and transparency should guide your development of the participant consent form.

5.12 Discussion Points

1. Discuss the type of study design you would like to use for your study. Define all relevant variables (responses, explanatory variables and secondary or mediating variables). Define the specific set of hypotheses you would like to test. Define the basic statistical analyses you will use to examine each hypothesis.
2. For your research study, discuss the issue of sample size. For the response of interest, will you be able to collect several hundred subjects? Or a very small number of subjects? Is the study a large-scale research study, or essentially a pilot study? To access enough sample size, will you have to include more than one data collection site? Discuss how sample size issues will affect (i) your study design, (ii)

TABLE 5.2

Study Design Guidance

Context	Study Design	Statistical and Data Analysis
Case study	No design needed	Describe individual case
Case series	No design needed	Describe collection of cases
Cohort Study	Observe Exposed and Non-exposed	Variety of methods available
Case-Control Study	Match controls to existing cases	Variety of methods available
Systemic Review	Choose studies with defined criteria	Meta-analysis

the data to be collected at each site, and (iii) the statistics you will be able to use to test your hypotheses of interest.

3. For your research study, using approximate numbers, fill in an initial NIH consort diagram.
4. Discuss the potential benefits and pitfalls of the following study designs: (i) case-control, (ii) cohort, and (iii) case series designs.
5. Discuss methods of restricting sample variation in your design and data collection. In particular, discuss how you may be able to (i) match cases and controls, (ii) recruit larger numbers of subjects.
6. Given the study design and overall statistical model, list out (i) the specific statistical hypotheses you will need to test in your research study and (ii) the specific statistical test you will use for each.

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6

Administrative Issues: Consent Guidelines, IRB, HIPAA, FDA

Research within institutions, both private and public, has developed into a carefully monitored process with many formalities and related paperwork.

Over the past 50 years this has evolved to require formalized training in regard to most aspects of the research process, including the recruitment and treatment of subjects and related data. This is especially true for medical research involving human subjects.

Medical research in the United States with human subjects tends to reflect federal Health Insurance Portability and Accountability Act (HIPAA) regulations [1], Federal Drug Agency (FDA) [2] and NIH Human Subjects regulations [3].

Often these various requirements and laws are reviewed in the training offered by your institution especially in regard to the proper administration of a research project, including patient rights and privacy considerations. The guidelines are carefully monitored by the IRB [4] of the institution.

Several areas of administrative focus are worth reviewing:

- Obtaining consent on the part of research subjects,
- The HIPAA guidelines covering access to electronic medical records and information
- IRB and its overseeing of research activities.
- Participation in drug or medical device related trials and related FDA review.

Additional references can be found in NIH documentation [5, 6] and other related sources [7, 8]. See for example the PubMed [9] listing.

6.1 Consent Guidelines

The definition of consent and the rules for obtaining consent have been carefully developed over many years, often in response to earlier settings where consent was not properly obtained [10].

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A consent form and process should follow some basic guidelines:

- The consent form for a specific study should clearly explain the research project and the information to be collected from each individual participant, including how this will be kept confidential, and how this information will be used.
- All participants have the right to drop out of a study at any time.
- All research must show some sort of benefit to the patient or society in general.
- In some settings, for example where children are to be involved in the study, consent must be obtained from legal guardians, usually a parent.
- In some extreme situations, for example stroke research in an emergency department setting, the patient may not be able to give consent, and time limitations restrict the ability to find a legal partner or guardian. In such cases consent may be acceptably limited if there is a benefit to the patient.
- A cover letter should explain the research project and the costs and benefits of being involved in the study.
- The consent document should be read and signed by all participants.
- An initial pre-consent may be required to contact individuals if they are to be contacted by phone. In medical settings this may be carried out in a clinic or doctor's office setting.
- Almost all studies will require a basic reference or control group. If the study is a case-control study, where cases have been identified and controls are to be matched to them, electronic hospital records are an excellent source of potential patients who can serve in a control group.
- All consent documentation must have IRB approval.

6.2 Specific Content of a Consent Document

Each potential research subject must be given a copy of the study consent document to review. This consent document will have several mandatory sections that individuals must read prior to signing the document:

- A clear statement that the study involves research, an explanation of the purpose of the research, the expected duration of the subject's participation, a description of the procedures to be followed, and identification of any procedures that are experimental.

- A description of any reasonably foreseeable risks or discomforts to the subject.
- A description of benefits to the subject or to others that may be expected from the research.
- A disclosure of any alternative courses of treatment that might be advantageous to the subject.
- A description of how confidentiality of records identifying the subject will be maintained.
- For research involving more than minimal risk, a description of possible compensation and medical treatments available if injury occurs.
- An explanation of whom to contact for answers to questions regarding the research, and whom to contact in the event of a research-related injury.
- A statement that participation is voluntary, refusal to participate involves no penalty or loss of benefits, and the subject may discontinue participation at any time.
- A statement that identifiers might be removed from the identifiable private information or biospecimens and the information or biospecimens used in future research studies.

6.3 Consent Form Development

- As a general rule: the more detailed the procedures, the more detailed the consent form.
- If you are applying a Randomized Control Trial (RCT) study design [11], the best type of design, there is much to document and record in terms of information and the consent form may include many pages.
- The form will be carefully reviewed as part of IRB review.

A suggested outline of the consent document is given:

Table 6.1

Consent Form Outline

Variable	Content
Title	Formal title of the research project
Where	Location of the study
When	Timeline of the study

(*Continued*)

TABLE 6.1 *(Continued)*

Consent Form Outline

Variable	Content
Who	The investigators carrying out the study and data analysis
Definition of consent	The voluntary aspect of consent should be emphasized
Background	Describe the treatment and why the research is important
Purpose	What is the goal of the study, what the researchers expect to learn, potential benefits to patients and society
Procedures	Listing of screening exams, all tests and procedures to be carried out
Risks	Discuss risks of each procedure involved in the treatment
Alternative Treatments	Discuss available treatments if patient does not participate in the study
Potential Costs and Benefits	List potential benefits to the subject and other patients having similar illness. List also benefits to medicine and medical research in general
Clinician/P.I. Financial Disclosure	List out if they are relevant
Payments to Participants	List out in detail, but not always allowed.
In case of Injury	Where subjects can receive treatment for injury or side effects
Patient Rights and Withdrawing from the Study	This should clearly be stated and understood
Confidentiality and Privacy	Procedures for safeguarding all individual research data should be clearly listed
Questions	Potential participants should be asked if they have questions

6.4 Remuneration

Some basic level of remuneration for the participants is fairly standard, as well as covering transportation and other miscellaneous costs. But it should not be allowed to unduly bias the study.

6.5 Example

For the example outlined in Chapter 2, detailed information will be collected from all participants and this will require a detailed consent document.

Patients might be recruited from doctor's offices, or various clinics or hospitals and consented in those settings.

The clinical assessments required for the study would typically be carried out in a central location or set of clinics and patients and controls would be reimbursed for travel and participation.

For the case-control study, cases would be recruited and then controls selected from existing electronic databases, matching on age range and excluding those subjects with health conditions not related to obesity, for example, excluding patients with cancer and other illnesses.

A detailed explanation of how the data would be collected, organized and kept confidential would also be required.

6.6 Institutional Review Board (IRB)

An IRB [4] is a review committee established to help protect the rights and welfare of human research subjects.

U.S. federal regulations require IRB review and approval for research involving human subjects if it is conducted, supported, or regulated by U.S. federal departments and agencies. Often institutions use their IRB to review all research activities.

We examine here what you can expect from an IRB review.

First, some relevant definitions:

A **research activity** must meet the definitions of both research and human subjects in order to be considered human subjects research.

Research is defined as: "A systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge" (Protection of Human Subjects 2018).

A **human subject** is defined as: A living individual about whom an investigator (whether professional or student) is conducting research.

Identifiable Private Information is private information for which the identity of the subject is or may readily be ascertained by the investigator or associated with the information (Protection of Human Subjects 2018).

An **Identifiable Biospecimen** is a biospecimen for which the identity of the subject is or may readily be ascertained by the investigator or associated with the biospecimen (Protection of Human Subjects 2018).

A **Clinical Trial** is a research study in which one or more human subjects are prospectively assigned to one or more interventions (which may include placebo or other control) to evaluate the effects of the interventions on biomedical or behavioral health-related outcomes (Protection of Human Subjects 2018).

According to **federal regulations an IRB can:**

- Approve or disapprove research.
- Require modifications.
- Conduct appropriate continuing reviews.
- Suspend or terminate approval.

There are minimum federal requirements regarding the information subject to IRB review:

- Risk/Benefit Analysis should be carried out.
- There should be an Informed Consent review.
- The benefits of the research should be distributed fairly among the community's various populations.
- There should be clear procedures to assure the confidentiality of all subject information collected.
- There should be a plan for Collection, Storage, and Analysis of Data.
- The qualifications of the principal investigator (PI) and collaborators should be reviewed.

6.7 There Are Three Possible Initial IRB Review Procedures

- Convened Committee Review
- Expedited Review
- Limited IRB Review of Select Exemption Categories

6.7.1 Convened Review

The review must be conducted at a convened meeting of the IRB, which may only occur monthly or quarterly. A majority of IRB members (a quorum) must be present at the meeting.

At least one member whose primary concerns are in nonscientific areas must be present at the meeting (if relevant, FDA policy recommends that a physician be present).

6.7.2 Expedited Review

May be used for minor changes in previously approved research and for research falling into specific expedited categories.

Expedited reviewers can:

- Approve research.
- Require modification.
- Defer review to the convened IRB.

To approve a research project, the IRB must determine that:

- Risks are minimized and reasonable regarding anticipated benefits.
- The selection of subjects is equitable.
- Informed consent will be sought, documented or appropriately waived.
- Adequate provisions exist for monitoring collected data.
- Adequate provisions to protect privacy of subjects and data confidentiality.

6.7.3 Continuing Review

The IRB must re-review studies that have greater than minimal risk not less than once per year.

The IRB may require reports for:

- Adverse events or unanticipated problems involving risks to subjects or others.
- Incidents of noncompliance.
- Deviations from an approved study plan and violations of the terms of approval.
- Data safety and monitoring report summaries.

6.8 Example

Some of the issues relevant to the study in Chapter 2 that may draw the interest of the IRB include:

- Consent forms.
- Recruitment of controls.
- Clinical assessments to be conducted.
- Genetic information to be collected.
- Data privacy issues.

- Training and professional background of researchers.
- Risk/Benefits analysis.
- Diversity of patients recruited.

6.9 Health Insurance Portability and Accountability Act (HIPAA)

Given the use of electronic records and the ease with which large amounts of information can be accessed, the U.S. Congress decided to pass a law that would govern the collection, protection, maintenance and access to medically related information. This is the HIPAA [1].

- The purpose of HIPAA is to protect a patient's health information while providing basic rights related to access and tracking how that information is used and shared.
- Covered entities, such as hospitals and most health care entities, health care clearinghouses and related business associates, are all obligated to comply with HIPAA.
- **Protected health information (PHI)** [1] is any information that identifies a person or could reasonably be used to identify a person AND is related to the person's past, present or future health care or payment for health care.
- The law defines 18 identifiers that, if removed, would render patient information de-identified and no longer subject to HIPAA. These are listed in Table 6.2:

TABLE 6.2

HIPAA Patient Identifiers

Names	Health plan numbers
Birth date	Account numbers
Admission or discharge date	Certificate/license numbers
Date of death	Vehicle VIN and license plate numbers
Telephone or fax numbers	Device identifiers or serial numbers
E-mails	URLS
Social security numbers	IP addresses
Medical record numbers	Finger and voice prints, biometrics
Geographical regions smaller than a state	Full face images

The HIPAA Privacy Rule establishes standards to enable patients to have greater access to their own medical records and more control over how their personal information is used and/or disclosed outside a covered entity.

- An organization may not use or disclose PHI except as permitted by the HIPAA Privacy Rule or as required by law. The HIPAA PHI disclosure rules affect day-to-day activities within a health care organization including talking with patients and co-workers, computer work and making phone calls.

HIPAA basically requires workers in health care organizations that are covered entities to use and **share only the minimum necessary patient information needed to complete their jobs**. This means that representatives of a covered entity should:

- Use and disclose only the parts of the patient's medical record needed to complete the task.
- Identify which members of its work force require certain PHI and limit access accordingly.
- Review requests for non-recurring disclosures on an individual basis under the criteria.

If you can link health information to a patient's identity, then the information is considered PHI and must be protected.

6.9.1 Patient Rights

Patients have the right to:

- View or request a copy of their medical information.
- Request an amendment to their medical records.
- Request confidential communications.
- Request accounting of disclosures.
- Request restrictions on how their information is used and disclosed.

6.9.2 Covered Entity

Every health care provider, regardless of size, who electronically transmits health information in connection with certain transactions, is a covered entity.

These transactions include claims, benefit eligibility inquiries, referral authorization requests, or other transactions for which HHS has established standards under the HIPAA Transactions Rule.

6.9.3 What Information Is PHI?

PHI. The Privacy Rule protects all "individually identifiable health information" held or transmitted by a covered entity or its business associate, in any form or media, whether electronic, paper, or oral. This includes patient identifiers.

Individually identifiable health information is information, including demographic data, that relates to the following:

- The individual's past, present, or future physical or mental health or condition.
- The provision of health care to the individual, or the past, present, or future payment for the provision of health care to the individual.
- Includes name, address, birth date, Social Security Number.

A covered entity must disclose PHI in only two situations:

- To individuals (or their personal representatives) specifically when they request access to, or an accounting of disclosures of, their PHI.
- To the U.S. Health and Human Services when it is undertaking a compliance investigation or review or enforcement action.

6.9.4 De-Identified Health Information

There are no restrictions on the use or disclosure of de-identified health information. De-identified health information neither identifies nor provides a reasonable basis to identify an individual.

There are two ways to de-identify information:

- Formal determination by a qualified statistician.
- Removal of specified identifiers of the individual and of the individual's relatives, household members, and employers.

6.10 Example

Chapter 2-related issues:

- Background information drawn from electronic databases to support clinical assessments of patients and controls.
- Accessing electronic databases to determine controls to match to cases.

- All data to be analyzed would be de-identified. Only the P.I. would have PHI level data.

6.11 Federal Drug Agency (FDA)

- The FDA [2] has authority over the evaluation and subsequent public use of health products.
- The FDA conducts reviews of drugs, biologics, and medical devices for safety and effectiveness before granting approval for marketing.
- The FDA comes to an approval decision using research conducted under an Investigational New Drug (IND) application or an Investigational Device Exemption (IDE), and reported to the FDA by the sponsor of the product.

6.11.1 Investigational New Drug (IND) Application

The FDA deals with research involving a drug or biologic that has not yet reached the marketplace or research that studies a new use of an existing marketed product requires an IND application [2].

The IND application is submitted by the sponsor to the FDA. A sponsor develops a research plan for a clinical investigation, which is then evaluated by the FDA. A sponsor can be a drug company, cooperative group, or even an individual physician. After review of the IND application, the FDA will allow human studies to proceed if it determines that risk of exposure to the drug is reasonable.

This determination is based upon:

- Data from prior animal or human testing.
- Methods of manufacturing.
- Plans for testing and reporting significant toxicities.
- A well-developed clinical research plan that minimizes risks to the subjects.

A sponsor cannot begin a clinical investigation until the IND is granted by the FDA and the IRB approves the study.

When sufficient data have been obtained to provide evidence of safety and efficacy, the sponsor submits the trial's results to the FDA as part of an NDA that requests permission to market the product in the United States.

The FDA then reviews the application. Given that it may take several years and much investment of resources to develop a pharmaceutical treatment, these reviews are very detailed.

Only after FDA approval of the NDA can the drug be sold to consumers.

6.12 Practical Issues to Consider Before Beginning the Research Project

For all of the above types of research considered, and any serious research project, the researcher needs to be able to handle all the necessary activities and obligations. This implies seriously considering what you can deal with in terms of time, support and workload.

Investigators should first carefully review the study protocol. The investigator should consider the ability of the research team and the organization to carry out the project to a successful conclusion.

Some questions should be asked:

- Does the investigator have adequate time to devote to study oversight and the work that he/she must perform?
- Are adequate personnel available to complete and process the required records?
- Is the proposed study budget adequate?
- Are facilities and equipment adequate to perform the study?
- Can subjects be recruited and complete the study during the timeframe desired by the sponsor?
- How will each aspect of the protocol procedures be executed?
- Is the timing of the procedures reasonable?
- Do the investigator and study staff members understand the requirements for data collection and reporting?
- Will the subjects' data be readily accessible for formal review?
- Can study-specific records and subject data be maintained for at least two years after the study has been completed?

6.13 Adequate Resources

If you are the primary researcher or principal investigator on your research project, then you are responsible for all aspects of the clinical trial. As such, you should consider the following resource issues carefully before beginning the project:

- You should be able to demonstrate (e.g., based on retrospective data or similar past studies at the same institution) a potential for recruiting the required number of consented subjects within your recruitment period.

- You should have available an adequate number of qualified staff and adequate facilities for the planned duration of the research project.
- You should ensure that all persons assisting with the trial are adequately informed on the protocol, the research intervention, and their research related duties and functions.
- You are responsible for supervising any on-site individual who has research project related duties and functions.

Be careful what you agree to do. If interacting with a local or federal agency, or a private or pharmaceutical company, you will be signing legally binding contracts.

6.14 Discussion Points

1. Outline and discuss the issues you expect to have in relation to the recruitment and consenting of subjects into your study. Do you have resources to provide compensation for individuals participating in the study? Can you list out possible reasons the resulting Sample may be biased?
2. Contact your local IRB and obtain the relevant documentation for applying to the IRB for permission to conduct your research project. Go over the relevant sections and outline the details of your project. Develop timeline for your project (this will be required by the IRB).
3. In relation to patient privacy and personal health information, how will you keep patient and/or subject data confidential? How will you ensure HIPAA restrictions are followed? What are the general data confidentiality issues affecting your research project?
4. List out the variables to be collected from each research subject. Which of them qualify as PHI?
5. In the section dealing with practical issues to consider before beginning your research project, answer each listed issues and related question given in the section in relation to your research project. Focus on the issue of resources, mentoring and adequately trained support.

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7

Qualitative and Quantitative Approaches to Research

Here we briefly discuss two basic types of data and related approaches to collecting information: quantitative and qualitative data and related studies.

These often require distinct methods for analysis of the data collected.

7.1 Qualitative Analysis

Qualitative research methods, often using interviews, focus groups and open-ended survey questions, tend to focus on responses that are not necessarily numerical in nature [1].

For example, a question asking the research subject to write a few paragraphs about their feelings in regard to their quality of care, is typical of the way in which qualitative data is collected.

Another approach are the responses given in a structured interview, with all participants receiving the same set of questions, where the possible answers do not follow a specific set of acceptable responses, but are typically open ended, encouraging more thoughtful and personalized responses.

Such approaches allow for broader, thematic analysis of the data, which are not easily accommodated in the setting of numerical data. That said, it is much easier to standardize and compare numerical results with existing results in the literature.

In some settings, video capture may yield insights based on the subjective opinions of observers. An example is the use of video capture and review in the determination of autistic diagnoses for children.

In each of these qualitative settings, the use of numbers or data values may not be of use in providing relevant and useful information.

These approaches to collecting information are often viewed as a way of making the overall statistical results, often based on averages, proportions and measures of variation, more personal, often focusing on personal experience, opinion and belief [2, 3].

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7.2 Quantitative Analysis

7.2.1 Statistical and Data Analysis

Quantitative research methods tend to be based on the collection of numerical data, typically given by continuous variables, categorical variables and combinations of the two. Graphical summaries are also often important. This requires consideration of the scales to be used for the data values in question. Data values can be analyzed as collected, or they can be adjusted in various ways, for example standardizing all the values so they can more easily be compared.

The interpretation of data values may often require the use of probability based models, and more formal structures, to model the randomness in the data and assess whether there is a significant pattern in the data, for example a difference between a group of cases and a group of controls.

This approach has been an important aspect of progress in science, helping to define and identify important patterns and mathematical patterns that summarize and make clear patterns in the raw data. These patterns are often linear, but can be more complicated, for example nonlinear [4].

Often data are collected in the context of a model-data combination with a mathematical model guiding the selection of variables in question. The mathematical models are then "fit" to the data, summarizing the underlying patterns in the data, but also provide a context within which specific model parameters, interpreted as population characteristics, can be tested for significance and relevance. They can also be estimated from the data.

An important aspect of all research is the study design chosen used to organize, collect and analyze the data. It will help the researcher choose the statistics that are most relevant, and provide a context for testing hypotheses and estimating population values [5].

Chapters 8–11 of this book provide a basic and accessible review of the some of the various quantitative methods available to analyze the collected sample data.

There are reporting standards that can be used to organize the reporting and dissemination of qualitative results. For example, The Consolidated Criteria for Reporting Qualitative Research [6], which is a 32-item checklist that applies more directly to data drawn from focus groups and interviews.

Note that when both quantitative and qualitative methods are used in the design of a study, this is usually referred to as a mixed-methods study [7].

7.3 Machine Learning and AI Methods

Also available for the quantitative analysis of data are recently developed machine learning algorithms [8]. These are primarily search algorithms,

which use specific structures, for example multi-nodal networks or classification trees, to detect and compare patterns in a dataset or larger database. They are extremely flexible and do not necessarily reflect the idea that there is an underlying, simple structure in the data about which random error is occurring. They use a simple random split of the data into training and testing portions to assess the resulting observed pattern in the data. Note that there is no formal prediction here, simply an assessment of whether the pattern in one portion of the data agrees with the respective pattern in the other half of the data [9].

There is a large number of such machine learning approaches, using many highly flexible formats with which to fit models ("train models") to the observed data. From the perspective of earlier approaches which viewed the data as pattern + random error, these approaches assume that much of the random error is useful in predicting data patterns. As such, these approaches can provide better fitting models to a specific set of data, but these results are closely tied to the specific source of the data (e.g., a specific hospital database) and do not easily generalize.

A review of these approaches is provided in Chapter 14.

7.4 Aspects of Qualitative Research

The advantage of qualitative studies is that they provide a more detailed set of data (non-numerical) from which a synthesis can be developed by the researcher, expressing the main themes mentioned by the subjects in the study. These may not be easily measurable by quantitative methods.

7.5 Some Qualitative Methods

Qualitative methods are typically based on less restrictive data collection methods compared to quantitative methods. These methods [10] include:

- **Individual participant interviews**: These are often structured (the same set of questions) so that each participant in the study is given the same set of questions. Interview sessions are also a useful setting for collecting information on sensitive subjects. The recording of responses here can take several forms, but is often, initially, the direct recording or transcription of the participant responses to each question.

- **Focus groups**: In some settings, the observation and recording of group dynamics and group discussion of issues, for example challenges in accessing healthcare may be the response of interest. These need to be carefully chosen to reflect accurately the overall population of interest.
- **Observation of participants**: Participant responses can be evaluated by the researcher or interviewer, either immediately or later, upon review of a video record of the meeting. In some settings nonverbal responses and reactions may be of interest, for example in patients with autism, or patients with physical limitations on verbal responses (e.g., stroke).
- **Additional Comments**: Comments that place the structured interviews in context, or comments that do not directly respond to the questions but given information on the community, school or other contexts relevant to the study may also be useful for the researcher.

7.5.1 Interviewer and Observer Potential Bias

As the information and related data are being collected from participants in a qualitative study often requires the active participation of the researcher, limiting the possible bias of the researcher or interviewer is important to achieving a good study design [11].

In the setting of focus groups, the researcher or interviewer will often need to respond to comments by participants in order to encourage group discussion and the development of group dynamics. In such settings, bias on the part of the researcher must be carefully limited.

Observational studies are often employed in the context of qualitative research studies. The use of multiple observers is important to avoid individual bias. This will imply the measurement of correlations and similarities between observers. A practical area where these issues arise is in the assessment of radiology graphs (x-rays for example) by several radiologists to determine a diagnosis.

7.5.2 Validation Criteria

There exist validation criteria for qualitative study approaches [12]. These can be used to review the qualitative study to be used. These are objectivity, internal validity, external validity and reliability. These focus on, respectively; confirmability, credibility, transferability and dependability. All of which reflect the more subjective nature of this type of research and the challenge in generalizing the findings beyond the sample cohort providing the responses.

There are some challenges as well in regard to the way in which subjects may alter their behavior in an interview or focus group setting. Two examples are well-known:

7.5.3 Hawthorne Effect

The Hawthorne effect arises when individuals who know they are being observed, alter their behavior [13]. This occurs even when small changes (better lighting for example) alter a well-known environment. This will often occur in interview or focus group settings.

7.5.4 Observer-Expectancy Effect

The observer-expectancy effect [14] often occurs in the setting of interviews where the subject provides answers they believe will satisfy the observer or interviewers desired effect. This may occur subconsciously and researchers need to be aware of the need to not imply a desired effect by carefully wording questions and standardizing their behavior throughout the study process.

7.6 Clinical Text or Doctor's Notes in Medical Research

Clinical text is also a component of electronic health records (EHR), often in a doctor's notes section. It may include discharge summaries, patient treatment plans, and patient progress notes, all of which can contain information that is useful for research purposes. However, this information may be not be standardized and recorded in a consistent manner. For example, fever may be reported as "fever," "above average temperature," or "elevated temperature." This makes the analysis difficult and often the data must be re-organized prior to being useful for analysis.

One possible approach to deal with nuanced clinical text may be to use open-source natural language processing tools. These can be programed to extract text relevant to, for example, disease stage, severity, progression of the disease or medical symptoms [15], which may not easily be captured by diagnostic codes. That said, it may be very useful for formal analysis to link these more general references, often the subjective output of the doctor in question, to specific ICD based diagnoses [16].

Note that such analyses are based on access to sources of information that are very private, both from the doctor and the patient perspectives. As such, they should be subject to very detailed HIPAA, IRB and other review. Confidentiality should be maintained during all aspects of data collection and data analysis.

7.7 Some Examples of Qualitative Based Data

- A question on a survey given to the parents of pediatric patients asks whether the parents feel about the level of care and overall treatment of the patient. The question is open ended, leaving an empty section for the parents or fill in. Here, the possible answers will be information given in a wide variety of sentence structures and possible topics related to patient care and treatment.
- A cohort of children is consented (parental consent) and brought into a clinic to be assessed for autism. After individual assessment using a response to an interview by the clinician or specialist, a video of each child in a standardized controlled setting is obtained and analyzed by a clinician. To avoid bias, several clinicians must assess each video, with a final diagnosis obtained when they come to an agreement.
- A cohort of young people (18–25) participating in an open house in the hospital are consented and given a survey to assess and measure their beliefs in regard to abortion rights and related access to care. While the survey contains many multiple-choice questions, several open-ended questions are provided to assess individual level experiences and feelings in regard to the issue.
- A cohort of health professionals attending a conference is consented, in a classroom environment, to assess an educational presentation by a chosen speaker. The assessment is both standardized and open ended. Questions focused on whether the participants have suggestions for the speaker to improve their presentation are open-ended.

7.8 Application to Chapter 2 Dataset

1. A patient satisfaction survey is administered at the end of the obesity study containing multiple choice and open-ended questions.
2. An open-ended question regarding adherence to the dietary program and potential reasons for higher levels of adherence is part of an exit interview given to all participants in the study.
3. A focus group of patients is asked to discuss their personal journeys and the difficulties of losing weight, both during the program and in previous attempts. This is captured as a discussion on video.

TABLE 7.1

Common Qualitative and Quantitative Methods.

Qualitative	Quantitative	Integrated
• Structured interviews • Unstructured interviews • Open-ended questions • Video capture • Image analysis • Focus groups • Individual interviews • Observation • Hawthorne effect • Observer expectancy • Clinical notes	• Statistical analysis • Data analysis • Statistical inference • Population based • Probability based • Numerical responses • Data-centric machine learning and AI approaches • Artificial neural networks • Random forest methods/ classification trees	• Qualitative methods • Quantitative methods • Categorical and continuous variables, often redefined • Scores based summaries • Personalized and overall assessments • Mixed methods

7.9 Quantitative Methods Summary

Much of the data analysis presented in this book is focused on quantitative analysis, the analysis of data that is primarily collected on a numerical scale. Some standard approaches to the analysis of such data include:

- Descriptive Statistics
- Graphics
- Statistical Inference
  - Estimation
  - Testing
  - Predictive Models
- Machine learning/AI
- Bayesian learning methods

7.10 Mixed Methods: Using Qualitative and Quantitative Methods

Using both quantitative and qualitative approaches together, so-called mixed methods [17], may provide a very rich and insightful dataset.

Qualitative methods are emphasized when developing and carrying out an initial pilot study to assess the relevance of a proposed larger study.

Researchers often survey a small sample of the overall population to ensure that their hypothesis of interest is relevant and understood by both participants and researchers.

Side-by-side: These approaches are often both used simultaneously in a research study. Quantitative methods for numerical responses, for example multiple choice Likert scales (e.g., low to high measured on a 1–5 numerical scale) can be used in a survey setting along with open-ended questions asking for general descriptions of how well the participant feels the study helped them deal with their condition. These can then be linked to sets of clinical and demographic data to create a wide-ranging and useful database for research.

Integrated together: The analysis of qualitative summaries can be later rescaled and analyzed using quantitative measures. For example, the number of times a specific set of keywords or text sequences arises in the responses to a specific set of questions. Individuals specifying a specific set of ideas or responses on open-ended survey questions can be grouped together and their continuous responses compared to a control group or other demographic group.

Initial pilot surveys of the population of interest, perhaps in the form of focus groups, can help researchers develop research goals and ideas that can then be examined using more detailed quantitative methods and formal study protocols. These more formal study protocols, in the planning stage of a larger subsequent study, will typically use a formal study design to investigate specific hypotheses. These include cohort studies, case-control studies, randomized clinical trials, systematic and meta-analytic reviews and case reports.

7.11 Examples of Use

- **Clinical**: In clinical studies it is sometimes difficult to objectively measure the pain a patient is feeling due to surgery or an invasive treatment. Many measures exist, but recording the actual description of pain given by the patient is important. To study the ability of Drug A to lower the level of pain, a researcher asks each post-surgery patient in a consented cohort of patients to discuss to describe the pain in detail. If certain key words are recorded in the open-ended format the patient is then placed in one of three pain groups: low, medium, high. This is then used to guide the

dosage given of Drug A. The time it takes to reduce the pain after the drug is given is recorded and analyzed as a function of dosage level. This analysis is then repeated within gender groups and across age groups.

- **Psychology**: A high rate of suicide is often observed in marginalized groups of people. After consenting a group of young people in a marginalized groups, each one is asked to fill in a brief survey focused on possible suicidal thoughts and availability of support. The researchers have pre-defined a set of specific words and answers that may indicate a young person is at high risk of suicide. Those young people meeting this criteria are classified as high risk and placed in an intervention program where a detailed treatment plan with various continuous measured variables is available.
- **Health Services**: A local hospital would like to improve services for walk-in patients. A prepared set of specific questions with multiple choice responses is given to each consented patient. The survey also asks each patient to discuss any issues, mentioned in the survey or not, that might be improved, giving the patient improved access to care and quality of care. These responses are used to develop an updated version of the survey instrument, formalizing some of the mentioned issues in the form of additional specific multiple choice questions.

A summary of some commonly used qualitative and quantitative methods is given in Table 7.1.

7.12 Application of Chapter 2 Dataset

In an attempt to help individuals avoid taking short-term pharmaceutical agents to lose weight, a comprehensive, individually-based exercise, diet and group discussion plan is made available to subjects in the study failing to lose weight in the initial study. To analyze the data using a primarily continuous variable, several dimensions of success are defined. These include self-reported attitudes, self-worth assessment, weight loss above 1 pound per week (yes/no) and other categorical variables.

As well, several open-ended questions dealing with personal habits, possible depression and how each person deals with such issues are included. The researchers have a defined set of categories that each patient is placed in, when certain sets of keywords are recorded in these questions. These

categorical qualitative responses are then weighted, summed and provide an overall score by which the success of the program is evaluated.

7.13 Discussion Points

1. Adding a qualitative component to your research study design and related data collection is always a possibility. This is most easily accomplished by adding an interview component for subjects. This might address subject feelings and perceptions of the research process, the treatment intervention itself, and related caregiver issues. For your own research project, discuss including a qualitative component to the protocol.
2. In the context of a planned research project, you have decided to include a qualitative component. Discuss which approaches you might use, including open-ended questions in a survey instrument, individual interviews and video based observations.
3. In a research study focused on autism that will use multiple hour video capture to study patient response to a treatment intervention, discuss how you might record relevant aspects of the data, to develop a more quantitative summary of observed behavior. For example, the number of times the patient walks in a circle, the number of loud vocal episodes, etc. Once you have these counts, they might be combined to develop a set of overall counts or scores, reflecting different aspects of behavior. Discuss also how qualitative review by individual observers and differences between reviewers might also lead to these observed values being biased.
4. Discuss use of focus groups in your research project to guide the collection of pilot data. In some settings, especially underserved populations, there may be existing biases in the set of potential subjects in regard to how they are treated by the health care system, for example in the setting of interaction with doctors. Focus groups may allow for the development of carefully focused sets of questions and give guidance on how these questions should be worded.
5. Many research studies are based on access to a clinical database, guiding the selection of cases and matched controls and providing baseline data for the study. This clinical quantitative database can be augmented with additional qualitative data, for example using a survey instrument to assess mental health after a surgical intervention.

Consider how you might use qualitative based follow-up in your research project.

6. Health services based assessments often require both qualitative and quantitative data. The data tends to be very broad and there are many instruments available that can be used to collect information on the overall health, mental health and general health information from the participating subjects. Make a list of the possible survey and assessment instruments that might be used to assess subjects in your research project.

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8

Basic Data Analysis and Graphics

8.1 Introduction

A researcher who wishes to develop a useful design for their experiment should have a good basic understanding of the statistical aspects of experimental design and data analysis.

These are the basic tools with which you will examine, study and interpret the data.

In this chapter we review some of the basic aspects, without much mathematical detail, and their application in examples.

8.2 What Are Statistics?

- A statistic is simply a function of the data.
- A statistic can be a number (sample mean, sample proportion, sample odds ratio, sample standard deviation, etc.). This is usually a function of the data values.
- A statistic can be a graphic (histogram, boxplot, scatterplot, etc.).
- A very important set of statistics are Measures of Variation which are often defined by the spread of the data about its mean or median. These include Standard Deviation, Standard Error, Inter-Quartile Range (75*th* percentile–25*th* percentile).
- Different statistics apply in different settings, in different study designs.
- Each statistic represents a piece of the overall information in the data.
- Sometimes a model is required to define the statistic of interest (e.g., the slope parameter of a straight line fitted to the data).
- Statistics also include measures of how well a model fits the data (goodness-of-fit diagnostics) [1].

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TABLE 8.1

Basic Definitions

Statistic	Interpretation
Mean	A measure of the middle of the data
Standard Deviation	Spread of the data about the mean
Median	The 50th percentile of the data values
IQR	75th percentile–25th percentile
Range	Maximum value–Minimum value
Histogram	Relative frequency plot
Skewness	Measure asymmetry of the histogram
Kurtosis	Measures probability of extreme values
Outliers	Data values exceptionally far from the mean
Proportions	Number of cases/Total number
Rates	Proportions defined over a time period
Odds	Odds of event A = P (A)/[1-P(A)]

8.3 What Is Biostatistics?

Biostatistics is basically statistics for specific areas of application in health related fields [2] including:

- Biology
- Medicine
- Infectious Disease
- Genomics
- Health Services Research
- Population Health
- Community Health

8.4 What Is Data Science?

Data science is a combination of statistics, computer science, probability models and database analysis. It has been extended over time to include various types of complex algorithms that can search existing databases and examine them for specific structures [3].

The data in question is very flexible, which differentiates it from more standard statistical approaches, including images, e-mails, text sequences, as well

as numerical data. The approaches can be supervised (linking a response of interest to a set of secondary variables) or unsupervised (examining all the data variables for clustering patterns).

This type of detailed database analysis is data-centric, meaning the conclusions drawn basically apply only to the database in question. There is often no attempt to use probability and population based methods and models to extend the relevance of the patterns found in the database. The methods include:

- Artificial Intelligence (AI)
- Data Mining
- Big Data
- Machine Learning
- Artificial neural networks
- Random forest methods
- Cluster analysis
- Applied statistical methods

These methods are discussed in more detail in [4] and Chapter 15.

8.5 Collecting Data

- The data you collect should be representative of the population you want to study.
- The variables you choose should be relevant to the demographics and specific measurements and responses you want to study.
- Note that data can be poorly collected.
- It can contain biased measurements (e.g., your measuring device was not accurate, or you sampled a non-representative sample from your population).
- If you simply go out and collect data values from a large database you run the risk of these and other problems being present in your data. You need to carefully consider how the data is being collected.
- To do quality research you need quality data. You may also need a moderately large sample size.

It is useful to consider some additional basic issues when putting together a research database. Often the database will be drawn from several sources, or downloaded from a larger database. This implies the need to develop

a set of keywords that will be used to narrow the database for research purposes.

It also implies the need to merge together various types of databases, for example, clinical, demographic and administrative, in order to obtain the set of variables to be analyzed.

Each database presents its own challenges, including understanding the definitions and scales used to define each variable. Further, the presence of missing data may present challenges in the use and interpretation of the data and any models developed for the data.

8.6 How to Find Electronic Sources of Data for Research?

- You can do electronic search in an existing database.
- Or find data in a clinic's databases.
- Or enter it yourself into Excel [5] or a data entry package such as REDCap [6].
- Often you will have to merge various databases together.
- If the database has been restricted or limited in some way over time, be sure to be very clear on reporting these limitations and how they affect your ability to interpret the data.

8.7 Descriptive Statistics, Graphics, and Basic Linear Models

Once we have a dataset, the question arises: how to organize and access the information in the data. How to turn data points into patterns relevant to your specific aims.

8.7.1 Descriptive Statistics

- The descriptive statistics used to summarize some of the information in the data, will depend on whether the variable is continuous or discrete (a counting variable).
- For a **continuous variable** descriptive statistics include: sample mean, median, standard deviation, Inter-Quartile Range (IQR), minimum, maximum, range and others.
- The sample mean and median give measures of the middle of the data (central location).

- The sample standard deviation, IQR and range give measures of variation about the central location (the spread of the data).
- For a **categorical variable**, standard statistics are often simply the count of the events of interest which are then expressed as proportions by dividing these numbers by the relevant totals [7].
- If proportions are collected through time or are interpreted on a monthly or yearly basis, they are typically called rates.
- If proportions are to be compared, relative risk measures and odds ratios are often the statistics of interest.

Table 8.2 gives an example of data summarization for the cohort data given in Chapter 2.

TABLE 8.2

Basic Descriptive Statistics

Variable	Sample size	Mean	Std Dev	Median	IQR
Bodyfat Index					
Pre-intervention	99	18.1	9.1	18.2	15.1
Post-intervention	99	14.3	8.8	13.7	14.5
Weight					
Pre-intervention	99	176.8	32.8	176.0	38.0
Post-intervention	99	170.9	32.8	168.3	39.1
Hip					
Pre-intervention	99	99.8	8.4	99.2	8.3
Post-intervention	99	95.6	7.2	95.0	8.6
Abdomen					
Pre-intervention	99	91.6	11.8	89.7	15.7
Post-intervention	99	86.7	11.7	85.2	15.3
Neck					
Pre-intervention	99	37.9	2.6	37.8	2.7
Post-intervention	99	35.3	2.8	35.2	2.5
Chest					
Pre-intervention	99	100.0	9.0	99.6	11.6
Post-intervention	99	95.6	9.1	95.1	11.4
Thigh					
Pre-intervention	99	59.4	5.8	58.9	8.1
Post-intervention	99	54.9	5.9	54.8	7.3
Age	99	44.8	14.3	46.0	23.0

8.7.2 Graphical Summaries

8.7.2.1 Continuous Data

Visual summaries are simple but often very important summaries of data information [8].

- **Histogram:** A histogram is a frequency plot, with the heights of the plot given by the relative frequency of data points found in various intervals of the horizontal axis. The pattern of the resulting frequency plot can be symmetric, skewed to one side, normally distributed, or something else. From the example in Chapter 2, for the baseline bodyfat variable in the cohort study, we have the histogram presented in Figure 8.1.

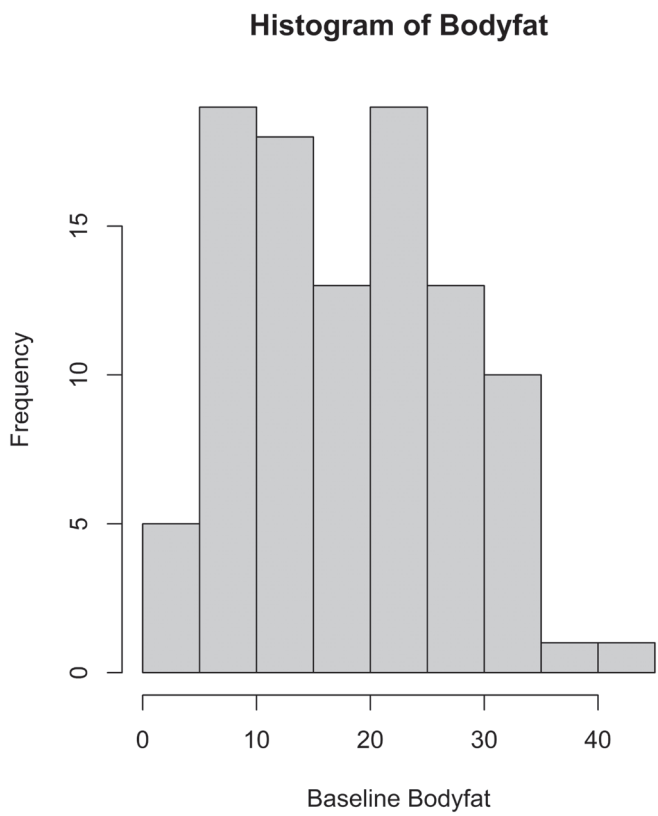


FIGURE 8.1
Histogram of Baseline Bodyfat.

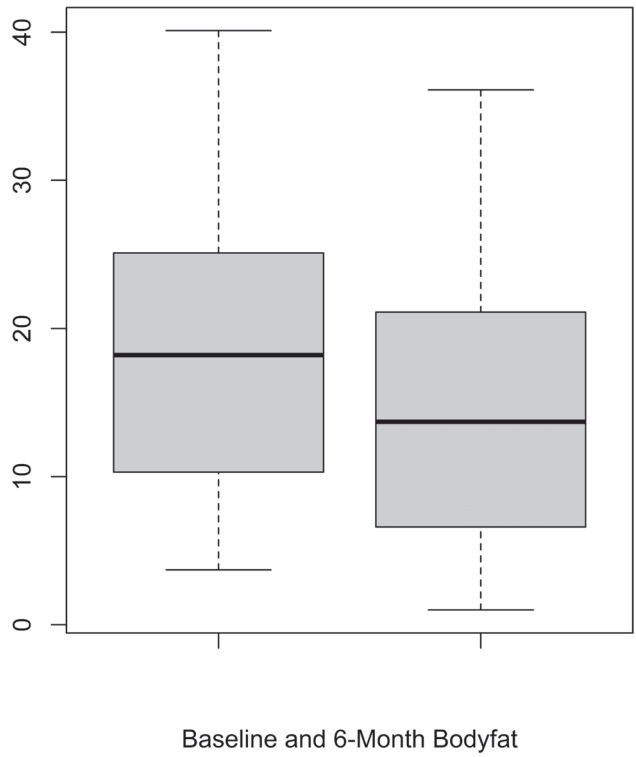


FIGURE 8.2
Boxplot of Baseline Bodyfat and Six-Month Follow-up.

- **Boxplot:** In a boxplot the length of the boxplot is given by the interquartile range (IQR) and the center line in the box is the median. Lines extending from the box often reach out to the minimum and maximum values. For the baseline bodyfat variable we have as boxplots for the baseline and six-month follow-up values (Figure 8.2).

Each of the graphics allow for interpretation of the observed data and can be examined for outliers, skewness and highly non-normal distributions. The graphics also give a visualization of the central location for the set of data values and the range of the data.

8.7.2.2 Categorical Data

Contingency tables can be useful to present summarized count data. For example, consider the case-control study given in the Chapter 2 dataset. A two by two table examining the association of treatment (cases/controls) with the presence of Gene A can be obtained. See Table 8.3.

TABLE 8.3

Cases and Controls versus Gene A (Yes/No)

	Gene A = no	Gene A = yes	Totals
Controls	26(59.1%)	18(40.9%)	44
Cases	15(50.0%)	15(50.0%)	30
Totals	41	33	74

TABLE 8.4

Mean(Standard Deviation) for Treatment by Gene A Levels.

	Gene A = no	Gene A = yes	All
Controls	0.86(1.52)	1.73(2.31)	1.14(1.83)
Cases	−4.28(1.96)	−4.60(1.53)	−4.41(1.78)
All	−1.07(3.02)	−1.19(3.76)	−1.11(3.28)

TABLE 8.5

Simulated Data for Simple Linear Model/Regression

y	3	5	7	9	12	13	11	15	17	18	21	22	23	22	21
x	1	2	2	3	4	5	4	6	7	5	6	8	9	7	8

Table 8.4 is used to express additional data, for example the mean and standard deviation for the baseline bodyfat variable for controls and cases versus Gene A (yes/no):

8.8 Relationships between Two Variables

8.8.1 Scatterplot and Fitting Straight Lines

To study the relationship between two continuous variables, begin with a scatterplot of the two variables of interest. Consider a simple example in Table 8.5.

The scatterplot of the data with fitted least squares regression line (y = 2.59 + 1.31x) is given in Figure 8.3.

Let's look at the basics of simple regression [9].

To fit straight line we are assuming the following model "fits" the data:

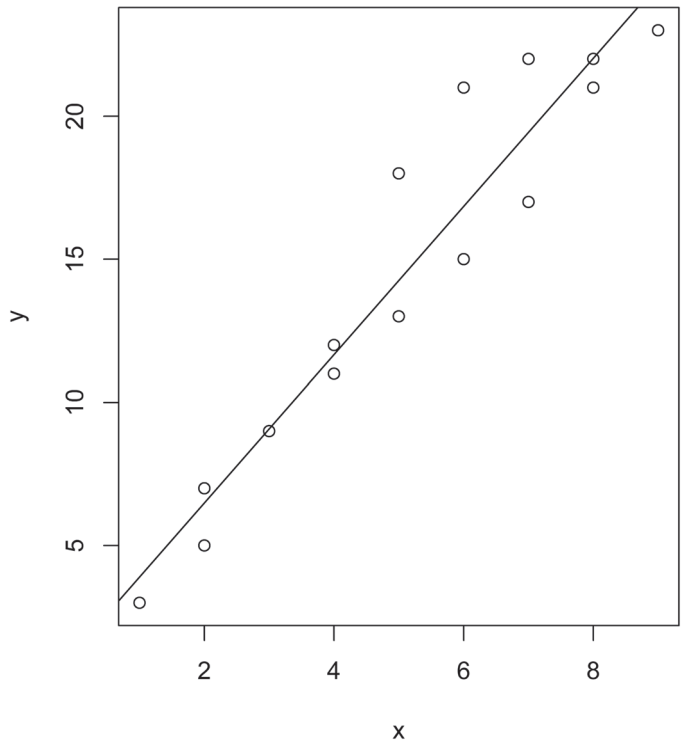


FIGURE 8.3
Plot of Data and Fitted Simple Linear Model.

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \tag{2}$$

for $i = 1, \ldots, n$. This assumes the basic pattern is a straight line plus random error ε_i.

- The method of least squares is typically used to "fit" the line to the data.
- The β_0 value is the intercept of the line and β_1 the slope of the line.
- Note that the slope β_1 is typically viewed as representing the relationship between y and x.
- For example, if the slope is zero, the line is flat, implying that as x increases, y does not change, that is there is no relationship between x and y. This gives us a hypothesis to test (is the slope = 0).

As the β values of the model are unknown here, they are estimated using the fit of the line to the observed data. Here the estimated intercept $= \hat{\beta}_0 = 2.59$ and the estimated slope $= \hat{\beta}_1 = 1.31$.

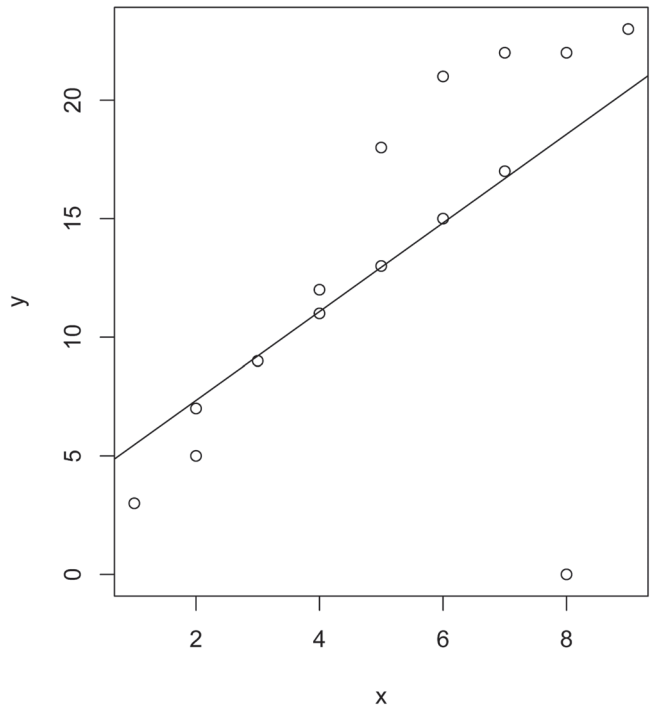


FIGURE 8.4
The Effects of An Outlier on Linear Model.

Consider a slight change to the data; let the value $y = 21$ in the original data be modified to the value $y = 0$. The fitted line is shown in Figure 8.3 and can be written $\hat{y} = 3.60 + 1.87x$. The percent of variation explained by the line, the R^2 value, decreases to 0.38 due to this single outlier. It is important to examine the data carefully for outliers or errors in data entry. This is shown in Figure 8.4.

8.9 Residuals

Once we fit the model, we can examine the accuracy of the model [10].

Write the fitted model as: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, where $\hat{\beta}_0$ is the estimated intercept and $\hat{\beta}_1$ the estimated slope of the fitted line. These are calculated using the least squares criteria. This gives a set of $\hat{y}_i$ values, called the predicted values of the model that lie on the regression line. These can be compared to the actual y_i values.

Let $e_i = y_i - \hat{y}_i$ be the errors or residuals of the observed data and model. These can be plotted to see if they support the assumption that the theoretical errors are normally distributed.

There are many ways to plot the residuals. A Q-Q plot is shown which is a plot of the percentiles of the observed data histogram versus the expected theoretical percentiles if we assume a normal distribution. The line of equality, a 45 degree line is shown. The histogram of the residuals is also shown. Both show some deviation from the assumed normal distribution, which is often the case in smaller samples. See Figure 8.5.

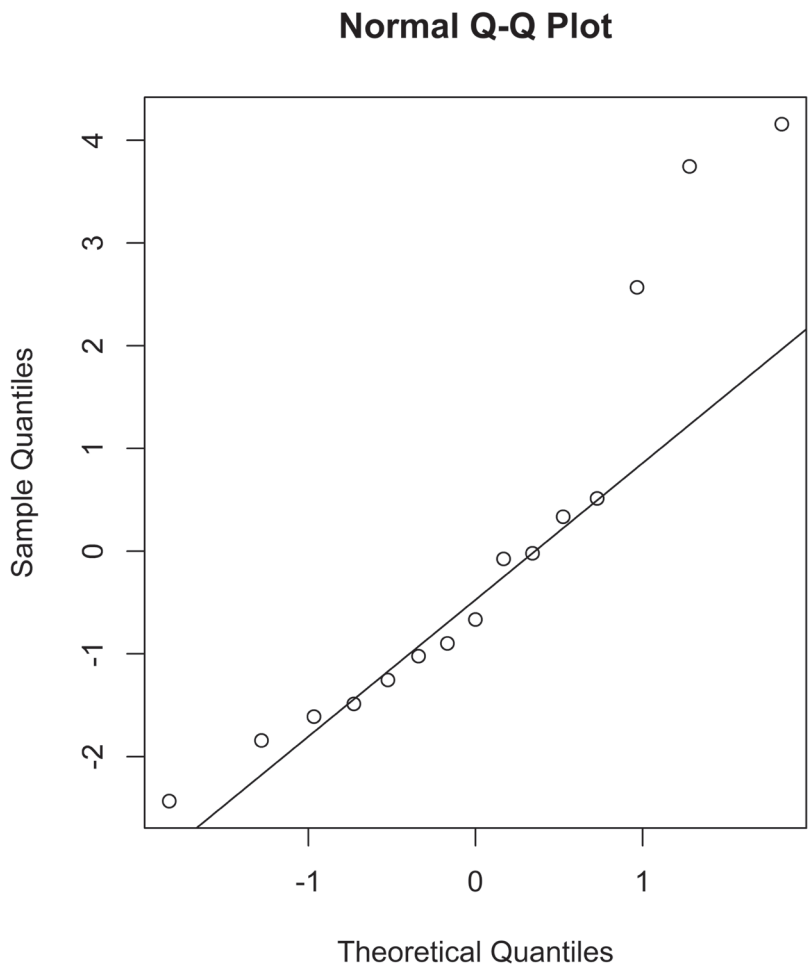


FIGURE 8.5
Residual Percentile Plot for Simple Linear Model.

8.9.1 Goodness of Fit

We can again also look at a single value that summarizes the overall accuracy of the linear model [1]. The R^2 statistic applies to all fitted linear models as a general measure of goodness-of-fit. It has the same general interpretation in a large linear model as in the simple regression model. In general:

$$R^2 = 1 - \frac{SSE/(n-p)}{SST/(n-1)} \tag{3}$$

where $SSE = \sum(y_i - \hat{y})^2$ and $SST = \sum(y_i - \bar{y})^2$.

The value of R^2 lies between 0 and 1. The closer to 1, the better the fit of the model to the data (the smaller the residuals or errors). Note that due to the geometry of least squares, there should be a constant in the linear model for R^2 to be meaningful.

As noted above, for the simple example considered here, the value was 0.91, which is close to the maximum possible value of 1.0. The line is close to the actual values and the slope of the line can be used to represent the relationship between y and x.

8.10 Correlation

In the regression setting above, y is assumed to be response variable and the x values are assumed to be fixed in advance.

If, however, both (y, x) are variable, a regression line may not be applicable. In such settings we can use the correlation statistic [11].

The correlation r between two continuous variables is given by:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}} \tag{4}$$

The correlation value can be interpreted. If it is zero, there is no relationship between y and x. This implies that on average, as x increase, y does not alter, on average. If the correlation is positive, the scatterplot of y versus x will trend upwards as x increases. If the correlation is negative, the scatterplot of y versus x will trend downwards as x increases. See Figure 8.6.

If we have more than 2 variables we can look at the correlation of each pair of variables. This is usually given in a correlation matrix. The set of bodyfat measurements in Chapter 2 give the following correlation matrix. See Table 8.6.

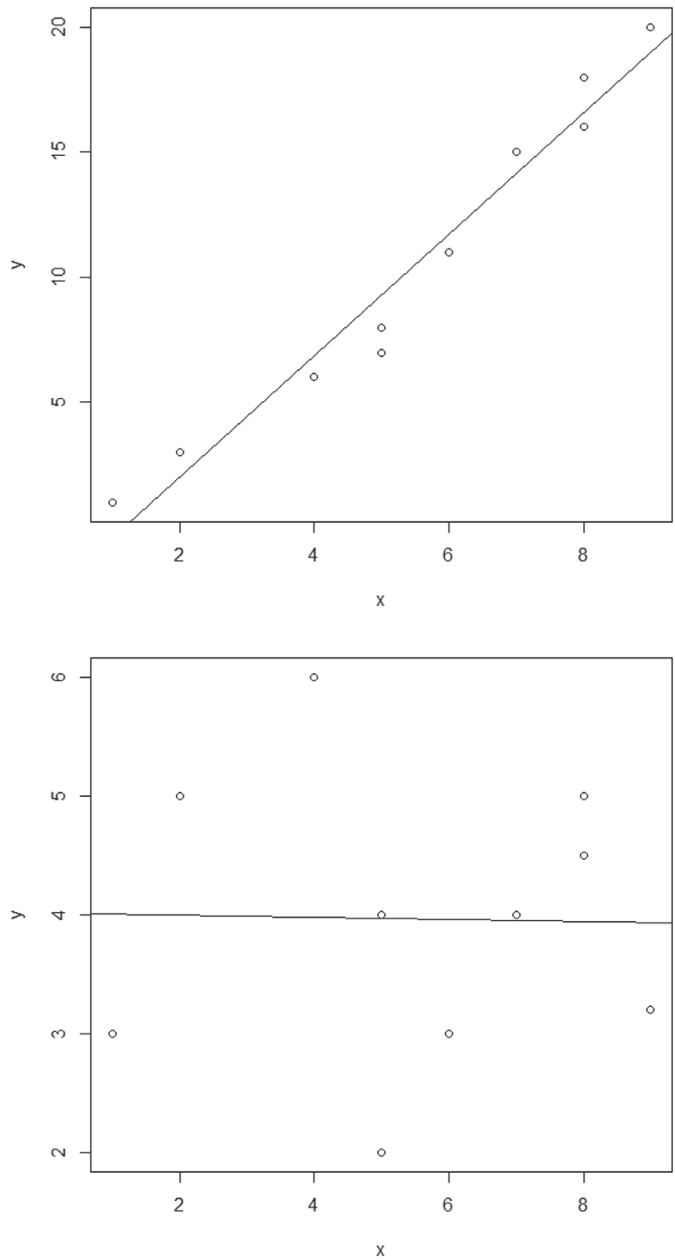


FIGURE 8.6
Interpreting Correlations.

Positive Correlation (r = 0.98)
Correlation = zero
Negative Correlation (r = −.98)

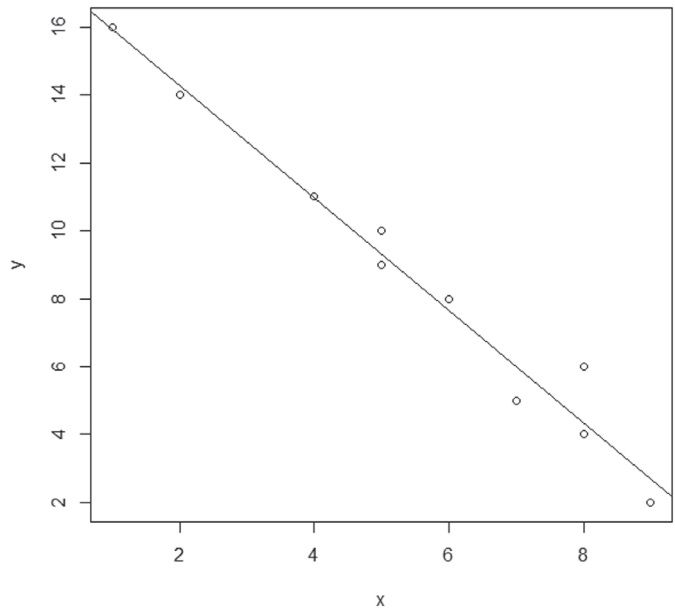


FIGURE 8.6
(Continued)

TABLE 8.6
Correlation Matrix for Chapter 2 Variables

	Bodyfat	Weight	Height	Neck	Chest	Abdomen	Hip
Weight	. 55						
Height	. 30	. 17					
Neck	. 45	. 84	. 20				
Chest	. 70	. 90	. 06	. 79			
Abdomen	. 89	. 87	-. 01	. 76	. 93		
Hip	. 58	. 94	. 01	. 76	. 85	. 88	
Thigh	. 55	. 89	-. 02	. 75	. 79	. 79	. 91

8.11 Adjusting for Secondary Variables

If the response of interest may be affected by the age of the subject, there may be a need to control for these secondary variables. This is often done by adding them to the original regression equation. For example:

$$y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i \tag{5}$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 Age_i + \varepsilon_i \tag{6}$$

This is called a multivariate linear model [9] and has the general form:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + \varepsilon_i \tag{7}$$

To use this extended model, note that there are some additional assumptions that should not be contradicted.

- The assumption of linearity of y versus each x_j. You can use graphics to examine this assumption.
- The errors ε_i in the model are often assumed to have a Normal distribution.
- The x_j variables should not be correlated with the ε_i random errors.
- The x_j variables should not be heavily correlated with each other.

This is further discussed in Chapter 13 examining the linear model.

8.12 Models and Data

When examining data that includes several variables, you will often need to assume a model to organize how you can interpret the relationships among the variables. The simplest model is the linear model. But not all relationships among variables are linear.

There are other models, for example nonlinear regression models, but unless you have a theory justifying such models, basic exploratory data analysis is often based on simple data summaries and basic linear models.

In medical research, nonlinear models do arise, reflecting underlying theory, especially in toxicology where the logistic curve is often used. Many other examples exist [12, 13].

We apply these concepts in the context of an asymptotic growth model applied to the BOD dataset found in Bates and Watts (1988). This displays curvature related properties and is given by:

$$y_i = \beta_1 \left(1 - \exp\left(-\beta_2 x_i\right)\right) + \varepsilon_i$$

The related dataset is given in Table 8.7.

A plot of the original data is shown in Figure 8.7.

TABLE 8.7
BOD Data

Demand	8.3	10.3	19	16	15.6	19.8
Time	1	2	3	4	5	7

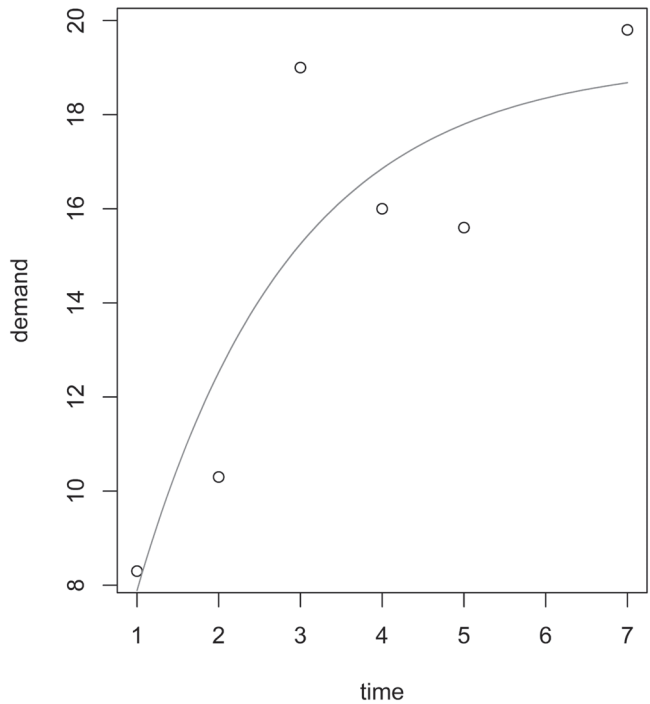


FIGURE 8.7
Plot of BOD Data and Fitted Model.

Note the presence of a threshold in the model, where the nonlinear aspect is obvious. This is typical of many toxicology related models.

8.13 Discussion Points

1. As part of planning your research study, it is important to estimate the sample sizes available for each variable of interest. In some settings the sample size for several variables may be limited and/or the data itself may be unstable, having outlying data values. For such

variables the use of percentile-based statistics, for example, medians, boxplots, IQR and non-parametric tests, may be necessary. Review the set of variables to be collected in your study and identify the set of variables requiring such statistics.

2. In the context of your study, list out the secondary or mediating variables you will need to include in data collection, for example, gender, age categories, urban/rural, insurance (yes/no). Describe how you will use these in the linear, logistic or survival analysis regression models you will be using in your study.
3. You can rescale a continuous variable into a categorical one, for example defining a variable having a value at or above 10 as "high" with less than 10 defined as "low." Sometimes the threshold is set equal to the median value for each variables ("median-split"). Examine the set of continuous variables in your study and how you might use such thresholds to create categorical versions of these variables for analysis.
4. If you are using a linear model, you should plot the response y versus each of the explanatory variables in the model, x_j to validate the assumption of linearity. Can you use the correlation value for each plot rather than the slope of the line as a way to evaluate the linearity of y versus each explanatory variable?
5. Residual plots are often available as part of the output when applying linear models. These plots, at a minimum, should reflect a random scatter of values and give an approximately normally distributed histogram. Do you have to examine residuals when you are in a setting where the sample size is large and the central limit theorem applies?

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9

Basic Probability Concepts

9.1 Introduction

Probability is pre-experimental in nature. It helps guide the planning of experiments. Using probability we can calculate an expected value for the experiment in question and measures of accuracy related to the expected values.

It is also the context within which we also interpret the data collected in our experiment or study.

It is interesting to note that probability is not technically based on the actual observed data. It is based on pre-experimental assumptions and probability distributions describing possible outcomes.

The basic definition for a probability function is straightforward [1].

(i) For a discrete variable, each probability must be greater than or equal to zero and all probabilities must sum to one.

$$P(X = x_i) \geq 0 \tag{1}$$

$$\sum_i P(X = x_i) = 1 \tag{2}$$

(ii) For a continuous variable, each defined interval must have a probability greater than or equal to zero and all probabilities must integrate to one (i.e. the area under the probability curve must sum to one).

$$f(x) \geq 0 \tag{3}$$

$$\int f(x)dx = 1 \tag{4}$$

9.2 Basic Sample Space Calculations

In general, probabilities are laid out for all possible outcomes. The set of all possible outcomes is called the sample space. Events are outcomes in the sample space and it is for these that probabilities are calculated.

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A simple example of a discrete (not continuous) sample space arises when we toss two coins, each marked H or T. The possible outcomes that comprise the sample space are: {(H, H), (H, T), (T, H), (T, T)}. If all of these events are equally likely, then each must have a probability of 1/4, so the probabilities sum to 1.

As an example of calculating the probability of an event, note that the event of seeing the same coin on both tosses is given by {(H, H), (T, T)} so the probability of seeing the same coin twice is 1/4 + 1/4 = 1/2. The probability of two heads alone {(H, H)} is 1/4.

A slightly more detailed example are the probabilities available when we toss two fair dice. Here there are 6 outcomes on each dice so 6 × 6 = 36 equally likely outcomes. These can be listed:

	1	**2**	**3**	**4**	**5**	**6**
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

(5)

Note that these outcomes are all equally likely because we have assumed the dice to be fair, with each possible outcome on each dice having a 1/6 chance of occurring.

If they are not fair, some events will be more likely than others. In real-world applications this type of effect may arise if our ability to consent patients is somehow limited and participation is not equally available to all possible participants.

We can also find the probability of various possible outcomes. Here, for example, we can find the probability of the event that the dice sum to seven: {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}. Each possible outcome has a 1/36 chance of occurring, so the probability of the event of summing to seven is 6/36, the number of elements in the event (6) divided by the total number of elements (36) in the sample space.

Similarly, P{Dice add up to 11 = {(6,5), (5,6)} = 2/36; P{Dice add up to two = {(1,1)} = 1/36.

9.3 Probability Formulas

For cases where we cannot list out the sample space, for example where the outcomes are continuous (blood pressures, incomes, age, volumes of lungs, Body-Mass-Index (BMI), weight, etc.) we usually use a formula for probability.

There are a small number of standard probability formulas that are useful in medical research [2]. They are applicable to specific situations. These are discussed in more detail below.

- Probability formulas are usually written with some unknown parameters θ or β in them.
- When used for statistical modeling and tests, these parameters are assumed to represent population characteristics (e.g., the population average blood pressure, the population odds ratio of getting sick once exposed to an infectious agent).
- It is these unknown parameters that we will both estimate and make assumptions about to test hypotheses.
- We usually are interested parameters representing typical population characteristics: means, standard deviations, proportions, odds ratios, medians, slopes of regression lines.

9.4 Discrete Distributions

Several standard probability distributions are listed here [2].

- The expectation $E[X]$ of the distribution is the middle of the probability distribution.
- Technically this is calculated using:

$$E[X] = \sum_i x_i P(X = x_i) \tag{6}$$

- in the discrete variable case and

$$E[X] = \int x f(x) dx \tag{7}$$

- in the continuous variable case.

- The variance of the distribution $Var[X]$ is the spread about the mean. This is calculated in a similar manner for discrete and continuous cases:

$$Var[X] = \sum_i (x_i - E[x])^2 P(X = x_i) \tag{8}$$

$$Var[X] = \int (x - E[x])^2 f(x) dx \tag{9}$$

9.4.1 Binomial Distribution

Assume there are n independent trials with two possible outcomes (e.g., success or failure), each trial having a probability p of success. Let $X = x$ be the total number of successes out of the n trials. The probability distribution for X can then be written for $0 < p < 1$ and $X = 0, 1, \ldots, n$, along with the mean $E[X]$ and variance $\mathrm{Var}[X]$:

$$f(x; p) = \binom{n}{x} p^x (1-p)^{n-x} \tag{10}$$

$$E[X] = np \tag{11}$$

$$Var[X] = np(1-p) \tag{12}$$

Figure 9.1 shows a typical Binomial distribution with $n = 20$ and $p = 0.3$.

9.4.2 Poisson Distribution

To apply the Poisson distribution, assume you will observe an event occurring within a given time frame (e.g., the number of people getting sick in the next month, or the number of people entering a hospital emergency room in the next 24 hours). Let X be the total number of events occurring, and λ be the average rate of occurrence within any 24 hour period. The probability distribution for X can be written for $\lambda > 0$ and $X = 1, 2, 3 \ldots$,

$$f(x; \lambda) = \lambda^x e^{-\lambda} / x! \tag{13}$$

$$E[X] = \lambda \tag{14}$$

$$Var[X] = \lambda \tag{15}$$

Figure 9.2 shows a typical Poisson distribution with $\lambda = 2$.

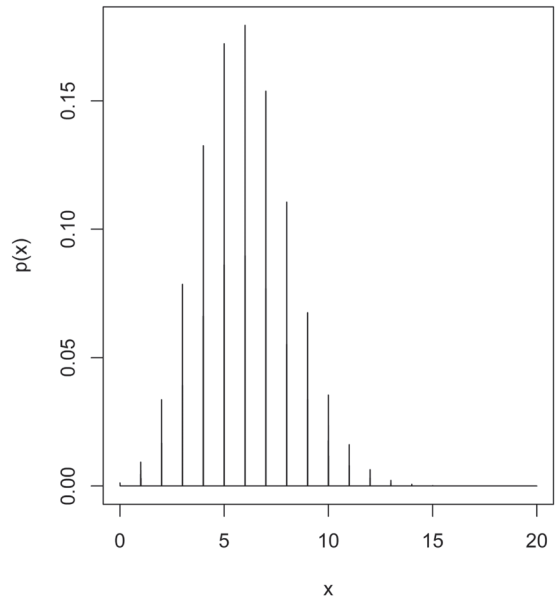


FIGURE 9.1
Binomial Distribution.

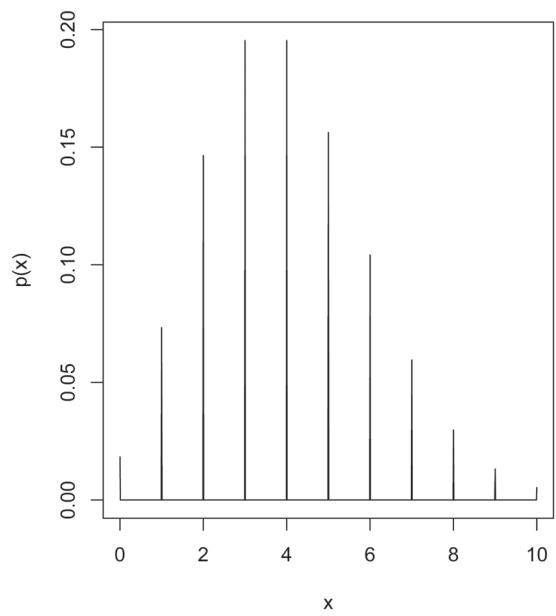


FIGURE 9.2
Poisson Distribution.

9.5 Continuous Distributions

9.5.1 Normal Distribution

The normal distribution for a continuous response $X \in (-\infty, \infty)$, with mean $\mu \in (-\infty, \infty)$ and variance $\sigma^2 > 0$ is denoted $N(\mu, \sigma^2)$ and can be written:

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} \tag{16}$$

$$E[X] = \mu \tag{17}$$

$$Var[X] = \sigma^2 \tag{18}$$

Note that standardized *Z*-scores are often used in medical research. This corresponds to using a $N(0,1)$ distribution. Typically if we subtract off μ and divide by σ for all data values, we say the data have been standardized. This often helps compare sets of data values measured in different units (e.g., in centimeters versus inches).

The standard normal distribution is given by:

$$f(z;0,1) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-z^2/2} \tag{19}$$

$$E[X] = 0 \tag{20}$$

$$Var[X] = 1 \tag{21}$$

Figure 9.3 shows a $N(0,1)$ density.

9.5.2 Chi-Square Distribution

The chi-square distribution with p degrees of freedom typically arises when we take p independent, $N(0,1)$ random variables, square each of them, and add them up. Many test statistics have this probability distribution. If a test statistic has a chi-square distribution, we say it is a chi-square test. There are many such hypothesis tests, for example tests of association and goodness-of-fit tests.

Assuming $x \geq 0, p = 1, 2, \ldots$

$$f(X;p) = \frac{1}{\Gamma(p/2)2^{p/2}} x^{(p/2)-1} e^{-x/2} \tag{22}$$

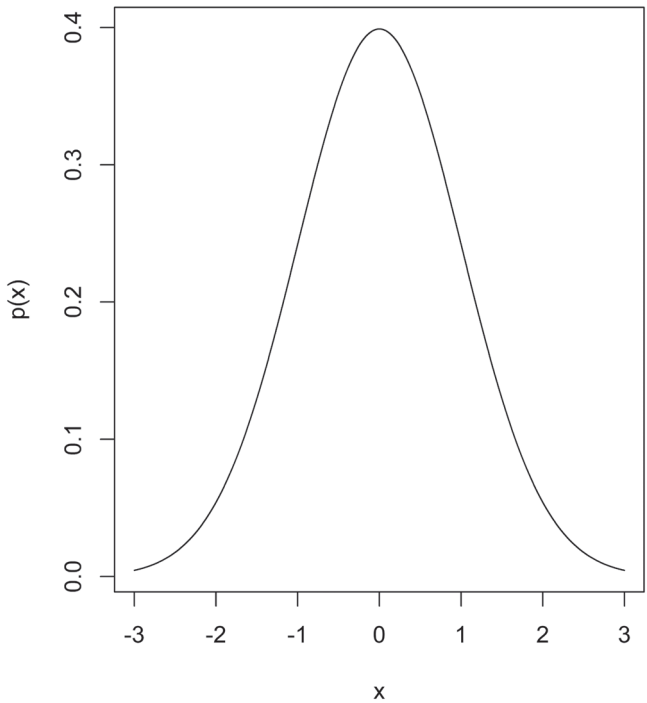


FIGURE 9.3
Standard Normal Distribution.

$$E[X] = p \tag{23}$$

$$Var[X] = 2p \tag{24}$$

Figure 9.4 shows a chi-square distribution with $p = 1$.

9.6 Some Practice Probability Calculations

- One hundred patients are to be randomly chosen from a large hospital and each will be assessed for exposure to the MRSA virus. Assume the chance of having the virus is p = 0.1 for each patient. Assume you can apply the binomial distribution with X = the number of patients with MRSA. What is the probability that more than 5 patients have MRSA? More than 10? What is the expected number of cases to be identified out of the 100 total? What is the variance here?

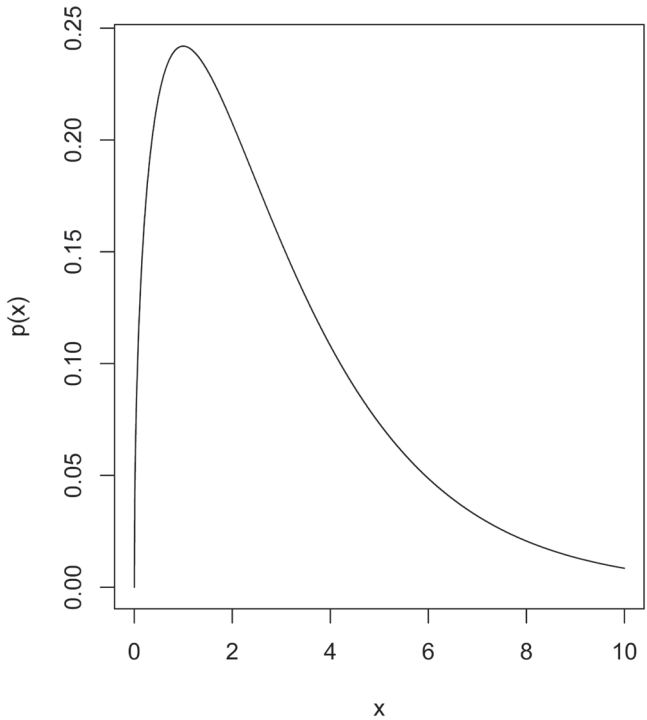


FIGURE 9.4
Chi-Square Distribution.

- An ecological study is conducted to examine the number of birds from species A that have a specific gene. Twenty birds are randomly obtained from a specific region and tested for the gene. It is assumed that the gene is fairly rare with probability p = 0.2. Assume you can apply the binomial distribution with X = the number of birds with the gene. What is the expected number of birds who have the gene? What is the expected number of birds who do not have the gene? What is the probability that more than 3 of the birds have the gene?
- Every year, over a specified area, the number of cases, X , of disease A is estimated. From past experience it is found that a Poisson distribution
- with $\lambda = 6$ can be used to estimate the probability of number of disease A cases occurring each year. For this upcoming year, find the expected number of cases of disease A, $E[X]$. Find the $P(X > 8)$. Find the $P(X \leq 2)$.
- You have set up a new emergency care facility. You would like to model the monthly visits (X) to the facility. To do so you assume a Poisson distribution for X and from past experience opening new

facilities assume that $\lambda = 6$. What is the $(X > 20)$? $P(X > 30)$? What is the variation of X, $Var[X]$.

- Assume a histogram of modified blood pressure values (X) for 1000 patients can be estimated using a $N(4,1)$ probability distribution. What is the probability $X > 5$? What is the probability that ≤ 2?
- Let a random measurement X have a $N(0,1)$ probability distribution. Remember that the normal distribution is symmetric about its mean. Show that $P(|X| > 1) = 2P(X > 1)$.
- Assume the distribution of a new diagnostic measure X follows a skewed distribution that can be approximated with a chi-square distribution with 5 degrees of freedom. What is the expected value and variance of X. Find the $P(X < 3)$.

9.7 Computer Based Resampling

The use of bootstrap resampling methods [3] have become commonplace in the application of statistical methods in much of science and medicine. The development of the bootstrap has provided a research tool for examining sampling distributions for a wide variety of statistical quantities. It has greatly increased the ability of applied and mathematical researchers to examine the statistical significance of many complex modeling structures [4, 5].

A caveat is that the approach is based on the observed data and all related statistical calculations are *de facto* conditional on the observed data.

This implies that difficulties with the original dataset, such as outliers, bias or non-representative sampling procedures, affect all resulting bootstrap related calculations.

Note that when using a mathematical model, it is through the observed residuals that the re-sampling aspect is usually generated, incorporating the fit of the model to the observed data. Other approaches to the resampling of datasets, especially in regression settings are possible [6].

9.8 Bootstrap

The basic bootstrap resampling procedure [3] is generally defined as:

1. Understand the basic study design, the data y to be collected, the model and hypothesis to be tested or population characteristic θ to be estimated.

2. Develop a test statistic $T(y)$, that can be used to estimate θ and can reflect any null hypothesis under consideration.
3. Define a related pivotal quantity $Q(y,\theta)$ that uses both θ and the statistic $T(y)$. Usually this is a standardized version of $T(y)$.
4. Collect the original sample data $y_1,\ldots,y_n$.
5. Generate a series of resamples, usually of the same sample size, by randomly selecting with replacement from the original sample data. This gives $y_1^*,\ldots,y_n^*$ which are statistically independent in terms of probability. Calculate the statistic of interest $T = T(y_1^*,\ldots,y_n^*)$. Repeat this procedure B times, typically with $1000 < B < 10{,}000$. This gives values $T_1,\ldots,T_B$ for the statistic T.
6. Obtain the empirical distribution (histogram of the $T_1,\ldots,T_B$ values) and related statistical properties of T. For example the bootstrap estimate of the population average is found by taking the average of the $T_1,\ldots,T_B$ values:

$$\hat{T}_{\text{Boot}} = \frac{1}{B}\sum_{j=1}^{B} T_j$$

7. The accuracy of the $\hat{T}_{\text{Boot}}$ estimator can be estimated by its standard error

$$S_B^2 = \frac{1}{B-1}\sum_{j=1}^{B}\left(T_j - \hat{T}_{Boot}\right)^2$$

8. Use the resampling distribution of $T(y)$ by itself to generate a 95% central confidence interval for θ.
9. If a p-value is required, assume a null hypothesis value for θ and generate the resampling distribution of $Q(y,\theta)$ values. Find the bootstrap p-value or tail area to the right of the original observed value for $T(y)$.

Note that when a model is available to guide the interpretation and analysis of the dataset, resampling is typically performed on the observed residual values. This respects the original fit of the observed data and the structure of the assumed model itself.

9.8.1 Example

For the study in Chapter 2, the mean baseline bodyfat decrease is resampled 1000 times from the 99 subjects in the cohort design study. The resulting bootstrap distribution for the sample mean is given in Figure 9.5. The central 95% interval is given by [3.7, 4.1].

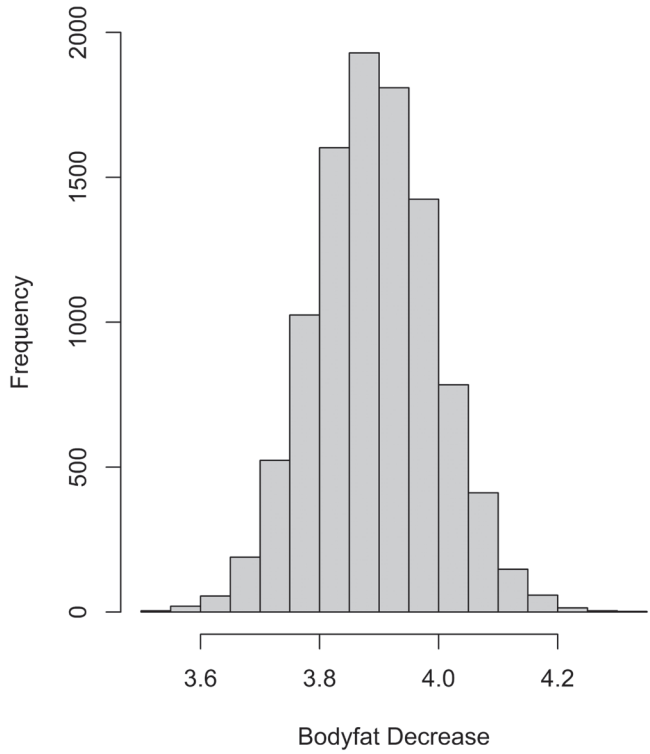


FIGURE 9.5
Bootstrap Distribution of the Sample Mean.

9.9 Pre-Existing Belief as a Source of Information

- If you have information in the form of expert opinion, there is a way to formally model this information and include it into a statistical analysis.
- This is called *Bayesian statistics*. A learning model [7].
- It takes your existing expert opinion (or belief) expressed as a probability distribution for θ (called the prior density) and updates it using the observed model-data combination the probability of the sample (seen as a function of and called the likelihood function). See Figure 9.6.
- The output of this is called a posterior distribution (your updated beliefs for the population characteristic of interest in the form of a probability distribution for θ, not the statistic T(x)).

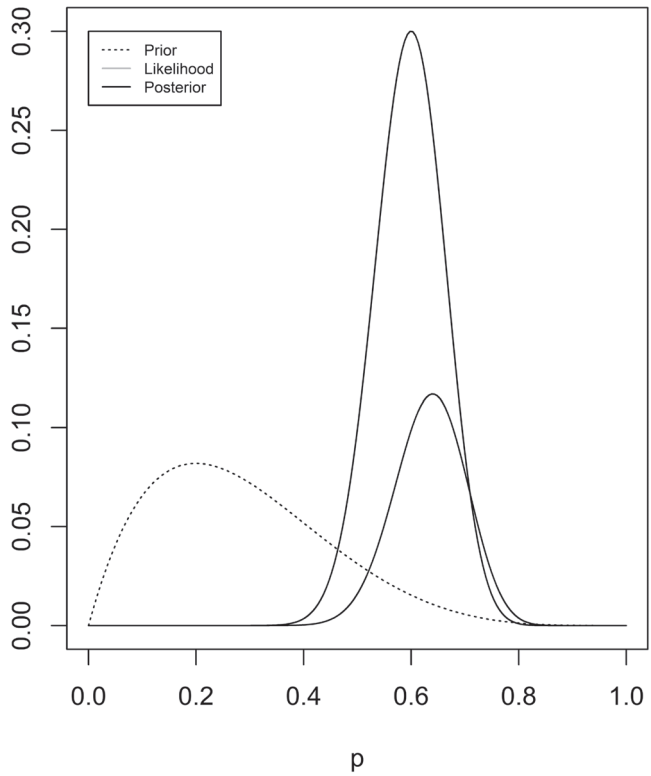


FIGURE 9.6
Prior Density, Likelihood and Posterior Density.

- The approach uses the same basic data based information as standard frequentist statistics, but from a different perspective.
- Here the "probability" scale is being used to describe your beliefs regarding the population characteristic θ of interest after you have seen some data.
- Available in statistical packages (R, STATA, other).
- This approach can typically be applied to standard modeling situations.

9.10 Summary

- Theoretical probability distributions provide expected values under various assumptions or hypotheses.

- They require assumptions.
- A set of standard distributions are available for many standard experimental or study designs.
- If we assume hypotheses regarding the model parameters are true, we can fit the resulting distributions to actual data and see if there are differences from what we expect. If the differences are too large, then the hypotheses are not supported by the observed data.
- We usually need to find the distributions of the statistics we are using in order to estimate or test hypotheses.
- The distributions of statistics called sampling distributions and are the context we use to test hypotheses within the framework of a probability model.
- *The distributions you will use will reflect the statistics necessary to carry out the hypothesis testing and estimation outlined in your study protocol.*

9.11 Discussion Points

1. Probability is important in trying to model sample variation in the context of your study design. The sampling distributions assumed provide a context for calculating expectations and deriving estimates. Discuss the distributions that seem relevant to your research study, focusing on the Poisson, binomial and Normal distributions. Are they useful, why or why not?
2. You have decided to toss three fair coins. What is the probability the three observed outcomes are NOT the same.
3. Discuss the potential use of Bayesian methods in your study. Find a publication in the literature for your area of research using Bayesian methods and summarize the findings and how they differ from a standard statistical analysis.
4. A new study for bird flu is going to sample 100 individuals who live in rural settings to estimate the incidence of the infection. Assuming the probability of having bird flu is 1/50, and the individuals sampled are
5. chosen independently, what is (i) the probability of finding more than 2 infected individuals, (ii) the expected number of infections. Use the binomial distribution.
6. If you have a smaller sample size, one option is to use the bootstrap to generate the sampling distributions you need for p-values and confidence intervals. Discuss the potential use of the bootstrap in

relation to your study design and planned sample size. Outline how you would do the relevant calculations. Is the bootstrap used in the literature for your area of research?

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10

Sampling Distributions

10.1 Introduction

Before we collect data for a study, we need to plan out how we will interpret useful information.

Typically, our goal is to say something intelligent regarding an unknown population mean value μ, for a population characteristic we are interested in, say average blood pressure or average T-cell count.

To do this, we use the following steps:

- Choose a useful statistic $T(x)$. For example, use the sample mean to estimate the population mean (μ).
- Work out the *sampling distribution* of $T(x)$. This can be easy or hard. These distributions are typically Normal, Student-t, chi-square, with some additional non-parametric distributions based on ranking the data and medians [1].
- Theoretical sampling distributions are the distributions used for statistical inference.
- Most of theoretical statistics is about working out these sampling distributions in different situations.
- Note that the randomization of the subjects used in the study design can be seen as a justification for the use of probability.
- We use these sampling distributions in two basic ways [2]:
  - Estimation of the parameter $\propto$ in the form of confidence intervals.
  - Testing of hypotheses for μ in the form of p-values.

10.2 Linking the Sample Mean and Population Mean

The sampling distribution for any statistic $T(x)$ is interpreted using the idea of repeated sampling. The experiment or study in question is imagined to be repeated, many times, and each time a value for $T(x)$ is obtained. This set of $T(x)$ values is then used to obtain a histogram or distribution that can

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be viewed as a probability distribution. It is this histogram that links $T(x)$ and μ.

Let us look at using the sample mean $\bar{x}$ to estimate and test hypotheses for the population mean μ.

- Fact: If each x_i is independent and is assumed to be drawn from a Normal distribution with mean μ and variance 1, then $\bar{x}$ has a sampling distribution that is Normally distributed with mean μ and variance $1/n$.
- This result is used to obtain confidence intervals to estimate μ, and p-values to test possible values for μ.
- Note that for this result to hold, we had to assume the underlying population distribution for the x_i values is Normally distributed.

10.3 Central Limit Theorem

A large sample shortcut exists for these calculations, in which we do not have to assume that we know the distribution for the population of possible values. It is called the central limit theorem [3].

Assumptions: Draw the samples independently from the same population using the same probability function $f(x)$ for all samples. Assume the distribution $f(x)$ has mean μ and variance $\sigma^2 > 0$.

The basic *Central Limit Theorem* for large samples uses the following argument:

- Imagine repeating the experiment or study many times.
- Each time, draw a sample and calculate the value of the statistic of interest $T(x)$ (e.g., $\bar{x}$).
- Now plot the histogram for these $T(x)$ values.
- The central limit theorem implies that this histogram will be a Normal distribution $N(\mu, \sigma^2/n)$ no matter the shape of $f(x)$.
- It is this histogram that links $T(x)$ and μ.
  - You can "see" this result by simulating data on a computer.
  - For this result to hold, the rule of thumb for the sample size is $n \geq 30$.
  - The Central Limit Theorem is over 350 years old.

- Note that this result is sensitive to sample size. If we select a sample that has too many subjects the variation σ^2 / n, which is a function of n, will collapse to zero, and the result is not very useful for determining p-values of confidence intervals. To avoid the effects of over-sampling, detailed statistical power calculations are often calculated, to choose the minimal level of n that will give use a result that is stable. See Figure 10.1 for a visual description of these effects.

10.4 Confidence Intervals for Estimation

To obtain a confidence interval for a parameter θ in general:

- Obtain the sampling distribution for the statistic $T(x)$. It will typically have the parameter of interest θ in its formula. Often $T(x)$ is chosen so that θ is the theoretical expected value for $T(x)$.

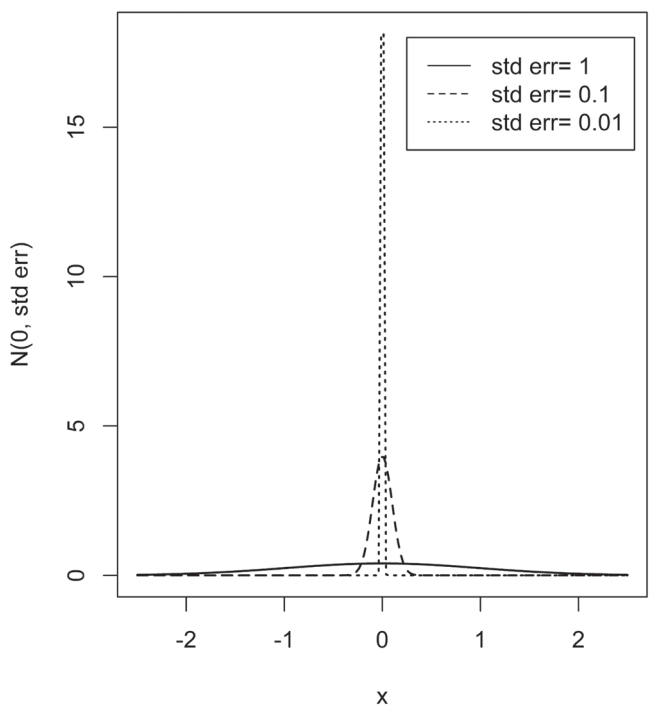


FIGURE 10.1
Normal Distribution for Different Levels of Variation.

- Typically $T(x)$ is standardized and its probability distribution worked out.
- Write down the probability statement for the middle 95% of the distribution. You will you get something like:

$$P[a < T(x) - \theta < b] = 0.95 \tag{1}$$

where a and b are constants.

- Now unscramble each inequality so you end up with:

$$P[L(x) < \theta < U(x)] = 0.95 \tag{2}$$

- $\left[L(x), U(x)\right]$ is called a 95% confidence interval for θ.

Note that the 95% confidence level formally has nothing to do with the actual data collected. Probability is pre-experimental. Usually, when we plug in the data, we assume the 95% number still applies. It is usually close and these types of intervals have been applied in medicine, science and industry for over 100 years.

10.4.1 Example: Confidence Interval for a Population Mean

- Assume you draw an unbiased, representative sample of $n = 36$ patients who have a given medical condition.
- An indication of the condition is a high level of chemical A.
- You measure the level of chemical A for each patient.
- Note that $n > 30$ so the Central Limit Theorem applies.
- In this setting the confidence interval for μ is given by:

$$\left[L(x) = \bar{x} - 2\left(s/\sqrt{n}\right), U(x) = \bar{x} + 2\left(s/\sqrt{n}\right)\right] \tag{3}$$

- Here the sample mean is $\bar{x} = 9.8$ and the sample standard deviation is $s = 3.2$. Standard error $= s/\sqrt{n} = 3.2/\sqrt{36} = 0.53$.
- The reported 95% confidence interval for the population mean is then given by: $\left[L(x), U(x)\right] = \left[9.8 - 2(0.53), 9.8 + 2(0.53)\right] = \left[8.74, 10.85\right]$.
- The population mean level of chemical A (μ) is estimated to lie between 8.74 and 10.85.

10.5 P-Values for Hypothesis Testing

The scientist R.A. Fisher developed the idea of tests of significance and the related p-value (he invented many statistical concepts, see [4]). He was the first to come up with a way of using probability to assess observed data outcomes. This has been further developed by many contributors.

Tests of significance: Use the sample information in the form of a standardized statistic $T(x)$ to test hypotheses regarding specific population characteristics. Assume the hypothesis is true and look for evidence to contradict this assumption [5].

P-value: Probability of being at (and further) than the observed value of $T(x)$.

- Basic Idea: The hypothesis of interest, the "null" hypothesis is assumed true (and usually represents the situation: the experiment DID NOT work, for example $\mu = 0$ or $\beta_1 = 0$). Now try to reject this hypothesis.
- If it is rejected, there is evidence the alternative is supported by the data (the experiment DID work).
- **Rule:** Reject the null hypothesis if the p-value is small. Usually this implies $p\text{–}value < 0.05$ or $p\text{–}value < 0.01$.
- We typically set up a test of significance in the same way: Assume your null hypothesis is true. Reject your null hypothesis if we observe a small p-value.
- Typically, the middle of a sampling distribution histogram will reflect the assumed null hypothesis and represents the EXPECTED value under the assumption the null hypothesis is true.
- Question: How far is the OBSERVED value of T(x) from this expected value?
- To measure this distance, we could count how many standard errors are between what we expect and what we observe. But we typically use the tail area to the right of the observed value for T(x). If this is very small, the observed value will be far from the expected value and we can reject the null hypothesis.
- **Question:** How small is small? 1/20 = 0.05 is the basic suggested level. In medicine a more conservative 1/100 = 0.01 is often used.

10.5.1 Example: T-Test for Population Mean

- For each practical situation, we need to decide on $T(x)$ and find its distribution.
- Assume we want to test a null hypotheses in regards to a specific population mean value μ, a one-sample " t-test."
- The standardized statistic (sometimes called the pivotal quantity) is usually taken to be:

$$T(X) = \frac{\bar{x} - \mu}{s / \sqrt{n}} \tag{4}$$

This has a Student-t sampling distribution with $(n-1)$ degrees of freedom. If the sample size is large $(n > 30)$ then it has an approximate $N(0,1)$ sampling distribution since the central limit theorem applies.

- **Example (cont'd):** Here $n = 36$ patients, the sample mean is 9.8 and the sample standard deviation is 3.2.
- The resulting standard error $s / \sqrt{n} = 0.53$.
- Test the hypothesis that $\mu = 7.0$.
- Here the observed value of the test statistic is given by:

$$T(x) = \frac{(9.8 - 7.0)}{0.53} = \frac{2.80}{0.53} = 5.28 \tag{5}$$

- Since the sample size is > 30, use the Central Limit Theorem approximation for calculations, so we do not have to specify the distribution for the values drawn from the population.
- The area under the $N(0,1)$ probability distribution at or beyond 5.28 is 0.00001. This is the p-value.
- **The p-value is small (**< 0.05**) so we reject the null hypothesis.**
- There is "evidence the population mean is not 7.0."
- Question: If the population mean is not 7.0, what value is it? You can report the 95% confidence interval [8.74, 10.85].

Some areas of research, for example physics, look primarily at the value 5.28 here. Since it is greater than 5.0 we reject the null hypothesis. The value 5.28 represents how many standard errors are between the observed sample mean (9.8) and the assumed population mean (7.0).

In physics the "5σ" distance is considered significant distance (it corresponds to a $\text{p} - \text{value} < 1.0x10^{-7}$). In medical research a cutoff closer to 2σ

(p-value < 0.05) is typically applied [6]. This is less strict due to the fact that in physics most experiments are carefully designed and controlled, while in medical research there is often less control when studying human subjects with varying family histories, diets, genetics etc. There is more variation to incorporate and model.

10.6 Some Questions

Q: Why do we use confidence intervals rather than single values to estimate a population characteristic?

A: To incorporate sampling variation.

Q: Is the 95% confidence interval really 95% all time?

A: No, it is an on-average concept.

Q: Do I need to know the underlying distribution for the measurement we are drawing from the population? For example, small sample, large sample.

A: If the sample size is > 30 you may be able to use the central limit theorem and avoid assumptions regarding the probability distribution for the variable of interest. In smaller samples, more assumptions are required.

Q: Are p-values (measures of statistical significance) always related to clinical significance?

A: The p-value is sensitive to standard error and this is sensitive to sample size. If the sample size is too large, then all p-values are small and lose their clinical meaning.

Q: How large does the sample size have to be for the CLT to apply?
A: For the sample mean of a single variable, $n = 30 \rightarrow 50$ is usually acceptable.

10.7 Count Data and the Chi-Square Test

The oldest statistical test is the chi-square test of association. It was developed by Karl Pearson, a population geneticist around 1901 [7].

Let's look at an example: 200 subjects are selected at random in a clinic, and they consent to have their hypertension and diabetes status examined. The following data is collected. See Table 10.1.

TABLE 10.1

Hypertension and Diabetes

	Diabetes Yes	Diabetes No	Total
Hypertension Yes	80	20	100
Hypertension No	20	80	100
Total	100	100	200

Q: Are these two variables associated?

- Re-phrase this in probability terms: *Q: Are the histograms along the rows very similar?*
- If they are, then Hypertension status (rows) does not affect the distribution of Diabetes status and these variables are not associated. The level of hypertension does not affect the probability distribution of Diabetes.
- If the distributions of Diabetes along each row are very different, then the level of hypertension does matter and hypertension is associated with diabetes.
- Note that *association does not necessarily imply causation.*
- To formally test this using a p-value, assume a null hypothesis of no association. This implies that the individuals in the study are ending up in each cell at random. So we expect 50 subjects in each cell (200/4).
- **Idea:** reject the hypothesis of no association if there is a pattern different from randomness (the histograms in each row differ).
- The formal probability model here is the multinomial distribution (4 cells, probabilities: p_1, p_2, p_3, p_4 with $n = 200$). Assume for the null hypothesis that all $p_i = 1/4$ (all cells are equally likely). So we expect 200 (1/4) = 50 in each cell. We measure the departure from this by the chi-square statistic.

The "Chi-square" test statistic is given by:

$$\sum_{i=1}^{4}\left[\left(\text{Observed}_i - \text{Expected}_i\right)^2 / \text{Expected}_i\right] = \sum_{i=1}^{4}\left[\left(\text{Observed}_i - np_i\right)^2 / np_i\right] \quad (7)$$

and this statistic has a chi-sq(1) sampling distribution. The observed value here is:

$$(80-50)^2/50 + (20-50)^2/50 + (80-50)^2/50 + (20-50)^2/50 = 72 \quad (8)$$

and the resulting p-value = P (Test Stat > 72) = 0.000001. Given the small p-value, we reject the null hypothesis of no association. This further implies that there is evidence of association in the current study.

NOTE: We never completely accept the outcome in testing hypotheses as we do not know what will happen in future samples. When reporting results, for example here, we might use the wording: "There is evidence to support a hypertension-diabetes association."

10.8 Discussion Points

1. List all tests you will be using in the data analysis for your research project and related specific aims and hypotheses. Also discuss the related sampling distributions. Is the sample large enough so that CLT will apply? What assumptions are required on the population based probability distribution for the response variable for the CLT to hold?
2. The CLT may help guide your sample size calculations. While power calculations are the context for sample size justification, the CLT can affect such power calculations and related assumptions [6].
3. Discuss the interpretation of a 95% confidence interval, both before the data is collected and after the data is collected, for example [8.7, 10.8].
4. The interpretation of the p-value is usually a required aspect of the study protocol. For example choosing a cutoff for significance of a p-value < 0.05 or < 0.01. Discuss the cutoff(s) you will be using in your research study.
5. Your study design and the specific hypotheses you will examine will guide the selection of your tests of hypotheses. List these out the hypotheses you will be testing and the respective statistical tests you will be using.

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11

Statistical Estimation and Hypothesis Testing

In this chapter, standard statistical methods relevant to observational and experimental designs are reviewed. Measuring differences between treatment and control groups in the presence of random variation is often the basic goal.

When we analyze sample data, we use probability concepts to link the observed sample results to the population level characteristics θ; for example, means, proportions, measures of association, odds ratios and other measures.

The basic model for most research settings is the linear model [1]. It provides guidance for the following situation: when there are many possible explanatory variables with which to explain a specific medical response, how do we identify which variables are most relevant? Some related standard models and testing approaches [2] discussed here include:

- Two-sample t-tests (comparing the mean difference between two samples; cases and controls)
- ANOVA (comparing mean differences between more than two groups)
- Non-parametric tests (t-tests and ANOVA tests based on ranked data for small sample cases or where the data does not follow a normal distribution. These tend to reflect median differences).
- Chi-square test of independence for count data and contingency tables, comparisons of proportions and rates.

For each setting, we look at p-values for the testing of hypotheses and related 95% confidence intervals for model parameter estimation.

Let us review some of the statistical models and related methods that form the basis of much standard statistical analysis.

All researchers using statistical methods should know them and view them as basic tools for understanding patterns in data, in the presence of random error.

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11.1 Proof via Contradiction

A key principle in any scientific testing of ideas if the use of indirect reasoning and proof [3]. This general approach to the testing of hypotheses is in accord with the way in which scientific inference has been taught for a very long time and corresponds to the philosophy underlying frequentist hypothesis testing. We prove nothing directly. Rather we assume the hypothesis we are testing is true and look to reject the hypothesis.

David Hume, Karl Popper and other philosophers of science have pointed out the need to prove results via contradiction, since we cannot say we have completely proved the hypothesis, as future samples are not yet known. We can say that we have seen evidence against the hypothesis based on the observed study or experiment.

Whether a single study should be the basis of such a decision is a question. These issues are the subject of a long debate both in the philosophy of science [4] and within statistics itself [5].

The Bayesian approach, which sees the information in the sample data and related model as providing an opportunity to learn and modify the importance given to the hypothesized population characteristic value, takes a different, more subjective approach to the definition of probability and related hypothesis testing [6].

11.2 Hypothesis Testing Using P-Values

We carry out statistical testing by looking at the pivotal quantity defined in Chapter 10 above carefully.

In a direct sense it measures the number of standard errors $\left(s/\sqrt{n}\right)$ that lie between the sample mean and hypothesized population mean value $\left(\overline{y}-\mu_0\right)$. If this difference is "large," typically 3 or higher, we reject the null hypothesis; that is it is not supported by the data observed in the context of the assumed model and data generating process.

To assess this difference using a probabilistic scale we look at the related p-value or tail area;

$$p-value = P(|T_1| > T_{obs})$$

and reject the null hypothesis if the p-value is small (typically less than 0.05) as this will correspond to a large difference in units of standard error. Note that here we are assuming an alternative of not equal, thus the use of absolute value.

Defining what is meant by a "small p-value" can be a challenge. RA Fisher, who developed the p-value concept, suggested using a value of one in 20 (0.05) as a useful cutoff point, but was flexible in this regard [2]. Note if the null hypothesis is actually correct and we simply observe a rare sample, then the p-value can be interpreted as an error rate.

11.3 One Sample T-Test: Testing a Single Mean Value

It is useful to note that test statistics typically reflect a specific underlying probability model. The (Student's) t-test allows for comparison of a single mean $\bar{y}$ to a specific hypothesized mean value μ.

The standardized statistic of interest that is the basis for statistical inference is given by:

$$\frac{(\bar{y}-\mu)}{s/\sqrt{n}}$$

This can be seen as corresponding to the following model: we will collect a sample of n individual measurements y_i independently from a common population, each with a probability distribution following a normal distribution with mean μ and unknown variance σ^2.

We write this as $y_i \sim N(\mu, \sigma^2)$ and look to test the hypothesis $H_0 : \mu = \mu_0$ where μ_0 is a fixed value (e.g., 0).

To do this, we assume the true value of the mean parameter is indeed μ_0 and look for evidence that this hypothesis is incorrect. This "proof by contradiction" approach to science has been applied for a long time and reflects the idea that we cannot predict what all samples will be in the future, so all we can say is that the assumed (null) hypothesis is not supported by the current data.

Typically, we choose a hypothesis $H_0 : \mu = \mu_0$ that represents "the experiment did not work." If we reject this hypothesis, then there is evidence the experiment did work.

To interpret the observed values, we can use the sampling theory result:

$$T_1 = \frac{(\bar{y}-\mu_0)}{s/\sqrt{n}} \sim t_{n-1}$$

where t_{n-1} denotes a Student-t distribution with $(n-1)$ degrees of freedom.

In larger samples $(n > 30)$, the t_{n-1} distribution can be approximated by the $N(0,1)$ distribution.

11.4 Pairwise T-Test

When the study design involves before and after measurements on the same individuals, the data is said to be paired. The "pre" measurements are often viewed as baseline reference or control values against which the "post" values are compared. The use of the same individuals provides a perfect comparison group; the same individuals twice.

Technically, we take the individual differences for each subject, (post – pre) and assume a null hypothesis of no difference (the mean difference = 0). We then apply a one-sample t-test to obtain the p-value.

To interpret the observed values, we can use the sampling theory result:

$$T_1 = \frac{\left(\overline{d} - 0_0\right)}{s/\sqrt{n}} \sim t_{n-1}$$

where $\overline{d}$ is the mean of the differences, n is the sample size, s is the standard deviation of the differences and t_{n-1} denotes a Student-t distribution with $(n-1)$ degrees of freedom.

11.4.1 Example

Consider assessing the mean change/decrease in body weight attained by subjects in the Chapter 2 cohort study. To look at the differences within subject, take the difference (follow-up value – baseline) for each subject, then test the null hypothesis that the mean difference is zero (the intervention did not work).

For the Chapter 2 cohort data, taking the n = 99 within subject differences in body weight gives a mean value of –5.84, with a standard deviation of 2.89. The p-value using the t-test with (99–1) degrees of freedom (well approximated by a N(0, 1) distribution) is < 0.001, assuming an alternative hypothesis of the mean difference is not equal to zero. This shows the reduction in weight due to the intervention is significant.

11.5 Two-Sample T-Test: Testing the Equality of Two Mean Values

To test differences between two independent samples, the t-test pivotal statistic is given by

$$T_2 = \frac{\left(\overline{y_1} - \overline{y_2}\right) - \left(\mu_1 - \mu_2\right)}{s_{pooled}}; s^2_{pooled} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

where s_1^2 and s_2^2 are the sample variances from each sample respectively. The statistic s_{pooled}^2 is a weighted average of the two sample variances.

This corresponds to a basic model; consider two samples of n_1 and n_2 individual measurements respectively and denote the respective observations y_{1i} and y_{2i}.

Assume these are collected independently, each respectively from a population with probability distribution $y_{ij} \sim N\left(\mu_i, \sigma^2\right)$, $i = 1, 2$ and $j = 1, ..., n_i$. We want to test the hypothesis of equality, $H_0 : \mu_1 - \mu_2 = 0$.

In this case, assuming the null hypothesis is true, the relevant sampling theory result for interpreting the observed values is;

$$T_2 = \frac{\left(\overline{y_1} - \overline{y_2}\right)}{s_{pooled}} \sim t_{n_1+n_2-2}$$

and the $p-value = P(|T_2| > T_{obs})$ which is the probability of the area at or beyond the observed value of T where an alternative of not equal is assumed. If the p-value < 0.05, we reject the null hypothesis; the data does not provide sufficient evidence to support the null hypothesis of equivalent mean values for the two populations being sampled.

11.5.1 Example

The mean difference/decrease in bodyfat values achieved by both cases and controls in the case-control study in Chapter 2 can be compared using a two-sample t-test. Letting the null hypothesis be no difference in the two populations means, we obtain:

Cases: sample size = 30, mean = −4.4, standard deviation = 1.8.

Controls: sample size = 44, mean = 1.1, standard deviation = 1.8.

This gives a p-value < 0.001 for the two-sample t-test. This implies there is evidence that mean of the cases, the intervention group, is significantly lower than the control group mean difference. Evidence that the intervention was successful. Related boxplots are given in Figure 11.1.

11.6 One-Way ANOVA: Testing the Equality of Several Means and Pairwise Differences

In the broader setting where there are more than two groups to be compared, R.A. Fisher developed a generalization of the t-test called the Analysis of

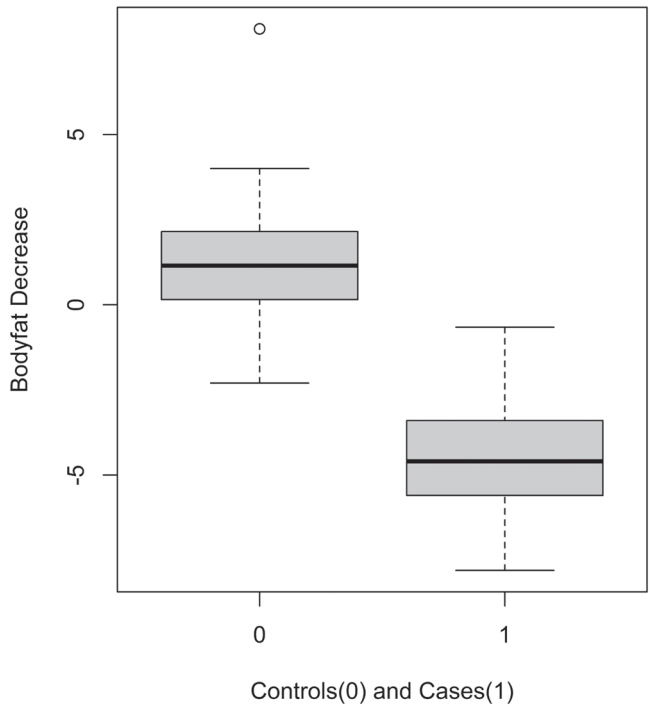


FIGURE 11.1
Boxplots of Bodyfat Differences for Cases and Controls.

Variance (ANOVA) [6]. In the case of basic 1-way ANOVA, we have k groups for one factor at several levels (e.g., dosage levels of a drug) and are testing the null hypothesis that there is no difference in the average response across the k dosage or group levels.

This is formally expressed as: $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$.

To test the null hypothesis the following statistical model is assumed;

$$y_{ij} = \alpha + \mu_j + \varepsilon_{ij} \sim N\left(0, \sigma^2\right)$$

Where y_{ij} is the response of the ith individual in the jth group, α is a baseline response level, μ_j the average response level due only to the jth treatment level and the ε_{ij} are random errors, assumed independent and uncorrelated with the levels of the factor in question. The variation in the problem is assumed to be similar at all k levels and modeled by a single parameter σ^2.

The pivotal quantity used here is typically the F-statistic, named for R.A. Fisher and given by;

$$F = \frac{\sum(\bar{y}_j - \bar{y})^2 / (k-1)}{\sum(y_{ij} - \bar{y}_j)^2 / (n-k)} \sim F_{k-1,n-k}$$

where $\bar{y}$ is the overall average and $\bar{y}_j$ is the average in the jth group and $n = \sum n_j$. This statistic has a $F_{k-1,n-k}$ distribution when the null hypothesis is true. The p-value is given by $P(F_{k-1,n-k} > F_{obs})$. A small p-value implies rejection of the null hypothesis.

Note that for both the t-test and F-test, the assumption that the errors follow a normal probability distribution is necessary if the sample is small. If the sample is large and the central limit theorem applies, the assumption of normal errors is not required.

ANOVA methods can be extended to include many factors, each with several levels and potential interactions between the levels. These extensions go under the name Design of Experiments.

This is further discussed in Chapter 14, examining linear multivariate models.

11.6.1 Multiple Comparisons

If we have many variables of interest and are going to test for differences between cases and controls for each variable, we have to be careful how to interpret the resulting set of p-values.

The set of hypotheses are sometimes seen as a family of hypotheses and the overall Type I error is set at 0.05.

To achieve this for a set of independent tests we need to adjust the Type I error (p-value cutoff) for each individual test, making them each smaller than 0.05.

For example, in the setting of a 1-way ANOVA example with 3 variables, there are 3 possible pairwise comparisons (1,2), (1,3), (2,3) and we use $0.05/3$ as the significance level for each two-sample t-test [7].

11.7 Chi-Square Tests

- The chi-square distribution is a common sampling distribution in statistical applications. If a test statistic has a chi-square sampling distribution, it is a called a chi-square test.
- See Chapter 11 for discussion of the chi-square test of association (independence) [8].
- There are also chi-square tests to compare two proportions.
- There are chi-square tests to compare the fit of a theoretical distribution to the data histogram, so-called goodness-of-fit tests.

11.7.1 Example

As shown in Chapter 13, in regard to the case-control study, the chi-square test of association of case status with Gene A status (yes/no) gives a p-value of 0.45. The null hypothesis of no association cannot be rejected and there is no evidence of an association between case status and the presence of Gene A.

11.8 Non-Parametrics: T-Tests and 1-Way ANOVA with Non-Normal Data

Let us assume the data given in Chapter 2 for the case-control study is not assumed to follow a normal distribution. It is possible to test various null hypotheses, usually based on median values and median value comparisons. This involves using a set of tests called non-parametric tests and these can replace the standard t-tests.

These tests can be run next to the standard tests above. They should give similar answers, though if there are serious outliers, they may differ. As noted in an earlier chapter, means are very sensitive to outliers, while medians are more robust.

The *Wilcoxon signed-rank test* [9] can be used to test of pairwise differences are significantly different from zero. This can be applied to the comparison of before and after weight measurements in the cohort study from Chapter 2. Taking the change/decrease for each individual as the data of interest, we obtain:

The median decrease = 5.0, range = 9.4, Wilcoxon p-value < 0.001. This implies that there is evidence the intervention significantly lowered bodyfat. Note that this agrees with the pairwise t-test given above.

The *Mann-Whitney test* [9] can be used to compare two groups, for example the cases and controls in the case-control study. The data for change in bodyfat is given by:

Cases: sample size = 44, median change = 1.15, inter-quartile range = 2.2.

Controls: sample size = 30, median change = −4.6, inter-quartile range = 2.35.

This gives a p-value < 0.001 for the Mann-Whitney non-parametric test. This implies there is evidence that mean of the cases, the intervention group, is significantly lower than the control group mean difference and evidence that the intervention was successful.

The *Kruskal Wallis test* [9] can be used in place of the 1-way ANOVA F-test.

The *Fisher exact test* [9], can be used in place of the chi-square test. As shown in Chapter 13, in regard to the case-control study, the Fisher exact test examining the association of case status with Gene A status (yes/no) gives a p-value of 0.62. The null hypothesis of no association cannot be rejected and there is evidence of no association between case status and the presence of Gene A. This agrees with the chi-square test above.

11.9 Additional Issues

11.9.1 Missing Data

If some of the data points in your database are missing, they may create patterns that affect the accuracy of your study design and the degree to which your data is representative of the population in question. It may also create a bias affecting the observed value of the statistics. If missing at random, these effects are typically minimal. However, if they are not missing at random, the pattern they reflect may be a cause of bias and a non-representative sample [10]. There are ways to test to see if the missingness is occurring at random.

Consider the following sets of data. The first is the complete data, the second is the same data with missing components.

1	7	20	15	16	18
3	4	3	7	9	2
11	13	15	8	10	2
1	8	3	21	30	42
7	8	15	22	23	31
8	9	7	11	13	14

1	7	-	15	-	-
3	4	3	-	9	2
11	-	15	8	10	2
1	8	3	-	-	-
7	8	15	-	-	-
8	9	7	11	13	14

The respective means and standard deviations are given in Table 11.1:

TABLE 11.1

Missing Data Statistics Comparison

	Mean	Standard Deviation	Median	IQR	Min	Max
Complete	12.1	9.2	9.5	8.8	1.0	42.0
Missing	8.0	4.8	8.0	9.0	1.0	16.0

A simple boxplot shows the difference in patterns, with a threshold limit at 15.0 limiting the values in the missing data component. This is a non-random condition, and it would be necessary to adjust the data analysis of the missing data component to reflect this condition. See Figure 11.2.

In complicated datasets and models, detecting patterns in missing data can be much more challenging.

11.9.2 Clinical Versus Statistical Significance

A practical consideration when using p-values to test hypotheses, is to note that probability calculations for many statistics are sensitive to sample size. The standard error in particular shrinks toward zero as the sample size gets large. This affects the p-value in that even small clinical differences will be identified as statistically significant.

Thus, statistical significance is not necessarily reflective of clinical significance.

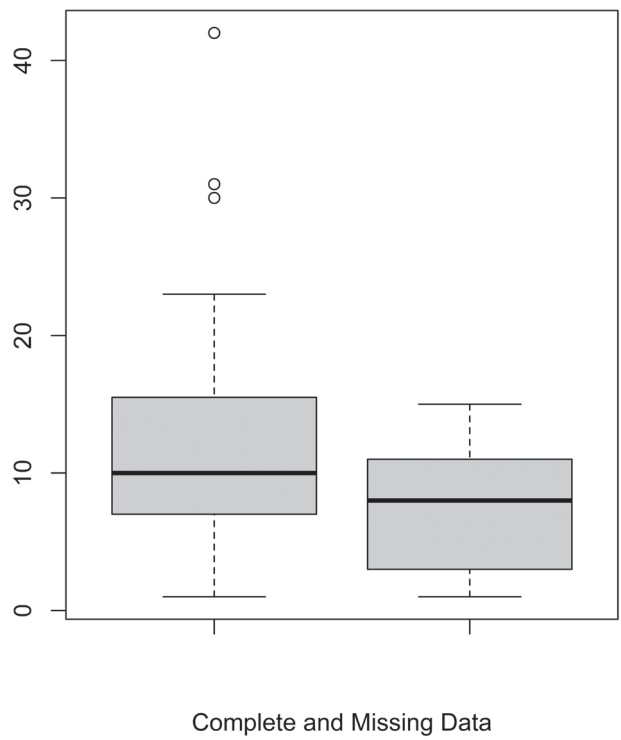


FIGURE 11.2
Dotplot of the Effect of Missing Data.

Note that having very large samples drawn from a homogeneous population is not necessarily a good thing. It is usually much more useful to collect moderate sized samples, carefully drawn at random.

Note that in very large sample sizes estimation may be a more useful strategy then using p-values. See for example [2].

11.10 Introduction to Sample Size Selection and Power Calculations

Power calculations guide the selection of the smallest useful sample that will support detection of meaningful clinical changes between treatment groups (so-called "effect sizes") [11].

The elements of the calculation include: Type I error (this is the cutoff you will use to interpret the p-value. This is usually 0.05), the sample size, the chosen test statistic and respective sampling distribution, the "effect size" (the clinically meaningful difference you are testing for, usually a 20% difference from controls), the null hypothesis of interest (usually "no change") and a measure of variation (usually the standard deviation of the sample distribution).

This type of calculation is usually required in formal grant applications and scientific protocols, often under the title "Sample Size Justification." Note that it also depends on the chosen study design, statistical model and null hypothesis.

The power calculation or sample size justification uses the idea of Type II error and related power; basically, the probability of rejecting correctly. Note that this does NOT imply the p-value is guaranteed to be less than 0.05 if a statistical test has high statistical power. It supports the idea that if the p-values is less than 0.05, you can have more faith in the significant result.

Rule of thumb: Choose your sample size to achieve 80% power for a 20% effect size (difference).

An Example: You are planning to study a new drug that raises T-cell count. There will be a treatment group and a control group. Same sample size for both. What sample size is justified here?

A review of the literature shows that we expect control group mean to be 15 with a standard error of 2. What difference in mean values will be viewed as a clinically useful effect size? Let us take a 20% increase (3 units) and use a two-sample t-test with Type I error = 0.05, standard deviation = 2, effect size = 3. The sample size in each group that gives power = 0.88 is n = 10. If we assume some people will dropout (say 20%) we will plan to use 12 subjects in each group.

Note that if standard deviation is assumed to be 4 then a sample size of 29 in each group gives a power of 80%. This demonstrates that as the variation increases it is more difficult to detect meaningful differences and requires more information (larger sample size). The power curve for this example is shown in Figure 11.3.

11.10.1 Challenges

Statistical power is sensitive to clinical effect size, Type I error, sample to sample variation (measured by the standard deviation) and sample size.

Sample size justification is a part of the planning process of a study. However, actual data collection can be difficult and attaining the planned sample size can be challenging.

Statistical power calculations require assumptions, for example the relevant standard deviation, usually drawn from the literature.

The effect size, the difference that is clinically meaningful, is typically set as a 20% difference from the control mean or proportion. At this level the respective power level should be at least 80%.

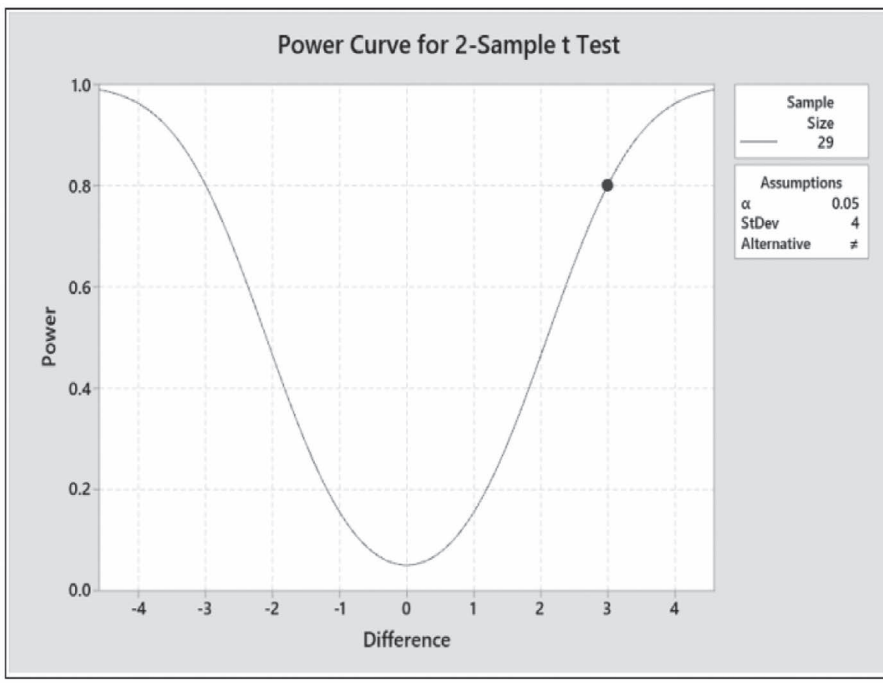


FIGURE 11.3
Power Curve Plot.

These calculations can be complicated in larger linear models, logistic regression, survival analysis (time to event analysis), epidemiologic models, longitudinal (follow-up) models.

11.11 Discussion Points

- List all the standard tests and related hypotheses you will need in your research project.
- Discuss when you would replace a standard test with its non-parametric version.
- Discuss why we use proof via contradiction. Give an example of a null hypothesis when comparing two average values, how you would reject the null hypothesis and what rejection of the null hypothesis implies.
- Power calculations require specific information. For your research project, focus on the primary aim and related hypothesis. (i) what is the assumed effect size, (ii) Type I error, (iii) assumed standard deviation (iv) specific test to be used.
- Assume you have a set of 10 variables and you will be comparing the set of average values for each pair of variables. You may have to adjust the cutoff for the p-values using a Bonferroni adjustment. If you compare 5 pairs of variables, what is the appropriate modified cutoff value for the set of p-values?

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12

Using Electronic Databases in Research

12.1 Overview

Accessing datasets drawn from an existing electronic health database (EHR) is a central aspect of data collection in many medical and other research contexts. It is often a primary resource when carrying out retrospective studies.

Such research data is usually collected in the context of a very large database, maintained by the medical, government or healthcare entity supporting the research project, or allowing access to the database [1].

An important aspect of using such databases is to realize that not all entered data is guaranteed to be accurate. Being able to research how and when the data was entered is essential when first reviewing the database as a potential research source.

Accessing the entire database is often not practical. As such, the specific time period and sets of variables to be studied should be carefully defined, often in the form of keyword searches to develop your research database, focused on your goals and hypotheses of interest.

Once your basic research database is developed, subsets of the patient data can usually be accessed using simple software. For example, randomly chosen treatment and control groups and other demographic sub-groups can be defined and compared.

The issue of missing data in large databases is often a challenge. These can arise for a wide variety of reasons, including entry error and individuals dropping out of the database, and can be a cause of bias in the actual data being analyzed, with the actual data not being representative of the overall population of interest, or even the overall database demographics.

There exist methods to randomly impute data, for example multiple imputation [2], to generate more complete statistical summaries and related analyses when missing data are present. Typically, a missing data percentage of 20%-30% is a maximum level for which accurate multiple imputation should be applied.

Figure 12.1 shows the process of obtaining the research database for analysis.

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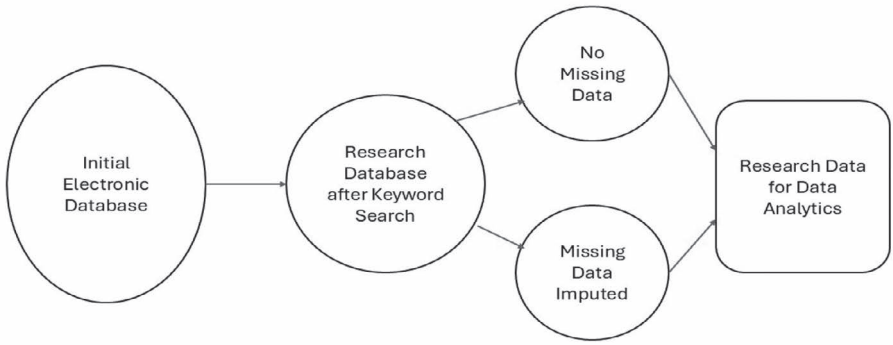


FIGURE 12.1
Developing an EHR-Based Research Database.

12.2 Clinical and Hospital Databases

When accessing patient databases in a healthcare environment, it is often necessary to follow the rules of accessing such data. HIPAA [3], for example, is a law that often applies to electronic health records (EHRs) and healthcare data in the United States when that data is accessed in patient settings such as large clinical databases or hospital databases. Personal health information (PHI) is often protected in the setting of human subject research, unless the database is de-identified, meaning that patient names, addresses, and other identifying information have been removed from the data. If accessed, patients typically must be informed when and how the data was accessed and for what purpose.

Accessing such data often requires taking HIPAA-related training courses [4], writing a formal protocol, obtaining permission to access the data, and having the protocol reviewed by the IRB of the institution within which you are carrying out your research project.

While the data can be accessed more easily if the selected data is de-identified, in smaller rural catchment areas or in the case of rare diseases, a stricter criteria may have to be met to preserve patient privacy.

The rules of HIPAA, including discussion of when the law applies, are further discussed elsewhere [5]. This is also further summarized and discussed in later chapters of this book.

12.3 Data Extraction

Each database that has been developed and curated will typically have its own set of procedures relevant to obtaining the data. Entire years of data may be available for download as an Excel file or other type of file.

These types of files will typically contain a very large number of variables, far more than is required. You will have to sort through the entire set to find your variable of interest, often using a data dictionary which describes each variable, what it measures and if it was re-scaled or standardized.

In some settings, for example the EPIC hospital utilization database [6], software will be available that allows for detailed keyword searches using packages such as the "Slicer-Dicer" package [7] to retrieve a more narrowly defined dataset relevant to your research topic. Such packages often require careful definition of the time-frame desired, the specific keywords or variables of interest, specific age groups, etc.

12.4 Electronic Health Record (EHR) Data

Electronic health records allow researchers to access patient data relevant to medical research. These databases often provide low-cost access to complex longitudinal data on large populations for medical research. EHRs can in certain settings be linked to contextual data using geographic information systems (GIS) and census data [8]. Subject to participant consent, they can also be combined with self-reported data, usually in the form of patient survey data, to address questions about related issues such as access to healthcare and the quality of care received. The combining of EHR quantitative data with qualitative survey based data is the basis of an integrated approach to the analysis of the delivery of many medical procedures and programs.

12.5 EHR Data Content

EHR data is mostly collected, initially, in the context of a clinical encounter. Rather than being driven by research goals, data are collected in relation to patient health status, administrative needs and may reflect variation in physician care practices and documentation procedures. EHRs used by different health systems may vary in the number of data domains (e.g., vital signs, laboratory data) they collect and how these domains are defined.

Physicians typically use diagnostic codes (ICD-10 for example) to report a patient's condition, document orders (i.e., medications, laboratory tests, imaging), and justify levels of service and related medical billing. Image and laboratory order codes may indicate the condition physician suspects is present (e.g., high cholesterol) or is monitoring (e.g., high blood glucose levels).

Interestingly, a single appearance of a diagnostic code does not necessarily indicate that the patient has the disease. This is due to the lack of accuracy in many diagnostic tests. For example, the positive predictive value of a diagnostic test for a disease may only be 75% in some tests. It may require additional information, for example, evaluation by a specialist, to validate the diagnosis.

The accuracy of case definition, especially for serious illness, for example diabetes or myocardial infarction, can be improved by examining whether the reported ICD-9 codes for the patient in question are stable over time. This involves careful study of the database and related data dictionary by the researchers.

Medication orders and related dosages and the scheduling of return visits, apart from proving useful information, may also represent a degree of physician concern. A challenge here is that providers may not use diagnostic codes consistently, and the meaning of any given code can vary among providers and across time, giving a level of variation that complicates analysis of patient data.

EHR research is often based on de-identified databases whose patient populations receive most or all of their care within a specific health system and thus have their data recorded within the single database. Researchers have employed EHRs for studies in, for example, health services research, disease surveillance, pharmacoepidemiology and genetic epidemiology and many other areas. See for example EPIC Cosmos applications [9].

Studies using EHR-based data include case series, nested case-control studies, and more broadly, prospective and retrospective patient cohorts. Researchers can use such databases to identify cases and also assess eligibility of cases or controls for individual-level or frequency matching in case-control and nested case-control studies. EHR databases are very useful for identifying controls for future prospective study.

EHRs represent databases in which patients may enter or leave care at any time. In studies examining, for example, time-related responses (e.g., time to cancer remission), a patient can only contribute person-time to the database record if they are under observation and are at risk for the outcome or disease of interest, with the patient's data entering the database if they have an encounter with a physician in the system.

Missing data in large EHR or hospital utilization databases are a challenge. Researchers may find it difficult to interpret gaps in care and time in the EHR data. When a patient lacks data, one cannot distinguish between patients who have left care, who have been well and have not sought care, or who have missed routine visits for other reasons, including being very sick, or have begun to use other providers outside the healthcare network. These issues will affect calculations related to person-time documentation and related exposure to disease, which is required, for example, for the estimation of accurate incidence rates.

12.6 Privacy and Consent

The data that is actually available for download from an EHR database may be affected by issues related to patient consent. For example, many health systems require that patients opt out of data-sharing if they do not want their EHR data used for research purposes. Some institutions require a formal opt-in patient consent, typically a signed form, which protects patient privacy, but requires more time to obtain. This may lead to selection bias in the resulting research database [10].

Another issue is the maintenance of patient-provider confidentiality. This is especially relevant when researchers merge datasets from multiple health systems, and it requires that reliable de-identification and security methods are in place for all included sources of data. As providers record PHI in many different formats, identifying information may persist in free text, for example doctor notes, despite efforts at de-identification.

Finally, although not unique to EHRs, electronic data storage may lend itself to data breaches such as laptop theft or the accidental emailing of data. However, EHR data can typically be encrypted and require role-based access and several level of authentication, as per HIPAA rules. Using EHR databases usually results in fewer opportunities for privacy breaches compared to manual chart review or other traditional data collection methods [11]. It is always the responsibility of researchers, clinicians, and policy makers to balance the need for representative samples with the protection of patient data confidentiality.

Note that EHR primary data collection can sometimes be used to inadvertently generate data indirectly on family members. Usually, due to confidentiality concerns, family members' data are not directly linked. However, matching algorithms based on names, dates and other information such as genetic expression data may be used to link, for example, mothers to their children.

For more discussion of PHI-related issues, see Chapter 6.

12.7 Time Aspect

Electronic databases may carry retrospective data from the last five, ten, 20 years or longer. This allows for longer-term study of trends in access to treatment, success of treatment, types of payment, and demographics. However, the databases should not be assumed to be of equivalent quality over longer time periods.

Researchers need to ensure that the database contains data that remains comparable across time periods. Different sets of hospitals with different specialities, standards of practice, and patient populations may have contributed at various times to the database and done so using different variable definitions.

Researchers need to be aware of these possible changes and carefully examine data dictionaries and related documentation to ensure the stability and accuracy of the generated analytic research database that will be employed in the study.

12.8 Retrospective Study

A very useful aspect of EHR databases is the possibility of traveling back in time to examine how patient responses and patient treatments have changed, or not changed, over time. However, going back in time, you may run into variables that have been re-scaled, variables that have been poorly collected, variables and diagnoses that have changed definition (e.g., autism). Missing data will often be an issue.

If the data is of high quality, retrospective-prospective studies are a possibility, for example beginning in 2010 and moving forward to 2015.

Of interest as well, is the possibility of time series and longitudinal study of data patterns over time. These include examining seasonal patterns over five, ten or 20 years, for example in the onset of asthma and other seasonal respiratory viruses.

12.9 Clinical Text or Doctor's Notes

A possible source of qualitative information is doctor's notes or clinical text that is a component of the EHR. It may include discharge summaries, suggested treatment plans, and progress notes, which can contain additional information about patients that is useful for research purposes. However, this information may be inconsistent as each doctor may use slightly different wording to report the same condition.

A possible approach when examining clinical text is to use open source natural language processing tools. These can extract sets of text relevant to defining similar disease stages, severity, and progressions of symptoms, data that may not be easily captured if using diagnostic codes. However, the issue of patient and doctor privacy is often a serious consideration when access to such data is requested and reviewed by an IRB.

12.10 Research Issues and Possibilities

EHR research can only study conditions that are routinely reported in a patient database, typically in the context of a patient visit to a clinician. EHR data collection may be sparse for less common conditions, for example, mild

asthma condition or early diabetes, conditions for which patients only rarely seek a physician's care.

EHR based studies can be used to advance clinical care and new precision medicine efforts [12]. If, for example, a child presents with shortness of breath, treatment could be better chosen if the clinician were aware, through linked geocoding, that the patient lived near a major industrial park and was subject to possible exposures known to cause respiratory problems.

More generally, EHR research may support the implementation of precision medicine to include important predictors of individual variability that lie outside the body and include occupational, environmental, and social determinants of health.

12.11 Assembling Research Cohorts and Multi-System Databases

Researchers can use EHRs to form standard cohorts and to assemble groups of patients with specific diseases. Kaiser Permanente in the United States has several EHR-based cohorts including the Diabetes Study of Northern California [13] that involves 20,000 patients with diabetes and has addressed wide-ranging data. The Centers for Disease Control carry out a state based national system of telephone surveys focused on health-related risk behavior for adults aged 18 and older (Behavioral Risk Factor Surveillance System) [14]. Canadian sources may also be used, such as the Saskatchewan health database [15], or European sources such as the Danish population health registry databases [16].

Data drawn from several health systems are can be merged or integrated to assemble large, multisystem cohorts. This is challenging and should be done carefully. For example, US examples include the EPIC system which uses, among many others, EHR delivery and birth data from many hospitals. The EPIC system focuses generally on hospital utilization data, and contains data on over half the U.S. population. It is the focus of many ongoing EHR-related data analyses and studies [17, 18].

A limitation on the development of such multi-system analyses is maintenance of patient privacy. As the set of variables increases or the number of related individuals included in a database increases, the ability to identify individual patients increases to a surprisingly high level [19].

12.12 Studying Disease Etiology

While disease status is often well documented in EHRs, the fundamental causes of the disease (e.g., social, behavioral, and environmental factors), are also often not well documented. As well, some relevant data are not retained,

including, for example, residential addresses over time (as only the current address is used for billing). This will limit the ability of a researcher to measure longitudinal exposure to various environmental pollutants in the context of, for example, asthma or autism-related studies.

If researchers require such additional information, the administration of survey based instruments and related information collection for the patient study cohort can be guided by patient lists generated from the EHR database.

It is possible to use correlated measures to indirectly measure specific socioeconomic aspects. For example, researchers have used health insurance status (e.g., commercial versus Medicaid) as a proxy for individual socioeconomic status (SES) [20], and have assigned census neighborhood SES on the basis of the median income or an index of deprivation in patients' communities [21]. This can also be done using the U.S. based COI measure (Community Opportunities Index, [22]).

12.13 Practical Issues

It is never trivial to download and interpret large amounts of electronic data. Just like data that is collected prospectively, in real time, there is a need to be very careful in regard to time intervals, the units and scales used to measure or report each set of variable measurements, keyword searches, the wording or ontology used in each data base, and the similarity of the definitions of each variable if the final, analytic variables are to be integrated or merged with other databases. These can all be challenging.

It is often the case that databases will be broken out into sets of variables all related to the same set of patients or individuals in the study.

When accessing electronic databases, it is necessary, as a researcher, to be sure that the individuals having data and information in the database have all consented, at some point, to having their data being made available to researchers looking for overall patterns in the data. If the data has been de-identified, this is not as important. But often researchers would like to access addresses to examine spatial patterns in the onset of a disease over time, as well as age and gender. These can be challenging to access as some patients may opt out of the study to preserve confidentiality.

12.14 Consent and HIPAA

The database being accessed by researchers must have been developed under a general consent by patients or subjects, respecting the rules of access defined by, for example, HIPAA or the relevant IRB.

There are often rules in seeking to obtain consent from individuals. You may not be permitted to simply call them up and ask for permission to use their data. Various institutions and types of institutions will have different requirements.

Of the best situation is to access de-identified data that is covered by a general consent on the part of the patient, obtained when they were first processed into the hospital. Not all institutions are focused on possible research use of data. You need to be aware of these issues at your own institution.

See Chapter 6 for a detailed discussion of HIPAA and the rules for accessing patient data.

12.15 Conceptual Issues Affecting Data Collection

The basic principles of statistical design apply to data collected from an existing electronic database. The collection of data needs to result in a sample that is unbiased and representative of the population from which the dataset is drawn, for example, the population of patients accessing the services of a specific hospital in the past five years.

There is a need to understand what population the dataset represents and the quality of the dataset, which may have been subject to modifications over the years, especially if the institution in question has changed ownership or altered their computational architectures or the set of variables they collect from each patient.

Often the decisions affecting a primarily administrative database are implemented with little consideration of the future research needs of medical and health-related researchers. It is not trivial to develop a related research database.

A primary tool for investigating existing databases and accessing relevant data is the use of keyword searches. This assumes the data has been organized so that keywords have been tagged and are observable to the search algorithms moving through the data. The quality of the data generated by keyword search will depend on the degree to which the dataset has been standardized, the degree to which variable names, for example, have consistent over time. If a hospital has dealt with various ownerships and insurance companies, this may be challenging to determine.

Multiple keywork searches and defined time limitations are important methods when pulling out analytic research databases from a larger database repository. Often, the way the keywords are defined will affect the data that is accessed. The specific keyword searches to be used should be listed in the study protocol and replicated to be sure the search algorithm is stable. This is particularly necessary when accessing an administrative database that was not developed for research.

It is also necessary to carefully review the definitions of the variables you are accessing in the database. Age, for example, may be measured in various ways (year, month, days), some of the data may be pre-processed and grouped, for example, the percent of children seeking mental health services in a given family or neighborhood. As well, certain variables, for example race and ethnicity can have multiple definitions, especially as individuals may belong to more than one group.

Each database will have its own complications. As such, be sure to access and carefully review the data dictionary that contains the definition of each variable. Most databases will have such supporting documentation.

12.16 Statistical and Data Analysis

Data analysis, even when using EHR data, is typically conducted within an assumed model, for example, a statistical linear model, that reflects the various sources of variation affecting the outcome of interest, which can be continuous (blood pressure, metabolic measurements) or binary (the onset of a disease or illness).

To be interpretable these measurements must be collected and defined in a standardized manner, consistent over time.

Apart from typical sources of variation such as between subject variability, the nature of EHR data, may increase the number of potential sources of variation affecting the variables in question.

- Instability over time: the cohort of patients may have been affected by administrative aspects, changes in insurance coverages, participation in healthcare networks. The cohort being examined may have been included in the database for reasons that are no longer relevant to the analysis and time period in question.
- Instability due to varying definitions of variables of interest. These will impact the data that collected into a research database generated by keyword searches. The keywords themselves or ICD codes may not be accurate, may have changed meaning over time, or have been implemented in the same, standardized manner, for all patients.

When these effects are present, interpretations of related statistical analyses become more difficult, and often studies will find obtain different significant factors in the statistical modeling, depending on the hospital, or system used to generate the EHR database for analysis.

In many EHR settings, there are a very large number of available variables for inclusion into data analyses. In such settings more recent,

data-centric approaches are very useful. These include machine learning and AI based approaches. These have already been applied in a wide variety of settings [23, 24].

12.17 Application to Chapter 2 Dataset

The availability of clinical EHR data when planning an intervention study such as the one defined and discussed in Chapter 2 is very helpful. It aids in the detection of cases through examination and review of the hospital database, and this also aids in the determination of matched controls. Here, obesity was the main requirement and the need to match the cases and controls. Mostly on age and gender. Secondary variables and comorbidities were also relevant to defining inclusion and exclusionary rules.

12.18 Discussion Points

1. If an electronic database is to be accessed as part of your research project, carry out a careful study of the types of data available in the dataset that are relevant to your project, for example demographics, clinical assessments, lab test results, diagnostic test results, genetics, follow-up data. Are these measured in a standard manner over the entire time period covered by your project?
2. If the electronic database you are using has a large percent of missing data, how will you interpret the missing data? For example, in a healthcare database missing data may represent (i) patients who have left care, (ii) patients who have gone to another healthcare network for care, (iii) patients have not sought care, (iv) a poor data entry system. Discuss how each of these might bias your overall results.
3. To properly use an electronic database, you will need to carefully develop a keyword search strategy, being aware that these will guide your access to the database. If examining diagnostic results in a healthcare database, you will need to accurately obtain the set of ICD numerical codes that index the many possible diagnostic results. List out the set of keywords you will use in your research project and the set of ICD codes, if relevant.
4. How will you protect the individual privacy of subjects in regard to their data? Develop a Data Privacy Plan for your research project. This may also have to be discussed in any research subject consent forms.

5. Time and the stability of any database to be used in your research project should be considered. Variables can be rescaled or reported differently over time, with changing health insurance patterns and different demographic patterns may affect the database. Discuss the potential effects of time on the stability of the research database you will be using in your project.

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13

Basic Epidemiology

Epidemiology is primarily the study of infectious disease and related concepts such as public and population health, preventative medicine, health outcomes, and community health and diagnostic testing [1].

A serious concern is the onset of disease, how it spreads, and the risk of onset after exposure to risk factors. The application of spatial assessment of disease spread through GPS technology has become common [2] as has the development of related statistical and machine learning predictive models [3].

Often, in the guise of preventive medicine, epidemiologic methods are focused on the prevention of disease and the prevention of exposures to known risk factors.

Historically, modern epidemiology is often assumed to begin with John Snow and detection of unclean water as the source of cholera outbreaks [4]. More recently, epidemiology is often big picture in its focus, identifying and fighting global health patterns [5]. Sometimes it is local in its focus, in the guise of community health.

- Everywhere, human beings share this world with other forms of life that, while microscopic, must be respected for their ability to cause infection and disease.
- Disease in the form of epidemics and pandemics will always come back. No society that pretends they do not exist will survive.
- Note that viruses often do not seek to destroy their home environment (you). They prefer you stay alive, living on and using you as a resource base.
- But new viruses have not adjusted to you as their local microbiome, and your immune system has not typed them in order to restrain them, and it is this lack of balance that we often call the disease state.
- Viruses that can often move between hosts easily, for example water borne viruses, which are often the most lethal.

13.1 Epidemiology Statistics: Count Data

Sometimes data is not a measurement, but a count of how many people are in a specific category.

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This type of data arises often in epidemiology and infectious disease.

For example, Cases: Yes/No, Controls: Yes/No, Exposed to Virus: Yes/No.

Typically count data is reported in relation to an overall count or denominator. This gives a percentage or proportion. If this is defined for a given time period, it is referred to as a rate.

Concepts that are important in epidemiology and often require statistical models [6] include:

- **Incidence, prevalence** of a disease. These are ways of expressing the proportion of individuals who have an infection or illness. Typically incidence is a short-term assessment and prevalence is defined over a longer term.
- **Spatial assessments** of how an infectious disease is spreading. These have become more common with the onset of GPS mapping abilities and provides useful graphics on where a disease is occurring and how it is spreading, often in real time.
- **Relative Risk** of disease onset after exposure to a specific risk factor. This is further discussed below.
- **Odds Ratio** (estimates relative risk in studies with relatively rare cases). This is further discussed below.
- **Diagnostic tests** examined by comparing the agreement between an existing gold standard test and a new test.
- **Sensitivity** of a test or predictive model (how accurately are true cases detected by the new test).
- **Specificity** of a test or predictive model (how accurately are non-cases detected by the new test).
- **False positives** and *False negatives* measure the lack of agreement between the gold standard test and the new test.
- **Concordance** is a measure of overall agreement between two tests; the proportion of time that two tests being compared both give positives PLUS the proportion of time the two tests both give negatives.
- **ROC curve** is the receiver operator characteristic curve and is a type of diagnostic for how accurate the predictive model is for a yes/no outcome. It is similar to the R^2 measure for continuous variables in linear models. The ROC curve plots sensitivity versus 1 – specificity for a variety of thresholds related to the probability of a yes outcome occurring. An example is given in Figure 13.1.
- **AUC** is the area under the *ROC* curve. It lies in (0,1) and the closer to 1, the better the agreement between the gold standard and new tests or the overall predictive accuracy of the model. It can also be used as a goodness-of-fit measure for logistic regression (as can all these diagnostic measures). Logistic regression is a way of modeling

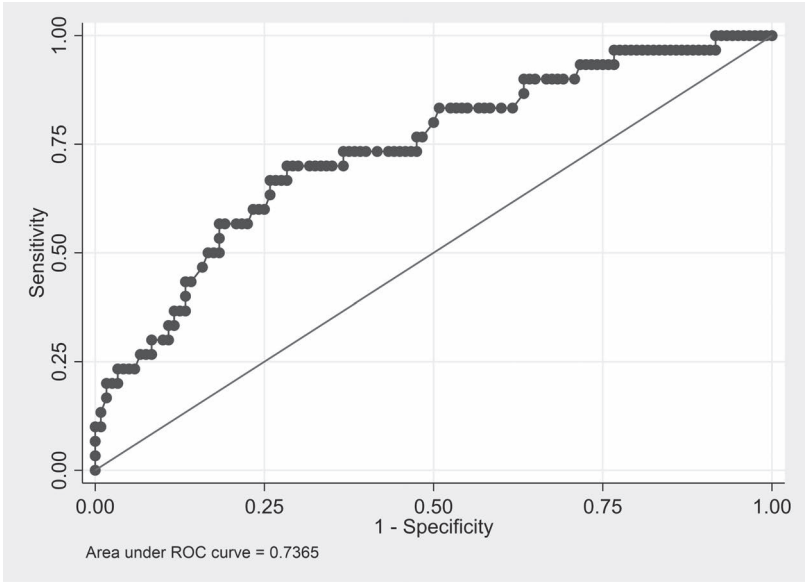


FIGURE 13.1
ROC Curve.

a binary (Yes/no, ill/not ill) outcome for each patient. The AUC measure does not tell the whole story in regard to the accuracy of the model. The sensitivity and specificity values should be also interpreted on their own.

13.2 Example: Diagnostic Testing

There is often a need to accurately test for the presence of an infectious agent so it can be appropriately treated. This process may involve the development of new diagnostic tests to replace older, gold standard tests.

Consider the following table of count data. Samples are drawn from 200 patients and each is tested twice. Once using a gold standard test, and a second time using a new test. The agreement of the two tests is of interest. See Table 13.1.

The reported statistics assessing agreement include:

- Sensitivity (true positive rate) = 60/70. This is the proportion of gold standard = yes values (70) that are also detected as yes by the new test (60).

TABLE 13.1

Diagnostic Test Comparison

	Gold Standard Test = Yes	Gold Standard Test = No	Total
New Test = Yes	60	40	100
New Test = No	10	90	100
Total	70	130	200

- Specificity (true negative rate) = 90/130. This is the proportion of gold standard = no values (130) that are also detected as no by the new test (90).
- Concordance = (60 + 90)/200. This is the overall proportion of accurately detected yes and no values, a measure of overall agreement between the tests.
- PPV = 60/100. Given the new test is positive, how often did it agree with the gold standard test?
- NPV = 90/100. Given the new test if negative, how often did it agree with the gold standard test? Note that 1 – NPV is the percent of individuals who are ill, even when the new test says they are free of the infection.
- False Positive rate = 40/130. The new test has identified these as positive cases, but according to the gold standard they are negative.
- False Negative rate = 10/70. The new test has identified these as negative cases, but according to the gold standard they are positive.
- The odds ratio=60(90)/10(40) = 13.5.

13.3 Odds Ratio and Relative Risk

13.3.1 Relative Risk

In case-control studies the relative risk can be defined and then approximated by the odds ratio (if we assume the controls were selected so that they reflect the exposure distribution in the population) [7]. Consider Table 13.2 (which represents a sample drawn from the population).

The relative risk of being a case once exposed, using the control group as a reference group is given by:

$$RR = (a/b)/(c/d) = ad/bc = OR.$$

TABLE 13.2

Case Status versus Exposure Status

	Exposed	Not Exposed	
Cases	a	b	a+b
Controls	c	d	c+d
	a+c	b+d	

Consider an example where 100 cases (infected) and 100 controls (non-infected) are examined. Some of each group was exposed to an infectious agent:

	Exposed	Not Exposed	All
Cases	80	20	100
Controls	50	50	100
All	1.30	70	200

Here, the relative risk is given by RR = (80/20)/(50/50) = 4.0.

13.3.2 Odds Ratio

- Odds are a way of rescaling probabilities.
- For example: 4 to 1 odds in favor of an outcome is really 4/(4 + 1) = 4/5 = 0.80 (80% probability the outcome will occur).
- In the other direction, odds can be derived using probabilities.
- For example:.8/(1 –.8) = 4 (the event is 4 times more likely to occur than not occur, a 4 to 1 ratio).
- When we have two separate groups to compare, and we want to compare their risks, we do this relatively by making a ratio of the odds of the event occurring (odds ratio).
- For example, if an individual from Group 1 has an 80% chance of getting sick and an individual from Group 2 has a 60% chance then the odds ratio is given by: OR =.8/(1 –.8) /.6/(1 –.6) = 2.6. This implies that the risk of getting sick is 2.6 times higher for an individual in Group 1 versus Group 2.
- Odds ratio is a relative comparison. It might over-exaggerate the importance of infection. For example, rare diseases can have large odds ratios as we compare the risk for cases versus controls. For example, 8/1000 / 2/1000 = 4. But in both Group 1 and Group 2, the occurrence of illness is very low.
- In large samples, the population odds ratio estimates the population relative risk.

Hypothesis Testing: Usually, we test to see if there is NO association:

$$H0 : PopulationOR = 1 Ha : PopulationOR \neq 1$$

Usually, we use a large sample, central limit theorem based result for the sample Odds Ratio statistic.

The results are easier to express on a logarithmic scale.

A *95% confidence interval* based on the sample OR in a 2-way table looks like:

$$log\,(OR) \pm 2\sqrt{(1/n_{11} + 1/n_{21} + 1/n_{21} + 1/n_{22})}$$

For the example table given above, this gives:

$$\log(4.0) \pm 2\sqrt{(1/80 + 1/20 + 1/50 + 1/50)}$$

Question: Is 1.0 in the interval?

If it is, there is evidence that the OR = 1 and therefore there is no significant difference in relative risk for cases and controls.

13.4 Case-Control Study Design Examples

A case-control study is a very common study design [8].

- In this setting the researcher typically has a group of individuals who are ill (cases) after exposure (or not) to a specific risk factor.
- To assess whether the risk factor is associated with the patients getting sick, one approach, given that you have identified the cases, is to select an appropriate control group and compare the rates of infection.
- The control group is often selected using existing records and is closely matched to the cases (similar age and gender distribution, for example).

13.4.1 Example 1: Asthma Study

A study was conducted to determine if there was an association between a rare form of asthma in children and exposure to chemicals emitted by a local production plant.

TABLE 13.3

Asthma Status and Exposure Status

	Exposure	No Exposure	Total
Asthma	17	579	596
No Asthma (Controls)	2	600	602
Total	19	1179	1198

TABLE 13.4

Case Status and Exposure Status

	Exposure		
	Yes	No	All
Cases	53	43	96
Controls	53	100	153

You obtain a listing of asthma cases in the local hospital aged 1 to 11 and determine those who lived within two miles of the production plant (exposed) and those who did not (unexposed) over the past ten years.

Using state and educational records, you identify a control group of individuals of the same age and assess their exposure status. Table 13.3 is obtained:

Here, the $RR = OR = 17(600)/2(579) = 8.8$. The risk is relatively much higher among the cases for those exposed versus those not exposed.

- Note a common limitation: there are only 19 actual cases. A very small number out of 1198 overall cases.
- The controls here provide a reference group, and should reflect the exposure distribution in the reference population.

13.4.2 Example 2: Baby Food Study

A certain type of baby food is thought to be related to a form of childhood illness. Cases are obtained from the hospital records for the past ten years and age-similar controls are randomly obtained as well. All parents in both groups are asked if the baby food was used in their household. Table 13.4 is obtained.

$$\text{RR} = \text{OR} = 53(100) \;/\; 53(43) = 100/43 = 2.2.$$

- The incidence rate is more than twice as high among the exposed as the unexposed. A typical cutoff level for a useful OR is 2.0.

- Note the retrospective and self-report nature of the reported data here. This is often viewed as implying the data have less validity.

13.5 Cohort Study Design Example

A cohort study typically observes over time various exposed groups of individuals, exposed to possible risk factors. Over time, the onset of the disease of interest is observed and interpreted [9].

A visitor is shown around Children's Hospital A, visiting various departments. It is later determined that the visitor had an infectious agent and 223 children have been exposed.

You are asked to carry out a cohort study and track the number of cases occurring and the cumulative incidence rate. Cumulative here refers to the increasing levels of exposure.

You are able to determine levels of exposure for each child in the cohort (A: least, up to D: greatest). The data are shown in Table 13.5.

- The cumulative here refers to the increasing exposure level.
- You can see that the higher levels of exposure (D) have led to higher levels of illness compared to level of exposure A.

Expressing this difference in the percentage of cases between the lowest and highest groups on an odds ratio scale gives:

$$\frac{(50/82)/[1-(50/82)]}{(1/40)/[1-(1/40)]} = 60.9$$

and the odds of being a case in the highest exposure groups is 60 times higher than in the lowest exposure group.

TABLE 13.5

Exposure Level and Number of Cases

Exposure Level	Denominator	Cases	%
A	40	1	1/40
B	77	12	12/77
C	24	10	10/24
D	82	50	50/82
Total	223	82	1

13.6 Confounding Variables

It is sometimes difficult to examine the relationship between say y and x, because other secondary variables are also relevant and need to be included in the interpretation of the relationship, for example diet, gender, BMI and the presence of secondary conditions [10].

This shows up when using statistics in the following way:

If you have a simple regression model, fit to the data using least squares:

$$y = ax_1$$

and you add in x_2, obtaining the fitted model:

$$y = ax_1 + bx_2$$

- If the x_1 and x_2 variables are independent (uncorrelated), the a value in each equation will be same.
- If it is not the same, this implies the x_1 and x_2 variables are correlated.
- This also implies that having x_2 in the model has affected the y <-> x_1 relationship.
- So x_2 is said to confound the y <-> x_1 relationship.
- To deal with this, if x_2 is a categorical variable, stratify the data on the levels of x_2.
- For example, look at $y = ax_1$ for $x_2 = 1$ and separately, $y = ax_1$ for $x_2 = 0$.

13.6.1 Example

A Case-Control study is conducted examining relationship between pediatric cancer and environmental exposure. Information is available regarding Gene A thought to limit the onset of pediatric cancer. See Table 13.6.

- The presence of the gene leads to a different odds ratio.
- Its presence affects the estimate of the odds ratio.
- The presence of gene A (yes/no) can be seen as confounding the effect between cancer and exposure.
- DO NOT combine the tables here.
- Report the data stratified by Gene A (yes/no).

13.6.2 Stratifying by Age

We use stratification to interpret results when confounding exists [11].

TABLE 13.6

Case Status versus Exposure Status by Gene A Present (yes/no)

Gene A Present (OR = 1/6)			
	Exposure		
	Yes	**No**	**All**
Case	20	30	50
Control	40	10	50
Gene A Not Present (OR = 2/3)			
	Exposure		
	Yes	**No**	**All**
Case	20	30	50
Control	25	25	50

It is also necessary if you want to compare samples drawn from populations having different underlying demographic distributions.

For example, comparing asthma in two counties, one of which has a much younger population. A fair comparison needs to adjust the numbers to reflect the relevant population structures.

- If an overall RR or OR is required, we reweight the data by sub-group and report the weighted average of the OR values.
- We often age-stratify the OR estimates to make them more comparable when comparing two different populations.
- This may involve re-weighting the observed values for both samples using the age distribution of a reference population (e.g., the U.S. population in 1970).
- Stratifying often lowers variation, making it easier to see differences if they exist.
- An older population has different health-related risk factors than a young one. This motivates age stratification.
- In general, we stratify on the variables thought most relevant to the disease or infection in question.

13.6.3 Example

You are studying a rare form of pediatric illness and obesity.

The overall data gives a 95% confidence interval for the overall OR = [0.069, 0.67].

Age stratified reporting of the same results gives Table 13.7:

TABLE 13.7

Case Status versus Obesity Status by Age Group

Age (4-6)	Obese	Non-Obese	All
Cases	2	35	37
Controls	11	30	41
RR = 0.16 [0.032, 0.76]			
Age (7-9)	**Obese**	**Non-Obese**	**All**
Cases	2	33	35
Controls	6	32	38
RR = 0.32 [0.61, 1.7]			

The OR is twice as high in the 7–9 age group compared to the 4–6 age group. The OR in the Age (7–9) group has one in the 95% confidence interval, implying no difference between cases and controls, while in the Age (4–6) and overall groups, the OR is clearly less than one.

13.7 Pre-Post Designs and Follow-Up in Cohort Settings

The most optimal form of controls is to choose a study design where the cases are their own controls.

This is typically referred to as a pre-post design, sometimes with further follow-up [12].

For each person, we measure the difference and compare the average difference between case and control groups.

If there are only two measurements taken for each subject then the differences can be viewed as a single set of independent measurements and a standard t-test applied.

Studies through time are often referred to as repeated measures or longitudinal studies. These can be technically difficult to analyze as the use of a common set of subjects with multiple measurements leads to correlation within subjects that must be addressed in the ANOVA or regression model. The specific type of correlation can be modelled in several different ways and there has been much work in this regard [13]. The simplest and most widely applicable approach is repeated measures ANOVA [14].

13.8 Example From Chapter 2

13.8.1 Cohort Study

While the change in weight achieved by patients in the cohort study over a six-month period is technically a continuous variable, it can be rescaled into a categorical variable. For example, consider Table 13.8 where the association between age being ≥ median value and weight decrease ≥ median value is shown to be significant.

13.8.2 Case-Control Study

In the case-control study, the association of case (yes/no) and whether Gene A is present (yes/no) can be examined. See Table 13.9, where the association between case status (yes/no) and the presence of Gene A (yes/no) is shown to be not significant.

TABLE 13.8

Age being ≥ median value versus weight decrease ≥ median value

	Weight Decrease ≥ median	Weight Decrease < median	All
Age ≥ median	51	5	56
Age < median	32	11	43
All	16	83	99

OR = 3.5. Chi-square test of association = 0.026, Fisher exact test = 0.031.

TABLE 13.9

Case status (yes/no) and the presence of Gene A (yes/no)

	Cases yes	Cases no	All
Gene A yes	12	14	26
Gene A no	18	30	48
All	30	44	74

OR = 1.4. Chi-square test of association = 0.45, and Fisher exact test = 0.62.

13.9 Discussion Points

- Discuss potential application of epidemiologic concepts relevant to your study. Are these considerations included in your study protocol? What statistics will you use to summarize epidemiologic information in your study?
- If there are community health aspects in your study protocol, consider the challenges that may exist in developing procedures and strategies to reach out into the community to collect information and conduct your study. What relationships exist to help you carry out and achieve these goals.
- Discuss how the odds ratio and relative risk concepts are related. When would you use the odds ratio to estimate the relative risk?
- Define how you might use a cohort based study design in your research study. Define the potential exposure variable. Define the time aspect of the study. Discuss the statistics you will use to assess the information and data collected in your research study.
- If you will be using a case-control study design, discuss how you will define and identify cases from existing clinical or other electronic databases. Discuss how you will also identify a relevant set of controls representing the exposure (yes/no) distribution in the reference population.
- Define and discuss how the concepts of confounding and potential stratification may affect the data and statistical analysis in your research study.

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14

Linear Models

- The qualitative aspects of medical research often require detailed review of the literature, related medical training and learning, and an understanding of both lab techniques and clinical insight.
- The quantitative aspects of medical research often require the use of study designs, database information, statistical models, and data analysis.
- Learning both the qualitative aspects of a research area and the related quantitative methods to collect and analyze related data is an ongoing process.

The statistical methods necessary to analyze data collected as part of a study often extend beyond the basic statistical methods required to compare treatment and control groups, for example, paired t-tests and two-sample t-tests.

Often multivariate models must be used to control for the effects of secondary variables affecting the interpretation of the response and explanatory variables of interest.

Here we review some of these approaches that usually go under the names multivariate regression, linear models and generalized linear models, focusing on linear and logistic regression models.

14.1 Simple Regression: Fitting Straight Lines

- We saw this as a simple and mostly descriptive model earlier in Chapter 8.
- Given data for y and x, we can plot the data values using a scatterplot.
- We can also fit and plot the least squares line through the data.
- The slope of the line will capture the relationship between y and x.
- We can then test if there is a relationship between y and x. This will be the case if we can reject the null hypothesis that the slope is zero.

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14.1.1 Example

Consider the following dataset comprising responses y and x.

y	3	5	7	9	12	13	11	15	17	18	21	22	23	22	21
x	1	2	2	3	4	5	4	6	7	5	6	8	9	7	8

- The plotted data points and fitted linear regression line are shown.
- The linear model seems appropriate and the goodness of fit is high ($R^2 = 91\%$).
- The fitted regression line is given by; y = 1.309 + 2.589x.
- The slope 2.589 implies that y will increase on average by 2.589 units for a one unit increase in x. Figure 14.1 shows the data and fitted line.
- The p-value to test the null hypothesis that the slope of the line = 0, is <0.001. Thus, there is a significant positive relationship between y and x.

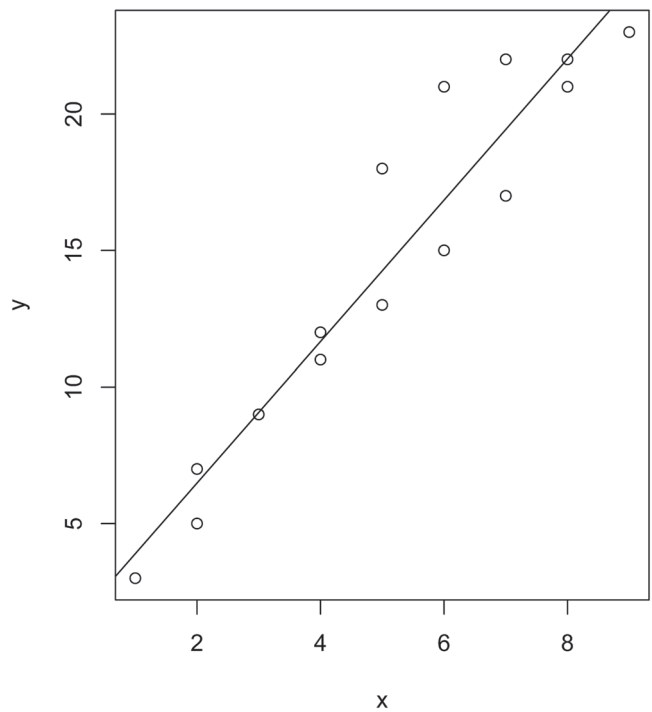


FIGURE 14.1
Simple Regression Plot.

14.2 Multivariate Regression

Statistical approaches to modeling apply probability-based models that reflect the design of your study/experiment and fit them to the observed data you have carefully collected. This context then allow for the testing of hypotheses, usually whether each of the variables in the model is significant in predicting the response of interest.

The linear model framework is strangely flexible and the basic ideas mostly simple.

Linear models are employed in settings where underlying relationships among several variables are to be investigated. They are extensions of simple regression models and are typically written;

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

where y_i is the response of interest (or dependent variable) for the ith individual measured on the jth variable, ε_i is the associated random error, typically independent and normally distributed (not required in larger samples) with mean 0 and common variance σ^2, the explanatory or independent variables can be grouped as columns, one for each measured variable, into a design matrix $X = \left[x_1, x_2, ..., x_p\right]$ with x_j column vectors each representing $i = 1..., n$ measurements on the fixed j^{th} explanatory variable [1].

The least squares estimates, $b_0, b_1, \ldots, b_p$ for $\beta_0, \beta_1, \ldots, \beta_p$ are obtained by applying the method of least squares to the multivariate model [2]. The observed variation in the model is then given by the residual sum of squared errors and is defined as:

$$SSE = \sum_{i=1}^{n} (y_i - b_0 - x_1 b_1 - x_2 b_2 - \cdots - x_p b_p)^2$$

Frequentist inference for specific β_j parameters, typically testing if these are equal to zero, can be carried out using least squares or the method of maximum likelihood estimation [3]. Least squares usually reflects the assumption of normally distributed, while the method of maximum likelihood can be used for a wider variety of assumptions.

14.3 Goodness of Fit

Linear models also have a goodness of fit measure available; the R^2 value. The R^2 value measures the relative decrease in unexplained variation or percent improvement in fit when the researcher uses the x_j variables in the model:

$$R^2 = (SST - SSE)/SST$$

Where $SST = \sum_{i=1}^{n}(y_i - \overline{y})^2$, is the variation in the response variable having nothing to do with the x_j variables. The R^2 value lies between 0 and 1. It is expected that if the researcher has chosen the x_j variables wisely, the SSE value (variation about the linear model) will be substantially less than the SST value (baseline variation in the response without the linear model). The closer the R^2 value lies to 1, the better the fit of the model [4].

Note that the stability of these models needs to be considered. As noted in an earlier chapter, outliers can greatly affect the estimated coefficients of a simple regression model and the R^2 value. These issues also affect multivariate models where it can be more difficult to detect outliers.

14.4 Example

For the case-control study given in Chapter 2, the basic finding of a difference between the groups in terms of decrease in bodyfat over the six months of the intervention is of interest. It might be the case that this result will not hold if other variables are considered in the model, for example, the presence of Gene A or the age of the subject.

In statistical terms, as the presence of these variables in the model may explain some of the variability in the data, the remaining variability related to the case-control groups may change, possibly altering the significance of the group variable.

Controlling for Gene A and age in the regression equation gives the multivariate regression model:

$$ChangeInBodyfat_i = \beta_0 + \beta_1 Group_i + \beta_2 GeneA_i + \beta_3 Age_i + \varepsilon_i$$

Fitting this model using least squares gives the following results shown in Table 14.1, usually laid out in an ANOVA table:

TABLE 14.1

Results for Multivariate Linear Model

	Coefficient	Standard Error	T-statistic	p-value
Group	−5.53	0.44	−12.61	< 0.001
Gene A	0.33	0.45	0.73	0.47
Age	−0.01	0.02	−0.64	0.53

Equation: Body Fat Differences = 1.43–5.53 Cases + 0.33 Gene A – 0.01 Age,

Overall p-value < 0.001, R-sq = 0.71

It can be seen from Table 14.1 that controlling for the presence of Gene A and age in the model does not affect the basic result that there is a significant difference between the treatment groups.

14.5 Additional Considerations

Additional topics to consider when using linear models include:

- **Comparing Pairwise Mean Differences**: If there are many groups to compare, for example comparing the group averages, we can do this two-at-a-time with pairwise comparisons based on two-sample t-tests or non-parametric Mann-Whitney tests. Typically, the p-values for these tests are adjusted for the family-wide error idea: multiple comparisons [5].
- **Interaction and Stratification**: Sometimes the y to x_1 relationship alters for different levels of x_2. This is called interaction and it can be detected via a simple statistical test. This is discussed in an earlier chapter. If interaction is significant, it may suggest a more **stratified analysis**, looking at the y to x_1 relationship for different subgroups defined by the levels of the secondary x_2 variable, for example gender or socioeconomic status [6]. These variables can also be viewed as mediating variables [7].
- **Confounding**: This is related to the idea of interaction, but tends to be applied in epidemiologic settings. Consider fitting a simple linear model to examine the y to x_1 relationship. Note the value of the coefficient for x_1. Now add a second variable to the model, x_2, and fit a linear model for y to x_1 and x_2. If the coefficient for x_1 in this second model is the same as in the first one, then the relationship between y and x_1 is not affected by the presence of x_2 in the model. This implies the y to x_1 relationship is NOT confounded by the presence of x_2 in the model and x_2 can be disregarded. If the coefficient of x_1 does change, then we say x_2 confounds the y to x_1 relationship and stratification on the levels of x_2 may be necessary.
- **Model Sensitivity**: The fitting of straight lines is sensitive to individual values that lie far from the center of the data, so-called outlier values [8]. These should be checked by looking at graphics of the collected random errors $e = yhat_i - y_i$ where $yhat_i$ is the predicted value from the linear model for the ith subject. The residual plot for the example above is shown in Figure 14.2. The observed errors or residuals closely follow a normal distribution.

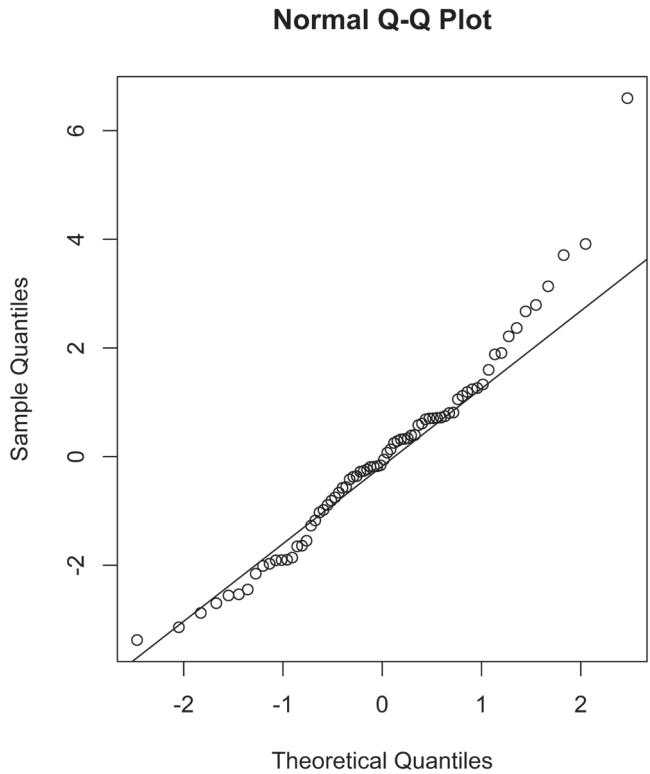


FIGURE 14.2
Residual Plots.

14.6 Generalized Linear Model and Logistic Regression

If the data do not follow a Normal distribution, the Generalized Linear Model extension can be used to incorporate other distributions (binomial, Poisson, exponential distributions, for example) [9].

This approach typically uses large sample results and the maximum likelihood method to fit the data to the model and carry out any required estimation and hypothesis testing.

The logistic regression model is a commonly applied regression model [10]. The distribution of the response variable is binomial (a binary yes/no outcome is the response, for example success/failure or above/below a given threshold) and the probability of success can be directly modeled.

The likelihood function is initially written;

$$L(\pi \mid y) = c \cdot \prod_{i=1}^{n} \pi^{y_i} \cdot (1-\pi)^{1-y_i}$$

where y_i is a $0-1$ binary response variable denoting success at the x_i setting, π is the probability of a success for each individual and the n individual binary responses are assumed to be independent Bernoulli random variables.

The probability of a success at each x_i setting is then modeled using the logistic function form;

$$\pi = \pi(x_i) = \frac{\exp(\sum_{j=0}^{k} \theta_j x_{ij})}{1+\exp(\sum_{j=0}^{k} \theta_j x_{ij})}$$

where $x_i = (x_{i0}, ..., x_{ik})'$ denotes the ith setting of the explanatory variables, θ_0 is the constant in the model and $x_{i0} = 1$.

This expression for π is then plugged into the likelihood function giving the result that the likelihood function is then a function of $\theta = (\theta_0, \theta_1 \ldots, \theta_k)$. The likelihood function is then maximized to provide maximum likelihood estimates for θ.

The model is often re-arranged to be a generalized linear model with a logit scaled response. For the example considered above this would be of the form:

$$\text{Logit(y)} = \beta_0 + \beta_1 \text{ group} + \beta_2 \text{ GeneA} + \beta_3 \text{ Age} + \varepsilon$$

The central limit theorem is then applied to obtain sampling distributions for the maximum likelihood function based statistics and an ANOVA table can be generated to examine the significance of the coefficient for each individual variable in the model, typically using p-values.

A new set of classification and diagnostic errors can be defined for the logistic regression model. These include the sensitivity and specificity of model-based predictions versus the actual data, along with false positives, false negatives, the ROC curve and the area under the ROC curve (AUC) [11]. The AUC value is similar in interpretation to the R^2 value, with a value closer to one implying a better fitting model.

Usually, an AUC of at least 0.7 is viewed as useful.

Linear and generalized linear models are some of the most common approaches to understanding relationships among observed variables in almost any research study that is quantitative.

14.7 Application to Chapter 2 Dataset

Taking the outcome of interest, in the cohort study, to be loss of bodyfat over the course of the six-month trial, there is an interest in determining if the subjects achieved the overall median (4.0) bodyfat loss at the 3 month time

point. A 3-month success variable is defined: If a subject achieves greater than or equal to the overall median loss by the 3-month time point the value is one, if they do not it is zero. This variable is the outcome of interest and has a binary distribution (0/1). In terms of predicting this outcome, the set of variables; age, weight, and socioeconomic status were used to build a logistic regression model.

The fitted logistic regression model results are given in Table 14.2. It is seen that socioeconomic status is significant, but age and weight are not.

The classification statistics for this model are insightful: The area under the curve, AUC = 81.5%, sensitivity = 98.8%, specificity = 6.3%, PPV = 84.5%, NPV = 50.0%, false positives = 93.8%, false negatives = 1.2%. The ROC curve is shown in Figure 14.3. Overall correct classification = 83.8%.

TABLE 14.2

Logistic Regression Model

	Coefficient	Standard Error	Z statistic	p-value
Age	0.046	0.027	1.72	0.085
Weight	0.014	0.013	1.08	0.281
Socioeconomic status	1.826	0.846	2.16	0.031

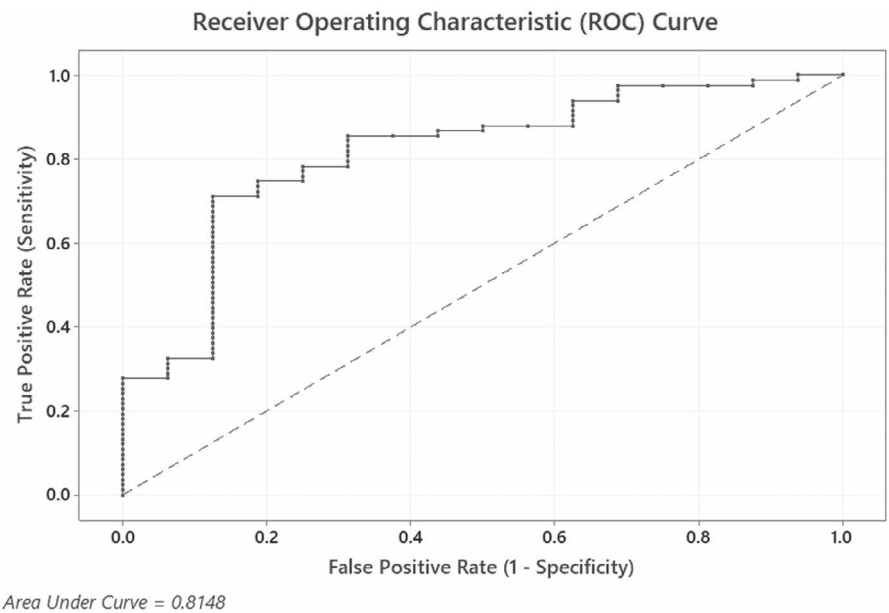


FIGURE 14.3

ROC Curve for the Logistic Regression Model.

While the AUC and ROC curve are acceptable, and sensitivity and PPV are high, the ability of the model to predict the non-achieving patients is very low with specificity = 6.3% and NPV = 50.0%. Thus the predictive accuracy of the model is limited. It is important when using logistic regression to examine all relevant diagnostics, not only the AUC, specificity and overall correct classification.

14.8 Machine Learning and Data Based Approaches

Recently, modeling approaches have been developed that are data dependent and highly complex, reflecting network structures rather than linear equations [12]. These include machine learning methods such as artificial neural networks, random forest models and other models, many of them in the form of proprietary apps that currently are available for use [13].

Many of these overlap with AI-related models, and are often motivated by the very large databases generated by genomics (500,000 SNPS on a single Affymetrix chip), Google, Facebook and other internet-related marketing companies.

These data based approaches do not primarily use probability models to assess significance for individual variables. They typically use a training/testing division of the dataset to examine if the model that best fits the training data, gives similar accuracy in the testing data. See Chapter 15 for a more detailed discussion.

This set of models, highly non-linear, continues to evolve. They reflect more flexible multi-nodal network structure and classification tree structure. They can be interesting to interpret [14].

14.9 Discussion Points

- To apply a multivariate linear model with a set of continuous variables, you need to assume the response follows a linear pattern when plotted against each of the x_i variables in the model. As well the residual values should have a histogram that is approximately Normal. Discuss how to carry this out with the set of variables in your research project.
- Discuss the levels of R^2 that are acceptable in linear models using the variables of interest in your research project. A typical level is 0.7.
- Which of the explanatory variables in your research project would you expect to be correlated? How would you include this information in the form of interactions terms in a linear model?

- Discuss the types of variables (continuous, categorical) you will include in your analysis. Discuss not only the overall model and the specific hypotheses you would like to test, but also any further stratifications of the data you will be examining, for example demographic variables.
- Consider a two-by-two table comparing the predicted yes/no values from a logistic regression model with the actual yes/no values. Find the sensitivity, specificity, PPV, NPV, false positives, false negatives for this result.

	Actual yes	**Actual no**	**Total**
Predicted yes	70	30	100
Predicted no	40	60	100
Total	110	90	200

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15

Introduction to Machine Learning and AI Data-Based Methods and Perspectives

Machine learning methods of data analysis are now very common in the analysis of large datasets, which are typically not collected in the context of a designed experiment. These methods have extended the flexibility of data modeling methods to include many different types of responses and data, including very detailed image-based responses in areas of medical research such as radiology and cancer research [1].

Such integrated databases, which can also be generated by genomic data chips, or internet or other data flows that provide, instantaneously, thousands of variables and hundreds of thousands of measurements. It is a challenge to define and interpret the information in such large collections of raw data [2].

Early on in the application of applied statistics, now mostly referred to as data science, data based methods such as linear discriminant functions, principal components, classification trees, cluster analysis, and other approaches providing data-centric aspects of scientific study, were developed, but often viewed as secondary and "hypothesis generating" [3]. They were often not the basis of formal statistical inference but allowed for examination of observed data structures.

More recently, under the various names of data mining, Big Data, Artificial Intelligence (AI), data science and machine learning, data-centric methods have slowly been developed using complex search algorithms based entirely on the assumption of the large databases themselves as the point of reference, not necessarily related to population-based probability models. These have been applied widely [4]. These models however, are often black boxes, giving overall model summaries related to the accuracy of predictive classification rather information on individual parameters such as those available in linear statistical models using, for example, the ANOVA table [5].

The databases to be examined are typically not subject to careful design criteria and often bring together data that may not easily be interpretable as an integrated whole. Such databases may require much processing to obtain a set of data that can be studied.

Whatever their quality, such databases are a key component in the area of AI and machine learning data analysis. These areas have generalized complex search algorithms and other algorithms to search, compare, and train/fit complex models to such datasets. Artificial neural networks (ANNs) are an example [6], as are random forest models [7].

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These approaches use highly flexible model structures, for example multinodal artificial neural networks (ANNs) and classification trees (random forest models), and assess the usefulness of these models by using one randomly selected portion of the dataset (training data) to develop a model and predict the same response in the remaining portion of the dataset (testing data). The assumption of an initial high quality database underlies the relevance of these approaches. Without an up-to-date and representative database these methods can be quite ineffective. These analyses themselves require updating to remain relevant. See Figure 15.1 for the elements being used to provide these data analyses.

When such data is not collected subject to a carefully design experiment or study protocol (though they can be), potential bias and the opportunity for misleading data patterns and related information abound [8]. As well, when the model is so focused on the particular dataset at hand, such models may not be easily generalizable (e.g., hospital utilization data or clinical data). It is possible to obtain differing results for the same set of variables collected at different hospitals or different time periods.

The stability of these data-centric methods is not easy to assess. The simplest and most interpretable approach, when possible, is the cross validation of the model (leaving out small pieces of the data and seeing if the model behaves differently), though this has challenges [9]. The final underlying machine learning/AI model or black box is usually nonlinear and recursive, difficult to interpret and offering limited insight into the contribution of individual variables.

For researchers wrestling with understanding the information and data patterns in large databases, there are now several paradigms and tools for detailed data analysis. Methods such as frequentist, ANNs and random forest

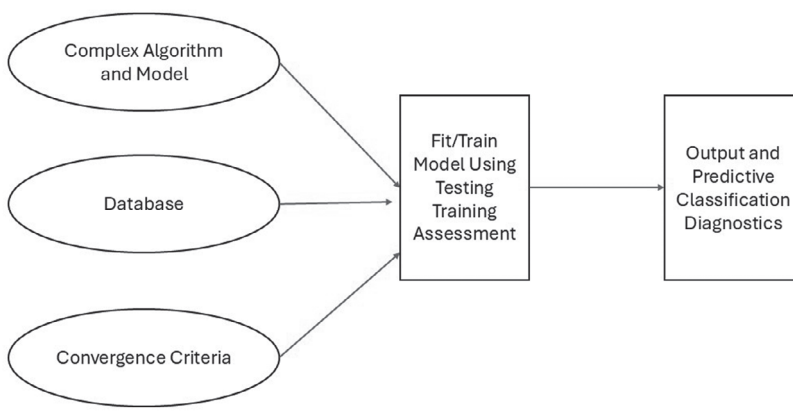


FIGURE 15.1
Basic Components of Machine Learning Based Data Analysis.

methods do not necessarily contradict or agree with each other, but in terms of the information generated may often overlap in terms of the accuracy of prediction. Each provides a different context and perspective on data-based information and its interpretation.

Note that frequentist probability and related laws of large numbers continue to play a role in the machine learning setting, as components of the technical aspects of the stochastic algorithms that converge to fit ANN models. As well the bootstrap approach of re-sampling the observed data is also prevalent in random forest classification analyses.

Note that sampling variation in the context of machine-learning approaches arises in the form of multiple training/testing random splits of the data and the reporting of the related range of classification-related diagnostic statistics.

The development process for a modern statistical analysis may now include the following possible analyses:

- Initial data summary (including graphics) and frequentist likelihood or Bayesian statistical analysis.
- Non-parametric or bootstrap frequentist analysis.
- Pre-processing of data variables and related dimension reduction.
- Singular Value Decomposition based cluster analysis, recursive partitioning, and random forests for the development of predictive classification trees.
- Neural network and random forest classification tree summary models based for predictive classification models (supervised learning).

More deeply algorithmic approaches are also available in the deep-learning AI context where complex logical and data structures can be initialized in the model and stochastic gradient approaches applied to maximization [10]. In fact, there are now a very wide ranging set of possible AI and machine learning approaches that can be applied and compared when examining a specific dataset [11]. Note that there is no truly optimal approach among them, just many overlapping assessments.

15.1 Overview of the ANN Approach

ANN models are usually built on a subset of the original data (training set) and validated on the remaining data (testing set). They often use a multiple hidden layer structure, using back-propagation, stochastic gradient descent methods and multiple replications to obtain convergence and fit/train a

predictive classification model. The sigmoid function serves as an activation function in the hidden and output layers so that the output of these models is a value in [0,1] which is then interpreted according to chosen cutoffs or threshold levels, very similar to logistic regression-related diagnostics. Typically an ROC curve with the area under the curve (AUC) and various diagnostics similar to sensitivity, specificity, false positives, and false negatives can be obtained from the outputs here, depicting how well the training model can predict in the testing data.

The learning/fitting process is often viewed as optimal when the average error (mean squared error) in the training set converges to a very small value [6]. Some measure of the sensitivity of the predictive classification result can be determined by assessing convergence of the algorithm and model using several randomized training and testing resamples, giving some measure of the stability of the model in regard to underlying sample variation, so-called bagging, and boosting [12].

A typical output showing the basic network structure of an ANN model is given in Figure 15.2 [13].

In terms of general interpretation, these data-centric models differ from statistical models which seek to generalize sample based results to a more

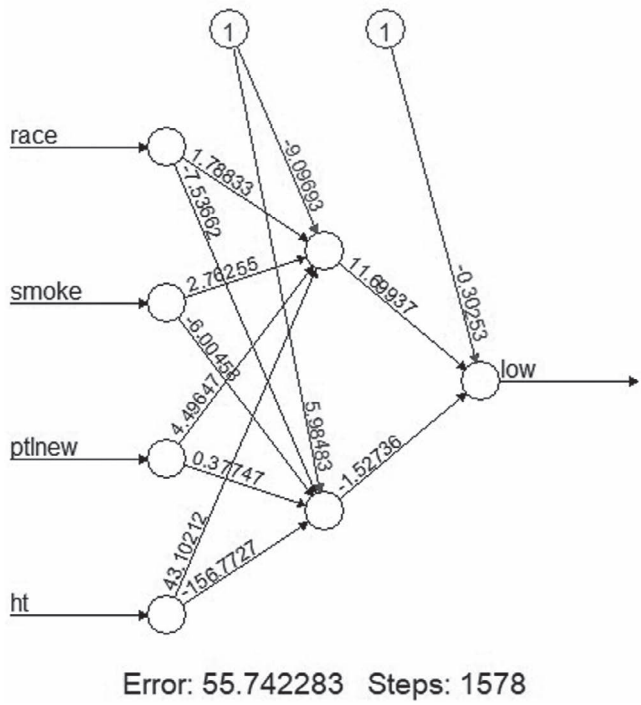


FIGURE 15.2
Artificial Neural Network for Predicting Low Birth Weight.

general population, based on probability models and related assumptions. Machine learning models are specific to the dataset and use a random splitting of the overall data to create a training dataset for fitting the model and a testing dataset for model evaluation. Machine learning results in general are heavily dependent on the hospital or clinic from which the dataset is being drawn.

15.2 Random Forest Methods

The random forest method and algorithm [14] uses a large number of classification trees to relate a specific response y to a set of explanatory variables X. Each classification tree is based on the observed data and yields a predicted class for the response y and the class with the greatest frequency becomes the chosen model prediction. The algorithm uses randomly selected resamples of the original data and generates a large number of such classification trees. These are combined and final predictions are made by taking the average of the predictions made by the set of classification trees. Individual parameters can be estimated and confidence intervals obtained. Bootstrap-related resampling is used to generate all required classification trees and related statistics.

Typically, the importance of a specific variable is measured by how the accuracy of the random forest decreases when the data for that variable is permuted while all others are left unchanged. A multi-way variable importance plot can be used to display individual explanatory variables and how they contribute to improvements in the mean squared error statistic (on the x-axis) and their importance scores (on the y-axis). The most influential variables appear at the top of the plot, with their importance decreasing as you move down the plot [15]. See Figure 15.3.

15.3 Challenges Due to Data Structures

As AI and machine learning methods are heavily dependent on the specific dataset being analyzed. As such they are sensitive to problems and issues affecting the data. These include the presence of latent variables, biased data collection, bias arising due to missing data patterns, outlier data values, highly correlated sets of variables. All these are known to affect statistical approaches to inference and there exist many diagnostic approaches to detect such data issues, but they also affect AI and machine learning methods, especially as they are heavily dependent on the data.

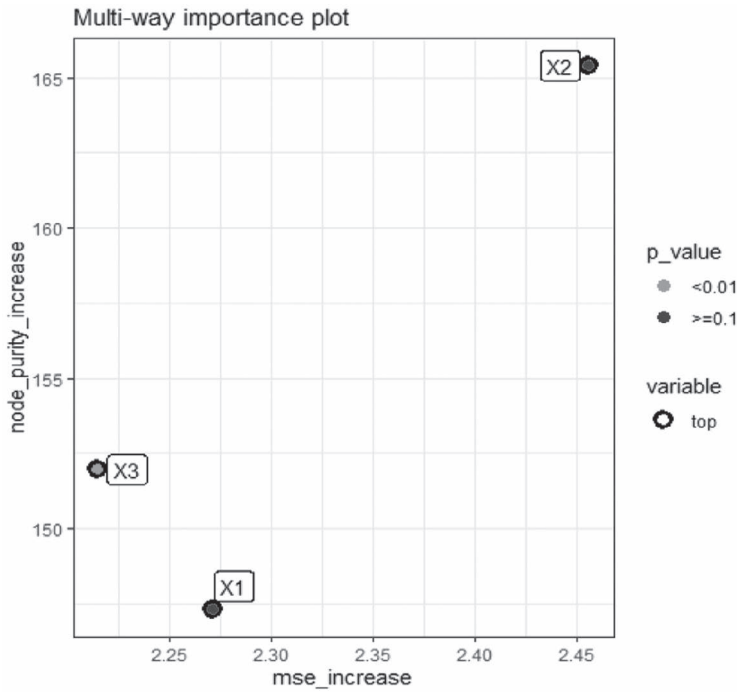


FIGURE 15.3
Multi-way Importance Plot.

15.3.1 Simpson's Paradox

Simpson's paradox or aggregation effects, are a specific type of data structure that arises in settings where unknown latent variables defining subdivisions in the dataset may yield data sub-groups in which the data patterns detected in the overall database do not hold, and indeed may display as the exact opposite of the overall patterns detected [16]. This can occur in both continuous and discrete data and can impact application of both population based statistical regression models and machine learning models [17].

The data variables stratifying the data in this manner may be available in the observed dataset, or remain uncollected, a reflection of the limits of the science or medicine in question. An example of this are studies that examine the onset of a disease (yes/no) without consideration of genetic variables which may give a small proportion of the subject's immunity and thus no chance of acquiring the disease. The inclusion of such individuals will bias the study, even though the specific genes giving rise to the immunity are unknown.

The presence of Simpson's paradox has little to do with probabilistic based modeling, rather it is a reflection of underlying data structure and

stratifications within the overall database. These considerations affect frequentist statistical approaches to modeling, as well as machine learning methods such as neural networks and random forest models.

In settings where there is a large dependence on data-analytic methods such as artificial neural networks or random forest models, the risk of such aggregation effects should be considered. The use of complex search algorithms on large datasets exposes them to the effects of such unknown data patterns. Also, the training/testing split of the data may be impacted to differing degrees by the effects of secondary latent or unknown variables.

Simpson's paradox may also be induced in a dataset due to missing data. If non-random missingness patterns are induced into the data, these may affect all the data variables in question, possibly leading to Simpson's paradox, as well as bias in the results. This again applies to both probability-based models and data-centric machine learning approaches.

15.3.2 Sensitivity to Outliers

Machine learning approaches such as artificial neural networks are highly nonlinear and can have challenging convergence patterns. Clusters of extreme values may affect the convergence and related predictive accuracy of the model. This is true as well of standard statistical models. The database being examined must be carefully cleaned prior to analysis and extreme, non-standard data values may have to be removed to obtain a stable predictive machine learning model.

15.4 Weakness in Small Samples and Comparisons with Standard Methods

Advocates of machine learning methods often cite improved model fit in large as a basis for selecting machine learning methods over standard statistical methods. This occurs primarily due to the high number of fitted parameters and the flexible nonlinear structure of such models. In smaller datasets this may not be the case [18]. The highly flexible nature of neural networks may require very large sample sizes to converge with high degrees of accuracy [19] and be stable.

ANN models, for example, are highly nonlinear, deliberately flexible to model more of the randomness in the observed data. But these models only assess whether the observed pattern in one portion of the data (training) is similar to the pattern in the rest of the data (testing). This is more a measure of agreement than prediction. Such models are more relevant to the overall population from which the dataset is drawn if the sample size is very large.

If deciding which approach to use in the analysis of research data, comparisons of data-centric methods versus statistical approaches should reflect an awareness of the importance of study design and the context in which the

data were collected. If examining a large, national survey based study, probability based adjustments such as weighting and random effects interpretations are standard elements of the analysis of such large scale survey data. These may need to be re-developed or incorporated directly into the context of data-centric machine learning methods in order to appropriately interpret the dataset itself and related data-centric inferences [13].

The flexibility of the ANN model with many hidden layers in the neural network and related free parameters make it easier to fit the neural network model to the observed data than more formal statistical models, but they also make it easier to fit any response to the explanatory data elements [20]. This may even give a fitted model that fits the overall data well, but yields poor predictions.

In the past, the use of statistical and probability models focused on simplicity and the assumption of a model + random error paradigm. The simplicity of the ANOVA hypothesis testing format and standard least squares-based analysis was supported by citing Occam's Razor and other principles favoring simplicity and interpretability. This template, for many years, has been the approach supporting advances in science, medicine, and most research fields. This is not the approach to modeling supported by application of highly complex, data-centric machine learning models.

15.5 Comparability of Results across Databases

Machine learning and AI methods depend heavily on the databases being examined. The underlying algorithms that power the analysis are basically complex search algorithms that have been modified in many ways.

Given how data-centric these approaches are, analyses conducted on the same set of variables or features in regard to the same illness, may not give the same results if the analyses are compared for different hospitals for example. The analysis is conditional on the database in question which limits the generalizability of the analysis.

15.5.1 The Effects of Time

As discussed in the chapter on Electronic Health Record Datasets, these databases may alter in real time, with both the actual data and the definition of the variables of interest altering over time.

As AI and machine learning approaches are heavily dependent on the database being examined, to be relevant, requiring the databases underlying their application to be regularly updated. Changes affecting the process by which the data is collected and entered into the database will affect the machine learning model and its predictive accuracy.

15.6 Interpreting the Importance of Specific Data Variables or Features

Machine learning models such as artificial neural networks are primarily network models with many nodes in their structure. This type of model is difficult to interpret in terms of the relative importance of individual variables in the model.

Statistical linear models provide a linear equation and associated ANOVA tables that can be interpreted in terms of the significance of each variable. Least squares provides stable estimation of the model parameters involved and p-values allow for the detection of significant variables. Often the reported linear model will be the one with only the individually significant variables reported.

A step forward in the interpretation of machine learning models has been the application of a game theoretic approach to report important individual variables. These Shapley diagrams [ref], provide guidance in the detection of important variables in a complex model. Note that the approach can also be applied to highly nonlinear statistical models as well.

The Shapley diagrams apply an algorithmic approach which views each variable as a participant in a game (predict the response) and examines, how often, on average, each variable is present in a useful predictive model. The bootstrap approach along with cross validation based on the variables themselves (dropping a variable out of the model and examining observed changes in the accuracy of the model) generates many possible datasets and for each the machine learning method is applied and predictive accuracy examined. The average importance of each variable is then calculated and the variables being considered in the analysis are ranked providing insight into the relative importance of each variable.

The details are quite involved and are not reviewed here. To some extent, this approach addresses concerns that the machine learning approaches are black boxes with limited interpretability [21,22].

15.7 Chapter 2: Potential Applications

The success of individuals in the weight loss program may be predicted by applying the ANN model based on various explanatory variables to predict the outcome of the program defined as a binary outcome (success/no success), or by applying a random forest model to the set of explanatory variables in relation to a continuous response such as the amount of weight lost. These results, in terms of predictive accuracy, can be compared to standard statistical results, noting the different context for such comparisons, the machine learning

approach limited to the specific dataset in hand, the statistical approach based on a wider, population based probability assessment.

15.8 Discussion Points

1. Machine learning and AI approaches to data analysis can be used in many settings where statistical methods are also used. How might you use such methods as a component of the data analysis for your research project. Examine, in particular, the possible use of artificial neural networks and random forest models.
2. Machine learning methods are heavily dependent on the observed dataset. If there are issues with the data, this will affect the usefulness of the analysis. Discuss the possible effects if Simpson's paradox applies to the dataset you are going to analyze.
3. The quality of the dataset you collect, following your research protocol, can be challenging. Through no fault of your own, there may be significant missing data. If it is missing at random, you may assume that the effects of the missing data are averaged out in your analysis. If the missing data are found to have a pattern, for example a higher percentage of females drop out of the study, or younger patients not filling in all the required data forms compared to older patients, then the effect of the missing data may be to bias the results of both statistical and machine learning based data analysis. Discuss the possibility of this occurring in your research study.
4. In regard to your research project, how stable do you expect the dataset to be over time? Do you expect new variables to become important, for example genetic information, over time, or will the dataset remain relevant over time? Do you expect the population of subjects or patients from which the sample is consented and drawn to remain consistent, or will changes in policy alter the population, for example hospital funding and health insurance related changes in regard to health care research.

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16

Basic Time to Event Analysis

The application of time to event or survival analysis [1] has become common in the investigation of time-related data in relation to the onset or treatment of many types of infections and diseases in humans and animal species. The analysis of time to event data in these settings may require the application of statistical nonparametric testing procedures [2].

More commonly, the analysis of time to event data uses survival models, for example the Cox proportional hazards regression model [3], to relate time to event with other explanatory variables. Also available is standard time series analysis [4], if the goal is simply to examine time-related patterns in the response variable itself.

16.1 Time to Event Modeling

Time as a response (time to infection, time to recovery, time to death) does not naturally follow a normal distribution. As such, time-related modeling is usually examined separately from standard linear models.

The key aspects of time to event models include [1]:

- **Time to event** as a response (time to recovery, time to death, time to success, etc.).
- **Censoring** of the data (the event of interest may occur outside the time frame of the study).
- **Kaplan-Meier** curves (graphically demonstrating the percent of those surviving in different groups, adjusted for censoring effects).
- **Wilcoxon Log-rank test** (to test for significant differences between the Kaplan-Meier curves corresponding to the various treatment groups).
- **Cox Proportional Hazards** regression model to model time to event in relation to a set of explanatory variables.
- Statistical estimation and testing for these approaches to modeling are based on large sample assessments of significance using versions of the **maximum likelihood** method.

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In some time frames, responses tend to become correlated within subject, due to multiple observations on individual subjects, seasonal effects, and other cyclical patterns. If time is available as a variable to be collected as part of any research project, time to an event of interest may become a useful response to model, subject to more complex assumptions, and more complex statistical models.

Survival analysis techniques offer experimental researchers a useful set of tools specifically developed for time to event response data.

16.2 Survival Analysis

Where the response of interest is time to an event, a set of statistical models and approaches have been developed that focus primarily on the risk of onset at any time as the central aspect in modelling time-related response variables. This risk or hazard rate [5] is often formally modelled using distributions such as the exponential distribution (which reflects an assumed constant risk of infection) and the Weibull distribution (which reflects a potentially increasing or decreasing risk of infection). Note that this type of analysis began as part of the insurance industry and is closely related to the ideas underlying actuarial tables.

Once the best fitting distribution and related model is chosen, overall differences in median time to onset of the event across the treatment groups can be compared on the hazard scale, often using maximum likelihood based approaches to estimate the parameters in the model.

The Cox proportional hazards model is the most flexible regression model appropriate for this purpose, and there are more formal regression models based on the exponential and Weibull distributions. These are often available in standard statistical packages.

The notion of censored data plays a role in the application of time to event methods. Some subjects, over the course of the experiment, may not become infected. For these individuals, infection time may lie beyond the endpoint of the experiment and will not be observed, so infection is effectively censored by the end of the study. To adjust for this, the probability of the unobserved infection time lying at or beyond the endpoint of the experiment is included in any calculations regarding risk by adjusting the overall probability distribution [6].

16.3 Typical Time to Event Distributions

- Time begins at t = 0; assuming a normal distribution for time responses is often not appropriate.
- We usually need another set of distributions, for example, the exponential, Weibull, log-normal distributions.

- To choose one of these distributions, compare the fit of the histogram of the data to these theoretical distributions using, for example, a chi-square test or graphical analysis. Choose the best fitting distribution.

16.4 Censored Data

- Usually, a research study is conducted within a fixed time period. As noted above, events might occur beyond the time frame of the research study. You will therefore observe an individual in the study for some amount of time, during which they are at risk of onset of the event (disease, infection, death etc.) This effect is called right-censoring of the data. It is often assumed that this occurring at random in your sample.
- There is also left censoring. This affects how the individual subjects in the study are possibly restricted when joining the study. You will also want to assume this occurs at random. But this type of censoring is fairly rare.
- The censoring effect can easily be incorporated into the probabilities underlying the likelihood function and accounted for in the maximum likelihood based calculations [7].

16.4.1 Censoring and Immunes

- It is also worth noting that censored data points (infection time beyond the end of the experiment) and immunes (probability zero of infection) will appear in the observed data similarly. Given the current limited understanding of genetics, this may have affected the analysis of some existing medical studies.
- How to formally include and interpret the infection times of non-responsive subjects in the analysis after the research data is collected is often a matter of interpretation and can affect your statistical results.

16.5 Kaplan-Meier Curves and the Survival Function

- Before testing formal hypotheses, some graphical depiction of survival times may be desired. The most common approach to estimating and displaying survival times is use of the Kaplan-Meier estimator [8].
- This is a very general nonparametric approach to graphically expressing the proportion of a sample that survives at any given time during the experiment.

- They can be adjusted to include censored data.
- These are non-parametric in nature and are usually fit to the data without considering a formal statistical model.
- Note that the Kaplan-Meier curves for each treatment group of interest should not intersect if you intend on using the Cox Proportional Hazards model as the risk of event onset should be proportional in the treatment groups.
- The Wilcoxon log rank test can be used to test for differences between the Kaplan-Meier curves.

If the actual survival function, the probability of remaining alive past time t for a given subject is given by:

$$S(t) = P(T > t) = 1 - F(t)$$

defined at times $t_1, ..., t_k$ where $F(t)$ is the cumulative distribution for infection times, the Kaplan-Meier estimate of the survival function is given by: the Kaplan-Meier estimate of the survival function is given by:

$$S_{KM}(t) = \prod_{j:t_j<t} \frac{n_j - d_j}{n_j}$$

where $t_1, ..., t_k$ are the set of distinct event times observed over the length of the experiment or study, d_j is the number of deaths at time t_j and n_j is the number of individuals alive (the event has not yet occurred) and uncensored just prior to time t_j. This can be adjusted for censoring effects.

This can also be plotted as a function of time and compared for each treatment group. Again, if these are parallel, there is some evidence for the proportional hazards assumption.

Once defined, Kaplan Meier curves can be used, for example, to estimate the times at which the event of interest has occurred for 50% of the sample (median survival time) in a given treatment group. Such approaches are standard in dose-response studies [9] where the percent survival is viewed as a function of dosage.

16.6 Hazard Rate

The hazard rate [4] (risk of infection at a specific point in time, given non-infection up to that point) can be used to define how fast the survival function is changing as a function of time:

$$h(t) = \frac{f(t)}{F(t)} = -\frac{d}{dt}\{\log(S(t))\}.$$

For example, with the exponential distribution we have $f(t) = \lambda e^{-\lambda t}$ and $S(t) = e^{-\lambda t}$. The hazard rate here can be shown to λ, a constant. This implies an assumption of constant risk or hazard if we assume the exponential distribution. While unlikely, this may sometimes be the case for clinical trials of limited duration.

As another example, an assumed Weibull distribution to describe time to an event has the form: $f(t) = \lambda\gamma t^{\gamma-1}\exp(-\lambda t^{\gamma})$ and its survivor function is given by $S(t) = \exp(-\lambda t^{\gamma})$. It follows that its hazard function is given by $h(t) = \lambda\gamma t^{\gamma-1}$. If $\gamma > 1$, the hazard or risk of infection is increasing over time and if $\gamma < 1$, it is decreasing. There are many possible survival functions and related hazards rates.

Maximum likelihood estimation is usually employed in these models to estimate the unknown parameters.

Once the unknown parameters are estimated, the potential distributions and related hazard functions are fit to the data and the best fitting distribution chosen [1].

If the choices are between Weibull and exponential distributions, this is formally accomplished by statistically testing to see if $\gamma = 1$ in the Weibull model. If this is not rejected, the assumed Weibull distribution can be replaced by the simpler exponential probability distribution.

The median survival time ($t(50)$) is simply the time t such that $S(t) = 0.5$. For a Weibull distribution this is given by:

$$t(50) = \{\frac{1}{\lambda}\log 2\}^{\frac{1}{\gamma}}$$

and setting $\gamma = 1$ gives $t(50) = 1/\lambda \log 2$ for an exponential distribution.

When testing for differences between treatments using a survival regression format, overall measures of fit based on the likelihood ratio test statistic have a large sample chi-square distribution and are usually reported as well as individual tests of significance for each variable based on large sample normality.

16.7 Time to Event Modeling: Modeling Strategy

- Focus on modeling the risk of event onset (e.g., infection, recovery, etc.) and related time to onset as the central aspect when modelling time to event variables.

- This risk or "hazard" is modelled using time-related distributions such as the exponential distribution (which reflects an assumed constant risk of the event occurring) and the Weibull distribution (which reflects potentially increasing or decreasing risk of the event occurring). The Cox proportional hazards model is also a possibility.
- Choose the distribution that fits the data best. Then look at overall differences in time to onset between treatment groups.
- If secondary variables such as demographic variables are required in the analysis, a regression type survival data model can be used to relate time as a response to these variables. The hazard rate can be modeled using a generalized linear model format.
- The Cox proportional hazards model is the most flexible regression model, but requires the assumption of proportional hazards between treatment groups. This can usually be observed by seeing if the Kaplan-Meier plots of the survival curves for each treatment group do not cross. There are also formal tests.

The Cox proportional hazards model can be written:

$$h_i(t) = \exp(\beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) h_0(t)$$

as the hazard rate for the ith subject where this risk is assumed to be related to covariates $x_1, \ldots, x_p$.

For the assumption of a Weibull distribution, the baseline (setting all $x_i = 0$) hazard rate is given by $h_0(t) = \lambda\gamma t^{\gamma-1}$. Plugging this in and using the result that the probability density is given by $f(t) = S(t)h(t)$, we can determine the density function modelling the time to event for the ith subject.

Typical graphical outputs beyond Kaplan-Meier curves include plots of the estimated log-scale survival curve versus the log-scale of the cumulative hazard function which can be examined for linearity. Residuals are also available for plotting for general assessment of the assumed regression model [10].

16.7.1 Interaction in Time to Event Models

- Sometimes, some of the x_j regression variables are correlated with each other (e.g., Diet and Obesity, Age category and T-cell count).
- Just as in the case of linear models, this type of correlation can be included in the statistical model by using interaction terms in the model.
- For example, $h(t_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 (x_{1i} + x_{2i}) + \varepsilon_i$, where $h(t_i)$ is the hazard rate for the ith subject and ε_i is random error.
- Note that before we can interpret the β_1 and β_2 parameters, we should to test if $\beta_4 = 0$.

16.7.2 Confounding in Time to Event Models

- This is basically the same idea as in standard linear models.
- To see if the variable x_2 confounds or affects the relationship between y and x_1, first fit the model using y and x_1. Then add in x_2 to the model equation and re-fit the parameters using maximum likelihood estimation. If the estimated value for x_1 alters, then x_2 is said to be a confounding variable, as the relationship of y and x_1 has been affected by the presence of x_2 in the model.
- If x_2 is a categorical variable (e.g., age group), you can stratify the analysis on the levels of x_2. This implies fitting a separate equation for each strata, for example, age group.

16.8 Example

A study is conducted to examine the effect of a specific treatment on a type of cancer. The variables collected are baseline temperature of the patient, an index measure of the size of the cancer mass and whether or not the patient received treatment (yes/no) versus ordinary standard of care. The response of interest is survival time over a 30-month period. The median survival time

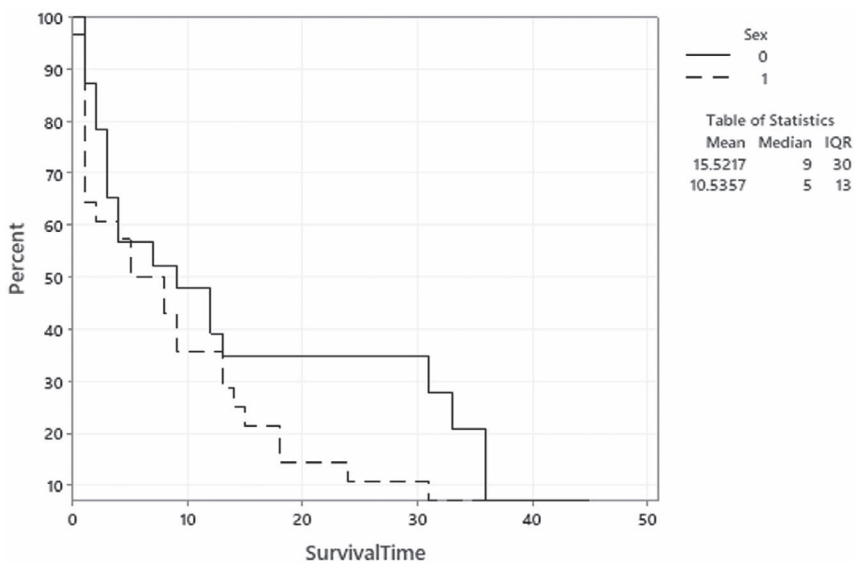


FIGURE 16.1
Kaplan Meier plots.

for the control group is 1.0 months with an IQR of 3.0, and the median survival time for the treatment group is 20.0, with an IQR of 15.0. The Kaplan Meier survival curves for the groups is given in Figure 16.1:

As the survival curves seem proportional, the analysis is conducted assuming a Cox proportional hazards regression model using Group, baseline index and baseline temperature. The p-value here for the group variable is <0.001 showing a significant difference between the groups in relation to survival times.

16.9 Discussion Points

- The use of time to an event as a response of interest can often be added to most standard analyses. Discuss (i) the possible use of such a response in your research project and (ii) how such a time-related response will be affected by the overall timeline of your project.
- If you are able to study a time to event response, is the possibility of right-censored data an issue? Discuss how you will define censored data values.
- The Cox proportional hazards model has several assumptions. Discuss how to verify the proportional hazards assumption if you choose to use the Cox regression model.
- When fitting a time to event model, there are primarily non-parametric models such as the Cox regression model, and parametric models such as exponential regression and Weibull regression. Discuss how you might decide which approach to use.

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17

Recent Topics for Consideration

- New areas of research often open up as technology and knowledge increase.
- They are often driven both by the new technologies themselves and the need to calibrate and interpret how these technologies, and the data they generate, can be interpreted and, for example, used to detect and help fight disease.
- Several recent areas are genetic and epigenetic data, the analysis of large genomic databases, and the definition and study of personalized medicine, for example, the microbiome and aspects of cancer research.
- Also relevant are new methods of analysis, for example machine learning and AI.
- A brief overview of these several connected new areas of medical research is presented here in the hope of encouraging the pursuit of new research areas.

17.1 Genetic and Epigenetic Data

The recent study of genetic data structures, which began with Southern blot electrogenesis technology and other simple DNA-related technologies, has developed into detailed approaches including single nucleotide polymorphism (SNPs), copy number variation, gene splicing and RNA-related deep sequencing [1] and much more [2].

Recently, the rise of epigenetics, the chemical triggers related to gene expression (chromatin, histone, DNA methylation for example) has led to yet another level of complexity, as have the growing number of networks of genes that govern cell developmental and maintenance-related activities [3, 4]. This increase in technology and related complexity will no doubt continue.

Many studies have been conducted to detect the genes and sets of genes related to the onset of disease, or chronic conditions, for example depression. This has led to the detection of many genes, perhaps too many, that are related to specific illnesses, infections and conditions. The OMIM online listing has many of these results [5]. The often overwhelming numbers of genes

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in relation to more frequent conditions such as diabetes and hypertension has led to limited treatments related to genetics. In fact, research has often been re-directed toward the examination of rare diseases that reflect known defects in specific genes [rare dis and genetics [6].

17.1.1 A Weak Baseline

As you have seen in this book, study design is very important when studying an illness or condition and having a useful control or reference group is often required to interpret results.

To detect the presence of an illness or genetic anomaly, one should know what the genetic structures look like for a healthy individual, the genetic baseline for a healthy control group.

The Personal Genome study in Canada [7] has shown that many healthy individuals have what current science would indicate is a serious deficiency in their genes.

From this study, it is obvious that in relation to the onset of illness and disease, genetics often does not currently provide a stable baseline of what can be defined as healthy.

Without such a reference, it is difficult to detect meaningful differences between healthy and non-healthy genetic patterns in individuals.

There is also the issue of immunity, and detecting genes that provide immunity [8] to specific diseases. This remains a mostly unexplored area of research.

17.1.2 Difficult Analyses

Genomic information in the future may be able to predict illness in currently healthy individuals. DNA sequencing and electronic health records may become standard elements in health care based research. Combining these with real-time monitoring of diet, exercise, blood pressure, heart rate, etc., may lead to better ways of preventing or treating chronic diseases.

But currently the lack correlation between specific genes and the onset of disease is a challenge.

Each of us may have genetic variants or mutations that may put us at increased or decreased risk of various diseases, and make us more or less responsive to specific treatments. But for most common diseases, as noted earlier, hundreds of genes with small effects have been identified, and it is hard to develop a clear picture of who is really at risk for what and when.

It had been thought that just a few genetic variants would be found that would account for most disease risk, but for common conditions such as diabetes, heart disease, and most cancers, no clear genetic pattern has been identified in the majority of cases.

For example, some cancers seem to have a random onset [9], with standard risk factors such as age, sex, body weight serving as better predictors of Type

2 diabetes, than, for example, a genetic expression-related score. Age, for example, is an important underlying factor as greater age generally implies a weaker immune system, one which is less capable of detecting and destroying cancer cells.

Currently the usual advice for those at risk for most illnesses is to exercise more, eat healthy and get enough sleep. This advice often remains the same whether or not an individual has genes that might put them at greater risk.

A finding of the Human Genome Project [10] was that while statistics suggest a genetic explanation for diseases and conditions running in families, individual genes or genetic variants do not often correlate with increased risk for the onset of disease.

An example of this is depression. Clear genetic effects in relation to the onset of depression have not been found that hold across a variety of study design settings [11], though it does run in families, perhaps as a learned behavior.

While distinct from cross-sectional genomic studies, population genetics argues that genes that can be seen as mutations or alleles (versions of the original gene) having a negative effect on the ability to have offspring will have a lower chance of moving forward into the next generation, and will soon disappear from the population. This is an argument against genetic causation for depression (and other conditions such as autism).

This does not stop researchers from examining genetics in these settings (e.g., the related genetics of the immune system).

Many attempts to use genetic variants to better target drug therapies have been conducted with limited success [12].

For rare diseases like cystic fibrosis, drugs have been developed using genetic information, but the repair of defective genes remains in its infancy, though CRISPR based methods [13] and improved gene therapy are hopeful approaches.

17.1.3 Example

For example, an important area of the application of genetics is in relation to the determination of pathways relevant to the treatment of cancer.

The Cancer Genome Atlas [14] and similar research has shown that each patient's cancer tumors harbor a unique set of genes related to growth, and these predict whether a specific cancer will respond to treatment.

With a long list of known pathways involved in the development and progression of cancer tumors, there may be an opportunity to match genomic changes in an individual's tumor with specific drugs limiting those changes and to develop new drugs for newly discovered genetic pathways.

A possible aid here is the application of machine learning and AI methods which can cycle through many possible scenarios. However, much of the onset of cancer has been shown to be at random and may simply reflect the

inability of the individual's immune system to destroy the cancerous cell. Thus, age is often correlated with the occurrence of cancer. But as with all things genetic, this only accounts for a small proportion of total cancer cases. When the rules of the game are not clear, the algorithmic search and optimization techniques underlying machine learning and AI may not be as useful as desired.

17.2 New Types of Data

The analysis of the human genome and its expression in relation to proteins and cell development has driven the development of much high dimensional data analysis. SNPs, CNVs, genetic splicing, gene switches, and chromosome elements occur in the tens of thousands in any given patient and present datasets that remain difficult to analyze and interpret.

What does it mean for hundreds of correlated genes that form a genetic network to be significant? What is a gene? The individual pieces or the entire network? The ability of DNA and RNA to generate a large number of spliced versions is often hard to track and the complexity is often not incorporated into genetic data based studies.

The onset of relatively inexpensive genomics has led to genetic assessments becoming a standard component of research studies, whether the main focus of the study is genetics or not.

17.2.1 Statistical Modeling Issues

When attempting to model such data, if the considerations mentioned above are not addressed in the basic statistical or mathematical model underlying the gene expression analysis, results can be misleading.

The non-inclusion of gene clusters, the non-inclusion of multi-stage expression patterns through time, the lack of appropriate scaling, can all affect the accuracy and relevance of the model to be employed, no matter the statistical analysis.

In larger datasets with many variables and levels of stratification this is even more relevant.

Recent advances in detecting the geometry of chromosomes in the cell have underlined the need to consider more complex models than are currently being employed in GWAS and related work. The 3-D nature of the information in the chromosome [15] may affect the automatic use of a linear model based analysis for expressions of genetic components of the chromosome.

Genetic data, imaging data, health outcomes and clinical data, spatial positioning data, internet based data; all are examples of settings where the flow of data is massive and the ability to analyze such a flow is typically restricted [16].

A very large number of variables and relatively few subjects are often the mark of such data and the goal of the analysis is typically to detect various patterns within the overall dataset.

Often standard methods of statistical analysis often do not hold up well in such settings.

Multiple comparison issues where a large number of case-control comparisons are conducted require careful application and interpretation [GWAS]. Indeed, multiple testing of one-at-a-time mean differences may be of limited use for understanding of genomic and epigenetic data structures or networks of genes to specific cell and phenotypic structures.

Correlation and embedded nonlinear relationships also cause difficulty for linear models. The application of linear models to correlated structures may not be appropriate, with any nonlinear functional relationships potentially going undetected and creating instabilities in the predictive model.

The analysis of genes remains challenging and very much an open field for interesting research study, if the technology and support is available.

17.3 Tools for Analysis

17.3.1 Artificial Neural Networks

A standard machine learning approach to relating a binary outcome to the features/variables in the data matrix is artificial neural networks (ANNs). Neural networks are methods of classification that are essentially nonlinear recursive functions.

In regards to supervised neural network modeling, these often use a multiple hidden layer model structure, applying back-propagation methods and multiple replications to fit the model. The sigmoid function serves as an activation function in the hidden and output layers.

The output for each element of the response variable is a yes/no classification value, similar to logistic regression. ROC, AUC and other measures of classification accuracy and error can be generated and compared to frequentist and other methods. See for example, Chapters 14 and 15.

17.3.2 Random Forest Models

Another commonly applied machine learning approach is random forest models.

The supervised random forest algorithm depends on a large number of individual classification trees. Each classification tree yields a predicted

class and the class with the greatest frequency becomes the model prediction.

The individual trees are generated using bootstrap resampling. The algorithm uses randomly selected samples (sampling with replacement) typically generating a large number of relatively uncorrelated classification trees (models).

These elements are then combined to provide a robust classification of the variable or response of interest.

Final predictions and model fitting are made by taking the average of the predictions made by each individual decision tree in the forest.

17.3.3 Deep Learning Algorithms

The development of deep learning approaches to supervised classification approaches requires the development of detailed computational and algorithmic architectures, sometimes borrowed from similar studies (e.g., the use of algorithms and structures to help develop COVID-related predictions).

Once the basic computational structures and related algorithms are initialized and shown to detect and incorporate data structure and patterns, they can be trained on a subset of the data and validated on a secondary testing component of the data.

There is a great deal of interest and application of AI and machine learning methods. They have interesting and useful aspects, but are sensitive to data structures and sample size [17–19].

17.4 Personalized Medicine: Genetics, Biomarkers, Microbiome, and AI

The increasing amount of information available on any given individual implies the possibility of a personalized approach to medicine.

Indeed, this may be required to deal with cancer and the destruction of cancer cells.

The NIH currently defines personalized medicine as an emerging practice of medicine that uses an individual's genetic profile to guide decisions made in regard to the prevention, diagnosis, and treatment of disease.

17.4.1 Personalized Medicine and the Study of Omics

This can be extended beyond the genome and may include biomarkers, and microbiome elements. The microbiome in the gut has literally millions of potential bacteria, parasites and a wide variety of organisms

Personalized medicine is based on the availability of a very large amount of data on each individual: genetic expressions, biomarkers, proteins, bacteria in the microbiome. This includes cancer cells which show a large amount of heterogeneity. This type of data seems to suggest that advances in medicine may be found by concentrating on the individual and the local environments within the individual.

In terms of available resources, this type of focus will lead to the spending of large amounts of time and resources on individuals who can support these approaches. This will further limit the availability of resources for the many who cannot support such personalized efforts.

There are serious arguments against such approaches. A key one is that human beings are remarkably inbred. Genetic studies suggest that as recently 20,000 years ago there were very few humans alive [20]. This limits the variation to be expected the population. In fact, several population bottlenecks seem to have occurred. This implies a high level of genetic similarity between people, implying that treatments that work on small numbers of representative subjects should work broadly in the population. This seems to have been the case for many medicines and vaccines.

There is also the practical issue of clinical validity. Methods based on personalized medicine may improve treatment, but by how much? Will this provide a useful cost-benefit result?

Omics data, very large databases, requires unique methods of analysis. The use of AI and ML here may be essential to detecting relevant patterns in such datasets. Algorithms that can loop through many possible scenarios representing specific hypotheses will be useful, but it remains very challenging to interpret such results.

Polygenic risk scores [21] have become the fashion for analyzing data where there are many variables (e.g., 20,000 gene expressions or 500,000 SNPs), but these are highly questionable in terms of interpretation if genes, immunological biomarkers and other organisms do not operate independently, but as part of a system of responses. If this is the case the individual genes, for example, are expressing as a system of correlated values. Using polygenic scores essentially adds up the same thing over and over when expression is numerically correlated. More complex approaches are required. The use of AI and ML here again may be essential to detecting relevant patterns in such datasets. Note also that these considerations raise the question of what practically is a gene? An individual set of elements on a chromosome, or a set of such elements, structured in ways that overlap and use the same elements in different ways.

The analysis of the microbiome, like genetics, is generating many individual results, but perhaps too many of them, leading to confusion as to how to proceed. This is type of area if challenging, but a very interesting one to investigate. There are advances to be made.

17.5 Replication and Study Designs

In new areas of research there is a need to carefully identify useful pathways to serious results. Gene-expression studies, for example, often have a poor track record of replicability. Lab scientists also assume that we have observed levels of variation in randomized clinical trials that could only have a genetic difference. Clinical trial researchers assume that differences observed by geneticists in the lab that must mean differences in the clinic, but the many secondary variables (demographics, lifestyle, exposure)

Geneticists aren't using a random sample of mice; they're using pure strains and looking at what happens when they introduce a change.

Similarly, randomized clinical trials aren't filled with random people. Instead, they're selecting people who meet very specific criteria for each trial.

17.5.1 Variability Within and Between Subjects

A second challenge with customizing medicine is understanding variability. While there are clearly genetic variations in how people respond to treatments, that does not necessarily imply that an individual's response is due to a genetic variation.

The presence of high levels of variability at both the subject and genetic level has often made it very difficult to apply standard statistical methods and obtain statistically significant results.

Variation and heterogeneity are serious issues and reflect limitations of the experimental design. In response to this in research related to cancer treatments and the microbiome, the use of personalized medicine based on the single individual of interest has become a focus.

17.6 Discussion Points

- Discuss how you might investigate a possible new area of research. Discuss possible keyword searches in a resource such as PubMed to guide your investigation. What areas of application might you be interested in applying new technologies and methods of modeling?
- Examine how personalized medicine has been applied in your field of research. Carry out a review of the literature for the past two years, summarizing how these methods have been applied, and how these methods and approaches are defined.

- If you have decided to use genetic methods in your study, discuss the challenges you will have to deal with, including: (i) access to samples (ii) consent issues (iii) limitations in regard to the understanding of genetics itself, (iv) methods of calculations and the set of statistics you will use to summarize genetic information.
- In the context of your research project, you have decided to address microbiome issues. What are the practical challenges to including a microbiome aspect in your research project? Summarize, briefly, examples of the use of microbiome in your field of research over the past two years.

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18

Conclusion

There are many aspects to a successful medical research project. The topics covered here, including the suggestions in regard to developing research project ideas, pulling together a research team (including finding a mentor), the type of project and study design that is possible in your research setting, securing the support and time you will need to accomplish the project, dealing with the administrative aspects, and making sure you have prepared the necessary skills and focus, will help give you an idea of which projects are practical within your time frame.

When leading a multidisciplinary team, you need to be aware of what each expert can give to you and what to reasonably expect from them.

If you are the primary researcher or P.I. of the research project, then the responsibility falls on you to move the project forward and keep everyone working together, on schedule.

A focus here has been the need for understanding the development of an adequate study design and related data collection procedure, data analysis, statistical inference such as p-values and confidence intervals, examples of statistical results and the concepts that underlie statistical analysis, as well as some more advanced statistical models such as time-to-event analysis.

If you are interacting with a data analyst, you should try to master some of these statistical concepts in order to be able to communicate the types of analyses you would like to report in your research project. Qualitative data is also a possibility and the use of open-ended survey questions to personize the data collection process.

The administrative aspect of carrying out research in an institutional setting has also been reviewed. This is often the most challenging aspect of the early part of the research project, but it is necessary.

Many suggestions have been given in regard to possible ideas and approaches you can use to develop projects that others will find of interest. Some new areas of research have been reviewed and introduced.

Hopefully, the outlines and discussions given here will help you move forward with your own research.

To summarize in bullet form, there is a basic chronology underlying most research projects.

Typically, within a two-to-three year time window, the following components, if applicable to your area(s) of research, will have to be considered, given some thought, and accomplished. This sequence assumes that you have the appropriate education and training for the following:

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- Research idea and project development.
- Identifying a mentor.
- Background search/literature review.
- Basic protocol written and justification/expected results.
- IRB and administrative review.
- Research team development.
- Search for funding.
- Initial pilot data.
- Final protocol.
- Final funding decisions.
- Practical implementation of the project.
- NIH consort diagram-related recruitment of subjects.
- Application of study design.
- Data collection.
- Completion of data collection phase.
- Data analysis.
- Statistical inference.
- Machine learning applications.
- Results.
- Presentations.
- Manuscript development.

If you are determined, you will be able to overcome all the challenges. But the most important aspects are developing a good idea and making sure you have the support and access to data that is necessary. The first few steps are always the most difficult, as decisions must be made that will affect the project for the next few years.

In terms of the research context, understanding your science, clearly defining hypotheses and selecting the appropriate study designs and related study protocol are key. If these are in place, the next challenge is carrying out the protocol, dealing with the practical challenges, as the best laid plans often have to deal with the realities of consenting adequate numbers of patients and dealing with the administrative aspects of the research project.

Give yourself adequate time and resources. Focus and it will eventually get done.



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